



Behavioral Economics of Well-Being: Scarcity, Social Rank, and Media Exposure among
Thai Adults in Bangkok

Group 3

Submitted by

Kanokphan Sirithienthong	6610750092
Kitgawin Kunsansuphakit	6610750122
Nathorn Apichaikriengkrai	6610750190
Nutkamon Pongprasit	6610750203
Thamakorn Thanaboon	6610750297
Posanat Sapianchai	6610750408
Roybunya Chonpura	6610750491
Kittiya Kaenchan	6610754161

Presented to

Assoc.Prof. Siwamong Dheera-aumpon, Ph.D.

Asst. Prof. Thanasin Tanompongphandh, Ph.D.

Research Methods in Economics 01101491 (Section 415)

Faculty of Economics, Kasetsart University

2nd Semester, Academic Year 2025

Abstract

As digital environments increasingly redefine perceptions of poverty from material lack to comparative visibility, understanding the psychological mechanisms that shape well-being has become essential. This study investigates the interactions between psychological scarcity, subjective social status, and passive upward social media exposure among working-age Thai adults in Bangkok, drawing on social comparison theory and the scarcity-as-bandwidth-tax framework. Using a cross-sectional online survey of 411 respondents aged 18–44, subjective well-being is measured via the WHO-5 Well-Being Index, psychological scarcity via an adapted CFPB Financial Well-Being Scale, subjective social status via the MacArthur Ladder, and passive exposure via a three-item scale adapted from Verduyn et al. (2015). OLS regression with heteroskedasticity-robust standard errors across three nested model specifications reveals that all three predictors are independently significant. Psychological scarcity is negatively associated with well-being ($\beta = -3.68$, $p < 0.001$), subjective social status is positively associated ($\beta = +3.27$, $p < 0.001$), and passive upward social media exposure is the strongest single predictor of well-being loss ($\beta = -9.62$, $p < 0.001$). Critically, passive exposure acts as a conditional amplifier of well-being inequality: the interaction term with subjective social status is negative and significant ($\beta = -2.678$, $p < 0.001$), confirming that heavy passive consumption of aspirational content progressively erodes the well-being benefit of higher perceived social rank, with the indifference point identified at SubStat ≈ 3.22 , beyond which passive exposure has a consistently negative effect on well-being. Notably, 49.4% of respondents scored at or below the WHO-5 clinical threshold, highlighting a widespread burden of impaired well-being in the sample. These findings support targeted interventions including digital literacy programmes, income-based relief, and embedded financial counselling to address the hidden psychological costs of social media penetration in Thailand.

Acknowledgments

Our group would like to express our deepest gratitude to Assoc. Prof. Siwapong Dheera-aumpon, Ph.D., our primary advisor, for his invaluable guidance, continuous encouragement, and insightful feedback throughout the course of this research. His expertise and support were instrumental in shaping both the direction and quality of this study.

We are also deeply thankful to Asst. Prof. Thanasin Tanompongphandh, Ph.D., for his dedicated mentorship, thoughtful feedback, and constructive critiques. His involvement from the initial stages of this project to its completion greatly enhanced our understanding and encouraged us to approach the research with greater critical thinking and diligence.

We would like to extend our appreciation to all participants who generously contributed their time and responses to the survey. Their participation was essential to the success and depth of this research.

Finally, we would like to thank our families and friends for their encouragement, patience, and unwavering support throughout this journey.

This research would not have been possible without the support and contributions of all those mentioned above. We are truly grateful.

Group 3 Members

Table of Contents

Chapter 1: Introduction	12
1.1 Introduction.....	12
1.2 Objective.....	14
1.3 Benefit.....	14
1.4 Scope.....	15
Chapter 2: Literature Review	17
2.1 Literature Review	17
2.2 Conceptual framework.....	28
Chapter 3: Research Method	31
3.1 Research Design and Data Collection	31
3.2 Sample Size Determination	32
3.3 Measurement Instruments.....	32
3.3.1 Dependent Variable: Subjective Well-Being (WHO-5 Index)	32
3.3.2 Independent Variable: Psychological Scarcity (PsySca)	34
3.3.3 Independent Variable: Subjective Social Status (SubStat)	36
3.3.4 Independent Variable: Passive Upward Social Media Exposure (PasExp)	38
3.3.5 Control Variable: Offline Social Support (OSSS-3).....	41
3.3.6 Socioeconomic and Demographic Control Variables	43
3.4 Data Analysis Strategy	53
3.4.1 Descriptive Statistics.....	53
3.4.2 Measurement Validation	54

3.4.3 Econometric Models	55
3.4.4 Robustness Checks.....	57
Chapter 4: Results and Interpretation	59
4.1 Sample Description and Variable Distributions	59
4.1.1 Subjective Well-Being (WHO-5).....	60
4.1.2 Subjective Social Status (SubStat).....	60
4.1.3 Passive Upward Social Media Exposure (PasExp) and Usage Hours	61
4.1.4 Psychological Scarcity and Financial Controls.....	63
4.1.5 High-Status vs. Low-Status Profile.....	65
4.2 Pre-Estimation Diagnostics	66
4.2.1 Heteroskedasticity	66
4.2.2 Residual Normality	67
4.2.3 Multicollinearity.....	67
4.2.4 Outliers and Leverage	67
4.3 Main Regression Results	69
4.3.1 Model 1: Baseline Specification	71
4.3.2 Model 2: Adding Occupation (Industry Group)	72
4.4 Model 3 (Moderation Analysis): The PasExp $\times$ SubStat Interaction	73
4.4.1 Interpretation of the Interaction Equation	73
4.4.2 M3 and M3c: Uncentred vs Mean-Centred Specifications	75
4.4.3 Marginal Effects Across the Full Ladder Range.....	76
4.4.4 Interaction Surface	79

4.5 Industry-Group Heterogeneity in Marginal Effects.....	79
4.6 Robustness Checks	84
4.7 Summary of Findings	85
Chapter 5: Conclusion	87
5.1 Hypothesis Evaluation and Theoretical Contributions	87
5.2 Policy and Practical Implications	90
5.2.1 Differentiated Strategy by Status Group	90
5.2.2 Actionable Policy Recommendations	92
5.3 Limitations and Future Directions	96
5.4 Conclusion	97
Reference.....	99
Appendix	105

List of Figures

Figure 1 Conceptual framework of the relationships between psychological scarcity, subjective social status, passive upward social media exposure (and their interaction), and subjective well-being (WHO-5), controlling for socioeconomic and demographic control variables.....	29
--	----

List of Charts

Chart 1 Satisfaction with Life and Income per Capita in Japan between 1958 and 1991	9
Chart 2 Distribution of core study variables (n = 411).	55
Chart 3 Distribution of WHO-5 Well-Being Index (n = 411). Mean = 53.62; 49.4% at or below clinical threshold (≤ 50).	56
Chart 4 Distribution of Subjective Social Status (MacArthur Ladder, 1–10). Mean = 6.24. ...	57
Chart 5 Distribution of Passive Upward Social Media Exposure (PasExp)	58
Chart 6 Distribution of Daily Social Media Hours (SOC_HRS).	58
Chart 7 Distributions of Psychological Scarcity (PsySca)	59
Chart 8 Distributions of CFPB Liquidity	60
Chart 9 Distributions of Household Income (HHInc)	61
Chart 10 Distributions of Offline Social Support (OSSS-3)	61
Chart 11 Residuals vs. Fitted values, M1 plain OLS. Structured rails reflect WHO-5's discrete 4-point increments; no systematic heteroskedastic funnel pattern is present that would invalidate robust-SE correction.	63
Chart 12 Leverage vs. Normalised Residual ² — M1. High-leverage points (top left quadrant) have near-zero residuals and do not constitute influential observations. Dashed lines show sample-mean thresholds.	65
Chart 13 Marginal effect of passive social media exposure (PasExp) on WHO-5 well-being across social rank values (M3). Dashed red line = zero (null effect). Confidence intervals computed via delta method.	76
Chart 14 Marginal effect of subjective social status (SubStat) on WHO-5 across passive exposure levels (M3). The protective benefit of higher social rank diminishes and eventually reverses at high exposure levels.	77
Chart 15 Predicted WHO-5 well-being scores across the PasExp $\times$ SubsStat interaction surface (M3). Colours represent predicted WHO-5 values; each line is a fixed PasExp level.	78
Chart 16 Marginal effect of SubStat on WHO-5 by industry group (M3)	80
Chart 17 Conditional marginal effects of PasExp	81

Chart 18 Marginal effects of PasExp (left), SubStat (centre), and PsySca (right) on WHO-5 by industry group (M3), evaluated at each group's conditional means. Red dashed line = zero; error bars = 95% CI.....	81
Chart 19 Predicted WHO-5 well-being scores by industry group at each group's conditional mean values of all predictors (M3).	82
Chart 20 Placement of industry groups in the PasExp–SubStat interaction space. Points represent individual respondents, with group centroids overlaid. Industries with higher mean SubStat and higher PasExp fall in the zone of greatest interaction-driven well-being loss.	83
Chart 21 Interaction surface of PasExp marginal effects across social rank for each industry group. The sign-change point ($\approx$ SubStat 3.22) is consistent across groups; the slope steepness differs by group mean SubStat.....	84

List of Tables

Table 1 The variable from the research paper	25
Table 2 The World Health Organization-Five Well-Being Index (WHO-5).....	30
Table 3 Psychological Scarcity (PsySca) items and response scale as administered	32
Table 4 Passive Upward Social Media Exposure (PasExp) items and response scale as administered.....	36
Table 5 Offline Social Support Scale (OSSS-3) items and response options as administered	40
Table 6 Household monthly income bracket classification and sample distribution (n = 411)	42
Table 7 Age group classification, dummy variable coding, and sample distribution (n = 411)	43
Table 8 Gender classification, dummy variable coding, and sample distribution (n = 411) ...	45
Table 9 Living arrangement classification, dummy variable coding, and sample distribution (n = 411).....	47
Table 10 Industry group classification, coding scheme, and sample distribution (n = 411) ...	49
Table 11 Descriptive statistics of study variables (n = 411).....	51
Table 12 Socioeconomic and psychological profile of low-status ($\text{SubStat} \leq 5$) vs. high-status ($\text{SubStat} \geq 6$) respondents.	62
Table 13 OLS regression results — M1 (Baseline), M2 (+Occupation), M3 (Interaction). Robust standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.10$. Reference categories: male, 18–24 years, living with family, Corporate/Professional sector.	69
Table 14 below presents both specifications side by side.....	73
Table 15 Marginal effect of PasExp on WHO-5 at each level of SubStat (M3, n = 411). Shading indicates sign-change zone ($\text{SubStat} \approx 3.22$). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$	76
Table 16 Conditional marginal effects of PasExp and SubStat on WHO-5 by industry group (M3), evaluated at each group's mean covariate values. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$	76

0.05. ME(PsySca) = -4.097*** (constant across groups as there is no PsySca interaction term).....	79
Table 17 Robustness of M3 core coefficients across alternative specifications. Standard errors in parentheses. *** p < 0.001; ** p < 0.01; * p < 0.05.....	85
Table 18 summarises the hypothesis evaluation against the empirical evidence.	88
Table 19 Differentiated intervention framework for aspirational seekers (low status) and loss-aversion individuals (high status).	91
Table 20 presents these recommendations alongside their empirical grounding and proposed implementation measures.....	93
Table 21 Measurement instruments, scoring procedures, and psychometric properties (n = 411)	116
Table 22 OSSS-3 original published scoring reference	117
Table 23 Summary of Key Themes in the Literature Review	129

Chapter 1: Introduction

1.1 Introduction

The digital era has redefined poverty. It is no longer understood only as a lack of material resources. Instead, poverty increasingly reflects the constant and comparative visibility of others' abundance on social media platforms. Understanding this psychological shift is critical because it fundamentally alters how happiness and mental well-being are formed. Recent evidence shows that subjective perceptions of economic sufficiency predict mental health more strongly than objective income alone.

Chart 1 shows the relationship between Satisfaction with Life and Real GDP per Capita in Japan between 1958 and 1991. It can be seen that even though real GDP per capita increased, life satisfaction remained largely unchanged. This finding highlights the importance of psychological scarcity, the persistent feeling of not having enough, which plays a central role in shaping well-being. Individuals experiencing material stress report significantly lower life satisfaction, and this relationship remains significant even after controlling for income level (Knies, 2022; Sekulova et al., 2023).

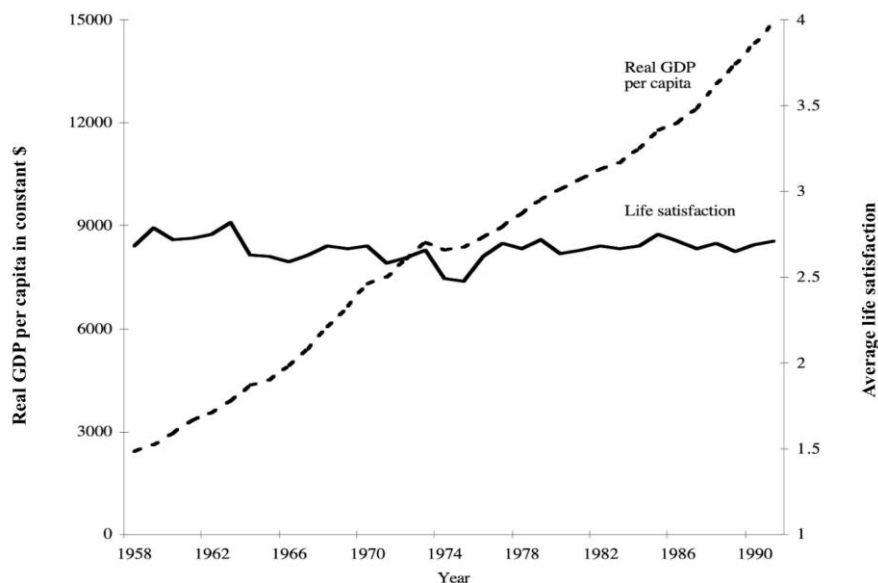


Chart 1. Satisfaction with Life and Income per Capita in Japan between 1958 and 1991
Sources: Frey and Stutzer (2002).

In parallel, subjective social status has emerged as a powerful determinant of mental health. Individuals who perceive themselves as occupying a lower social rank consistently exhibit poorer psychological outcomes. This pattern holds regardless of their objective socioeconomic position (Gross-Manos & Bradshaw, 2022).

These findings align with social comparison theory, which proposes that individuals evaluate their own worth through comparison with others (Festinger, 1954). Thailand provides a compelling context for this phenomenon. According to the Digital 2025 report, the country has approximately 51.0 million social media users, representing 71.1 percent of the total population (Digital 2025, 2025). In this hyper-connected society, social media platforms intensify upward social comparison by repeatedly exposing users to curated and idealized lifestyles. Prior research demonstrates that passive consumption of online content has a stronger negative association with subjective well-being than active engagement. This is because it reinforces feelings of inadequacy and relative deprivation (Verduyn et al., 2017).

Although urban development in Thailand has largely improved access to basic physical needs, disparities in mental well-being are increasingly driven by psychological mechanisms. These disparities are no longer explained by material deprivation alone. Economic insecurity continues to undermine well-being even in digitally enabled labour markets, with perceived stability proving more important than absolute income (Li et al., 2025). Despite this shift, most existing studies continue to emphasise income or total screen time. Therefore, overlook the critical interaction between psychological scarcity, perceived social rank, and passive upward social media exposure. These interactions play a key role in shaping mental health outcomes.

Addressing this gap, the present study investigates the relationship between psychological scarcity, subjective social status, and mental well-being among Thai adults. Specifically, it examines whether passive upward social media exposure amplifies the effects of psychological scarcity and subjective social status on well-being. By focusing on these social psychological pathways, the research aims to clarify how digital environments reshape mental health inequality beyond traditional economic indicators.

1.2 Objective

This study aims to examine whether passive upward social media exposure erodes the psychological buffer of ignorance and negatively affects mental well-being among Thai adults by amplifying the adverse effects of psychological scarcity and lowering subjective social status. Specifically, the research investigates the direct relationship between psychological scarcity and subjective well-being. As measured by the WHO-5 index, among Thai adults who use social media. It also examines the direct relationship between subjective social status and subjective well-being within the same population.

Furthermore, the study tests whether passive upward social media exposure moderates and strengthens the negative effect of low subjective social status on subjective well-being. It also assesses whether the same moderating effect applies to the relationship between psychological scarcity and subjective well-being.

Finally, it determines the extent to which psychological scarcity and subjective social status mediate or explains the observed association between passive upward social media exposure and subjective well-being. The analysis controls objective socioeconomic factors such as income, occupation, and living arrangement. It also accounts for demographic variables, including age, gender, and social support.

1.3 Benefit

This study offers meaningful contributions across academic, practical, and policy domains by exploring how passive upward social media exposure interacts with psychological scarcity and subjective social status. These interactions help explain their influence on mental well-being among Thai adults.

Academically, the research adds new empirical evidence to the literature on digital social comparison by testing a moderated pathway that extends classic theories of relative deprivation and social comparison to the Thai context. It strengthens understanding of how psychological mechanisms operate in digitally connected societies, particularly where

objective economic conditions may no longer fully explain well-being disparities. By validating key scales such as the WHO-5, CFPB Financial Well-Being Scale, and MacArthur Ladder in a Thai adult sample. The study also supports future cross-cultural research on subjective well-being in emerging digital economies.

Practically, the findings provide actionable insights for mental health professionals, university counselors, and student support services to identify individuals at higher risk of well-being decline due to frequent exposure to curated wealth and status content. The results can inform the development of targeted interventions, such as digital literacy programmes or mindfulness workshops. These interventions can encourage mindful scrolling habits and reduce passive comparison tendencies. Parents, educators, and community leaders can also benefit from recognizing early signs of psychological strain linked to perceived social rank and scarcity, enabling more effective guidance and emotional support for young adults navigating digital environments. On the policy level, the evidence can guide Thailand's Ministry of Digital Economy and Society and Ministry of Public Health in incorporating awareness of psychological scarcity and upward comparison into national digital well-being initiatives and mental health promotion strategies. The study highlights the need for regulatory or platform level measures, such as content labeling or reduced algorithmic emphasis on luxury lifestyles. To mitigate negative mental health externalities in a country with high social media penetration. Ultimately, by demonstrating that well-being inequality is increasingly perceptual and digital rather than purely material. The research contributes to Thailand's broader sustainable development priorities around quality of life, mental health equity, and responsible digital inclusion.

1.4 Scope

This study targets Thai adults aged 18 years and older who actively use at least one social media platform, such as Facebook, Instagram, TikTok, X, or YouTube, at least once per week and currently reside in Thailand. This population is selected because it represents the demographic group with the highest social media penetration in the country and is most likely to encounter passive upward exposure to wealth and status content.

The sampling method combines non-probability convenience sampling with snowball sampling via online channels, including university Line groups, Facebook, and Instagram posts. Also, personal networks, and Thai online communities, with participants encouraged to forward the survey link to eligible contacts. The minimum sample size is 384 respondents, calculated using Cochran's formula for an unknown population at a 95% confidence level and 5% Margin of error with maximum variability assumed ($p = 0.5$). However, to ensure sufficient statistical power for detecting the interaction term in moderated regression, the target sample size is increased to 400 or more respondents.

Data collection will be conducted through a self-administered online questionnaire created in Google Forms in the Thai language, with an estimated completion time of 3–5 minutes and is planned for March to April 2026. The questionnaire includes screening questions to confirm eligibility, demographic information, and the core measures of psychological scarcity, subjective social status, passive upward social media exposure, and subjective well-being.

The study focuses on social-psychological mechanisms rather than clinical personality traits or detailed objective material deprivation beyond basic income controls. It also excludes total screen time as a variable due to its high correlation with passive exposure and weaker predictive power in prior research. It also does not include clinical diagnoses of depression or anxiety, as the focus remains on hedonic subjective well-being measured over the past two weeks via the WHO-5 index. As a cross-sectional survey relying on self-report data and non-probability sampling, the design limits causal inference and generalizability, with potential risks of common method bias, urban response bias, and reverse causality. These limitations are acknowledged, and the study is positioned as exploratory and associational rather than definitive.

Chapter 2: Literature Review

2.1 Literature Review

Festinger (1954) proposed the Social Comparison Theory to explain how individuals evaluate their opinions and abilities. The theory posits that in the absence of objective benchmarks, individuals rely on comparisons with others to assess themselves. This process is particularly relevant in ambiguous contexts where direct evaluation is not possible. The theory further suggests that such comparisons influence self-perception and behavior. These findings establish the fundamental psychological mechanism underlying social comparison, providing the theoretical basis for understanding how passive exposure to others on social media can influence subjective social status and well-being in the proposed framework.

Kahneman and Tversky's (1979) Prospect Theory establishes that individuals evaluate outcomes relative to a reference point and feel losses more acutely than equivalent gains (loss aversion). Applied to digital social comparison, idealized media content establishes a salient upward reference point that can shift individuals into a perceived loss domain, reducing well-being. This framework perfectly explains heterogeneous responses to social media. It implies that identical content may trigger aspirational motivation for some people. And the harmful loss of aversion depends on their baseline social position.

Frey and Stutzer (2001) reviewed the role of happiness research in economic analysis, highlighting the limitations of traditional economic indicators in explaining well-being. The study showed that factors such as unemployment, governance quality, and social conditions have substantial effects on happiness beyond income levels. The authors also discussed paradoxes such as the lack of long-term increases in happiness despite economic growth. These findings suggest that economic variables alone cannot fully explain well-being, emphasizing the importance of incorporating psychological and institutional factors. This supports the need for a multidimensional model that includes subjective perceptions alongside objective indicators.

Howell and Howell (2008) conducted a meta-analysis of 111 samples to examine the link between economic status and well-being in 54 developing nations. The results showed that the relationship between income and life satisfaction is the strongest in low-income economies. This effect is particularly significant for samples with the lowest levels of education. The study found that wealth acts as a stronger predictor of well-being than current income flows. These findings suggest that financial resources are primarily used to satisfy basic biological needs in these contexts. This validates objective sufficiency as a critical foundation for happiness within the proposed model. It proves that material security must be established before digital factors can be analyzed as mediators.

Kahneman and Deaton (2010) examined the relationship between income and two distinct dimensions of subjective well-being using data from more than 450,000 respondents in the United States. The study differentiated between life evaluation and emotional well-being, employing large-scale survey analysis from the Gallup-Healthways Well-Being Index. The results showed that income is strongly associated with higher life evaluation, but its effect on emotional well-being plateaus is beyond approximately \$75,000 annually. Lower income levels, however, significantly increase emotional distress related to life hardships. These findings suggest that while income is essential for improving life evaluation, it has limited influence on daily emotional experiences. This supports the argument that objective economic sufficiency is necessary but insufficient for overall well-being, reinforcing the need to consider psychological and perceptual mechanisms in the proposed model.

Mullainathan and Shafir (2013) developed a framework on scarcity and its effects on thinking and decision-making. They show that scarcity, especially financial scarcity, reduces cognitive bandwidth and makes individuals focus more on the short term. As a result, people tend to make poorer decisions and experience higher psychological stress. The authors also argue that scarcity is not only a material condition but also a psychological state that shapes behavior. This framework helps explain psychological scarcity and shows how it links objective economic conditions to subjective well-being, supporting its role as a key mediator in the proposed model.

Euteneuer (2014) reviewed the empirical evidence linking Subjective Social Status (SSS) to physical and mental health. Using a narrative review approach to recent cross-sectional and longitudinal findings. Focused on the MacArthur Scale of Subjective Social Status. Included biological risk factors such as inflammation and susceptibility to infection. Evaluated the predictive power of SSS beyond traditional objective markers like income and education. To identify biological pathways connecting perceived social rank to disease. To assess the role of psychological pain arising from inequality. Also highlight the need for prospective research on status dynamics. The results show that low SSS consistently predicts poorer self-rated health and increased biological risk. SSS remains a significant predictor even when controlling objective status. Highlighting that the perception of inequality embeds biologically to influence health outcomes.

Hoebel et al. (2017) investigated the relationship between objective and subjective socioeconomic status (SES) and depressive symptoms in German adults. Logistic regression models revealed that lower subjective status predicted depression independently of income, education, and occupational status. The study is limited by its cross-sectional design, which prevents determining whether depression leads to lower perceived status or vice versa. Nevertheless, it underscores the unique contribution of self-perceived social rank to mental health inequalities beyond material conditions.

Huang (2017) examined the relationship between time spent on social network sites (SNS) and psychological well-being. Using 67 independent samples involving 19,652 participants. A meta-analysis method was used to synthesize findings across studies. Included indicators of self-esteem, life satisfaction, loneliness, and depression. Evaluated correlations between SNS usage intensity and mental health outcomes. To investigate the validity of the Facebook depression hypothesis. To explore potential moderators such as age, gender, and publication type. Also to determine if heavy social media usage is a primary driver of distress. The results show a negligible overall correlation between time spent on SNS and well-being. Associations with positive indicators were near zero, while negative indicators showed only

weak links. The study suggests that the quantity of time spent online is not a strong predictor of psychological issues.

Barua and Tejavivaddhana (2019) conducted a systematic review and meta-analysis to examine the impact of the Sufficiency Economy Philosophy on well-being among the Thai population. The study synthesized findings from multiple studies across domains including education, health, and economic development. The results showed a consistent positive relationship between SEP application and well-being outcomes, with significant effect sizes observed in the meta-analysis. These findings confirm that SEP contributes to sustainable well-being by promoting balanced and resilient lifestyles. This provides empirical support for the inclusion of cultural and contextual factors in understanding well-being beyond purely economic measures.

Navarro-Carrillo et al. (2020) revisited the relationship between socioeconomic status (SES) and psychological well-being. Using two independent samples comprising 1,256 adults in Spain. Cross-sectional surveys and regression analyses were employed. Included objective indicators like income and education alongside the subjective MacArthur Scale. Evaluated Ryff's six dimensions of psychological well-being (PWB). To compare the predictive power of objective versus subjective social status. To determine if specific dimensions like autonomy or purpose are differentially affected by status. Also clarify inconsistent findings in previous SES research. The results show that subjective SES is a robust predictor of all PWB dimensions. In contrast, objective SES showed negligible associations with well-being dimensions. The study confirms that an individual's perceived social standing is more consequential for psychological functioning than their actual material wealth.

Boer et al. (2020) explored distinctions between intense and problematic social media use among adolescents across 29 countries using HBSC survey data. They showed that problematic use was consistently linked to poorer mental, school, and social well-being. The impact of intense use varies depending on national norms. Vulnerability was greater among girls and younger adolescents with family affluence. The authors cautioned that the cross-sectional design limits causal inference. Low well-being may drive problematic use.

Their emphasis on differentiating passive scrolling from active communication further supports theories that passive exposure is a harmful element in social comparison.

Tan et al. (2020) conducted a meta-analysis of 357 studies to compare the strength of objective versus subjective socioeconomic status (SSS) in predicting well-being. Results showed that SSS had a significantly stronger association with health and well-being than objective measures like income or education. A limitation is the high heterogeneity among the included studies, suggesting that the relationship varies by context. Ultimately, this comprehensive review provides the definitive statistical justification for prioritizing the MacArthur Ladder over absolute income in psychological research.

Kross et al. (2021) reviewed the evolution of research linking social media use to emotional well-being. Using a theoretical synthesis of findings from the past 15 years. The review critiques the binary active versus passive usage framework. Included an analysis of three interacting domains: user characteristics, platform affordances, and social contexts. The authors proposed a value-based model to explain conflicting results in previous literature. To determine why early studies often found small, negative effects while newer data shows complexity. To identify how specific design features influence users. Also recommend methodological improvements for future research. The results indicate that social media effects are not universally positive or negative but are highly context dependent. The study concludes that the golden age of simple answers is over and calls for methodological pluralism to capture the nuances of the digital ecosystem.

Fung et al. (2022) evaluated the Chinese version of the WHO-5 well-being index using two cross-sectional studies of mainland university students. EFA and CFA analyses confirmed that the scale is unidimensional, internally consistent, and strongly correlated with established well-being measures. However, the reliance on a single, predominantly female sample limits generalizability to broader populations. Ultimately, the study fills a critical gap by providing the first rigorous psychometric validation of the tool for the Mainland Chinese context.

Gross-Manos and Bradshaw (2022) examined the relationship between material well-being and life satisfaction among children in 35 countries. They used data from the third wave

of the Children's Worlds school-based survey. The authors applied multilevel and correlational analyses at both the macro country and individual levels. The results showed that subjective material deprivation and multidimensional material well-being were strongly associated with national life satisfaction. However, the Family Affluence Scale was not significantly related to well-being at the macro level. At the individual level, the strength of these relationships differed across countries. These findings confirm that material security is an important part of well-being even among children. This supports the role of objective sufficiency in the proposed model. The study also highlights the value of subjective and multidimensional measures over simple wealth indicators.

Guan et al. (2022) systematically reviewed 40 studies to examine the association between financial stress and depression in adults. The analysis confirmed that financial stress is a robust predictor of depression, often mediating the impact of low income. However, the reliance on cross-sectional data in many included studies limits the ability to draw strong causal inferences. Ultimately, the review establishes that subjective financial perception is a more consistent mental health determinant than objective economic indicators.

Knies (2022) analyzed longitudinal data to examine how family income and material deprivation affect children's life satisfaction. The study used a large, nationally representative survey of children aged 10–15 in England. The author applied longitudinal regression models, including fixed- and random-effects analyses, to separate within-child and between child effects. The results showed that higher family income and lower material deprivation were associated with greater well-being. The impact of material conditions differed by the age group. Children aged 12–15 experienced larger declines in life satisfaction from material deprivation than younger children. Income effects remained small but statistically significant after controlling individual differences. These findings support the role of objective sufficiency as a foundation for well-being during adolescence. The study also shows that material status becomes more psychologically important as children move through their social development.

Samra et al. (2022) investigated whether social comparisons mediate the link between problematic social media use and depression in young adults. Experimental exposure to upward

comparison images confirmed that negative self-evaluations are a proximal mechanism driving depressive symptoms. The study sample was relatively small and skewed towards university students, potentially limiting wider applicability. Ultimately, the work identifies upward comparison as a specific cognitive mechanism. This mechanism transforms digital usage into psychological distress.

Simonse et al. (2022) analyzed longitudinal data to determine the role of financial stress in mental health changes during the COVID-19 pandemic. Fixed effects models demonstrated that changes in financial stress strongly predicted fluctuations in mental health, while objective income changes did not. A key limitation is the short timeframe of the study, which may not capture the long-term mental health effects of financial instability. The study validates that subjective financial stress is a more sensitive and immediate predictor of well-being than objective financial shocks.

Skogen et al. (2022) assessed the association between subjective socioeconomic status (SSS) and negative experiences on social media among adolescents. Regression analyses indicated a clear social gradient, where lower SSS adolescents reported significantly more negative incidents and fewer positive ones. The study relied on self-reported data for both status and negative experiences, introducing potential common method bias. Nonetheless, it suggests that low social status acts as a vulnerability factor that worsens the quality of digital interactions.

Twenge et al. (2022) revisited Orben and Przybylski's (2019) influential work on screen time and adolescent mental health. They challenged the earlier conclusion that the effects were negligible. They refined the analytical approach including distinguishing social media from general screen use, separating analyses by gender, and reconsidering the role of demographic controls. They highlighted how methodological choices shape outcomes. Their re-analysis underscored that collapsing screen activities and overlooking gender differences obscures meaningful patterns, particularly for female adolescents. The study emphasizes that age and gender should be treated as defining conditions in models of well-being rather than variables

to be controlled away, thereby reframing how social media effects are estimated in adolescent populations.

Kraft and Kraft (2023) explored how positive and negative effects mediate the relationship between subjective socioeconomic status (SSS) and health. Structural equation modeling showed that negative effects significantly mediates the link between low SSS and both physical and mental health. A limitation is the use of single-item measures for some variables, which may reduce psychometric precision compared to multi-item scales. This research highlights that the health impact of social status is largely driven by the emotional response to one's rank.

Sekulova et al. (2023) examined the impact of a basic income experiment on life satisfaction in Barcelona. The study followed individuals experiencing material deprivation who joined a two-year support scheme. The authors used regression and panel data analyses to assess changes in well-being over time. The results showed that financial support improved subjective well-being. However, ongoing stress from food and material deprivation predicted life satisfaction better than income alone. The findings indicate that socio-economic vulnerability leaves a lasting psychological effect. This reduces long-term life satisfaction. The study suggests that well-being depends strongly on stable economic and social security. This supports the role of psychological security as a key mediator in the proposed model. The research also shows that reducing material stress is essential for sustainable happiness.

Zhang et al. (2023) investigated the underlying mechanisms linking social media usage to student well-being. Using a sample of 1,004 college students from 135 universities in China. A moderated mediation model was tested using structural equation modeling. Included measures of social media intensity, self-esteem, and online social support. Evaluated cyberbullying as a critical moderator. To determine if social media directly affects psychological (PWB) and subjective well-being (SWB). To explore if self-esteem and social support mediate this relationship. Also to assess how cyberbullying victimization buffers positive outcomes. The results show that social media usage does not directly predict well-being but positively influences it indirectly through self-esteem and support. However,

cyberbullying weakens the beneficial impact of self-esteem on well-being. The findings suggest that while digital engagement can foster resources, online risks like bullying significantly dampen these benefits.

Zsila and Reyes (2023) provided an editorial review synthesizing current research on the dual impacts of social media on mental health. They argued that while platforms offer connection, they concurrently foster detrimental comparison, isolation, and pressure to conform to idealized standards. The paper is a narrative overview rather than an empirical study, lacking new data to test specific hypotheses. It serves as a foundational conceptual framework for understanding the double-edged sword nature of digital social environments.

Cai et al. (2025) investigated the role of psychological security in the link between income and happiness using a sample of 6,953 Chinese urban residents. Analysis of variance and bootstrap methods revealed that while income directly boosts life satisfaction, its effect is significantly mediated and moderated by psychological security. However, the cross-sectional design limits the ability to determine whether security causes happiness or vice versa. Ultimately, the study confirms that financial gains do not maximize well-being without a foundation of psychological safety, supporting the inclusion of psychological scarcity in well-being models.

Esmail (2025) analyzed the drivers of Gross National Happiness in Egypt using macro level data from 2000 to 2024. An autoregressive distributed lag model demonstrated that digital access and governance quality predict well-being more accurately than economic growth alone. A key limitation is the low R-squared value, suggesting that critical micro level psychological variables were missing from the macroeconomic model. Ultimately, this unexplained variance highlights the necessity of including individual level mediators, such as subjective social status, to fully understand well-being in developing economies.

Kliem et al. (2025) evaluated the psychometric quality of the WHO-5 Well-Being Index. Using a representative sample of 2,515 adults in Germany to provide updated population norms. Confirmatory factor analysis and measurement invariance testing were employed. Included item-level statistics and comparisons against norms from 2004. Construct validity

was evaluated through correlations with depression, anxiety, somatic symptoms, and loneliness. The study aimed to validate the instrument across different demographic groups. The results confirm that the WHO-5 is a unidimensional, reliable, and valid measure comparable across gender and age. Notably, the updated norms reveal a decrease in general well-being compared to two decades ago. The findings support the continued use of the WHO-5 for mental health screening under current social conditions.

Li et al. (2025) explored the impact of platform exploitation in China, focusing on the subjective well-being of digital gig workers. Using stratified sampling, they revealed that exploitation negatively impacts well-being. Using structural equation modeling (SEM) in AMOS with bootstrapping (5,000 samples), they tested a chained mediation model showing the effect that operates through reduced job rewards and job security. This finding provides a precise mechanism for the digital paradox, confirming that digital environments can erode security even when employment exists. However, the study relied on data from a single app, limiting generalizability. Future research should expand from gig workers to general social media users to test whether algorithmic exploitation harms subjective social status broadly.

Liu and Zhao (2025) examined the relationship between social media use and personal relative deprivation (PRD) among urban Chinese residents. A moderated mediation analysis found that social media intensity predicts PRD, with social class identity acting as a significant moderator. However, the study's focus on urban residents may not generalize to rural populations with different digital consumption patterns. The findings provide crucial evidence that digital interactions can amplify feelings of deprivation depending on one's psychological identification with social class.

Sakkhamduang et al. (2025) examined how the Sufficiency Economy Philosophy (SEP) can be applied in education to promote student well-being. Using a mixed-method approach, the study found that integrating SEP principles improved both understanding and behavior related to sustainable living. Students showed greater resilience, ethical decision making, and adaptability. Overall, the findings suggest that SEP shapes long-term psychological

development and highlights the importance of cultural and educational interventions in supporting well-being.

Samutachak et al. (2025) examined how the Sufficiency Economy Philosophy (SEP) supports resilience during COVID-19 in Thailand. The study found that key principles that are moderation, reasonableness, and prudence help individuals cope with economic shocks. Those who followed SEP showed greater stability in both material conditions and psychological well-being. Overall, the findings suggest that SEP acts as a protective framework, reducing financial stress and perceived inequality.

Concurrently, the digital environment has fundamentally restructured the mechanisms of social comparison. By expanding reference groups from local communities to global algorithms, social media creates a landscape of hyper salient comparison. Specifically, passive upward social media exposure refers to the act of observing idealized lifestyles without interaction. This process removes the nuance of reality. As a result, it intensifies feelings of relative deprivation and widens the psychological gap between the actual self and idealized others.

Consequently, this study posits that the relationship between social status and well-being is no longer linear but moderated by digital consumption. We hypothesize that high levels of passive exposure act as an amplifier, making low subjective status significantly more detrimental to mental health.

Variable	M1 (Baseline)	M2 (+Occupation)	M3 (+ Interaction)
PasExp	✓	✓	✓
PsySca	✓	✓	✓
SubStat	✓	✓	✓

HHInc	✓	✓	✓
SOC_HRS	✓	✓	✓
OSSS-3	✓	✓	✓
Age dummies	✓	✓	✓
Gender dummies	✓	✓	✓
Living Arrangement (LiveArrg)	✓	✓	✓
Occupation (WrkIndy_grp)	-	✓	✓
PasExp × SubStat	-	-	✓

Table 1. The variable from the research paper

Note: The variables that were significant in each paper are denoted by the ✓ symbol

To test this, this research employs a moderated regression framework that integrates psychological scarcity and subjective social status as primary predictors, while introducing an interaction term (SubStat x PasExp). This approach allows us to isolate the specific digital price of status, controlling objective income, and demographics to prove that the crisis of well-being is structural and psychological, rather than purely financial. A summary of the key themes and studies reviewed is provided in Appendix C.

2.2 Conceptual framework

This study proposes a conceptual framework that integrates social comparison theory, psychological scarcity theory, and subjective social status research to explain how passive upward social media exposure influences mental well-being among Thai adults. The framework posits that digital visibility of wealth erodes the pre-digital psychological buffer of ignorance by making upward comparisons constantly and unavoidable. Psychological scarcity,

defined as the persistent feeling of having too little relative to perceived needs, and subjective social status, defined as perceived rank on the social ladder, serve as the primary independent variables that directly affect subjective well-being, measured by the WHO-5 Well-Being Index. Passive upward social media exposure functions as the key moderator, hypothesized to amplify the negative effects of low psychological scarcity and low subjective social status on well-being through an interaction effect. Control variables, including absolute monthly household income, occupation status, living arrangement, and social support measured via the OSSS-3 scale, age, and gender, are included to account for objective socioeconomic and demographic influences.

The relationships are visually represented in Figure 1, which shows the independent variables (Psychological Scarcity, Subjective Social Status, and Passive Upward Social Media Exposure) with direct arrows pointing to the dependent variable (The World Health Organization's Five Well-Being). The interaction term is depicted as an additional arrow from Passive Upward Social Media Exposure and Subjective Social Status converging toward the dependent variable, illustrating the moderating effect. Control variables are shown as separate arrows also pointing to the dependent variable.

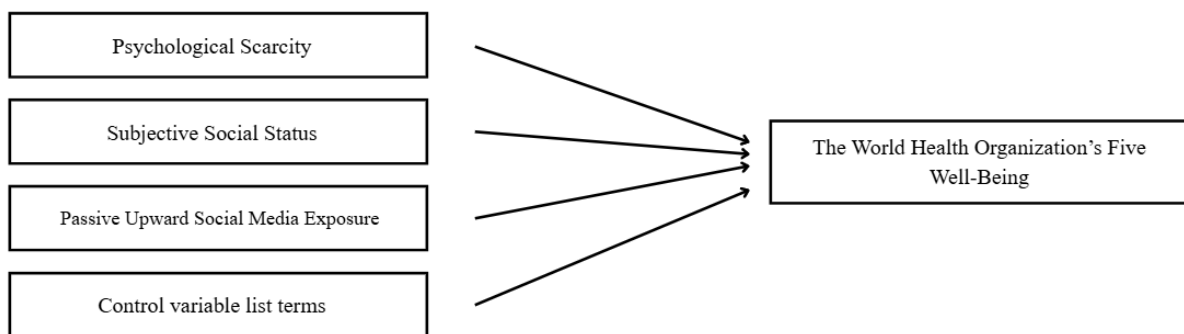


Figure 1 Conceptual framework of the relationships between psychological scarcity, subjective social status, passive upward social media exposure (and their interaction), and subjective well-being (WHO-5), controlling for socioeconomic and demographic control variables.

This framework is grounded in the reviewed literature, where objective economic conditions provide a foundation for well-being, but subjective perceptions and digital

comparison mechanisms exert stronger independent and interactive effects in modern contexts. The model allows testing of direct effects and the specific moderating role of passive exposure, offering a structured way to isolate the psychological and perceptual drivers of well-being beyond material resources.

Chapter 3: Research Method

3.1 Research Design and Data Collection

This study employs a cross-sectional quantitative design. It examines how psychological scarcity, subjective social status, and passive upward social media exposure collectively influence subjective well-being among working-age Thai adults residing in Bangkok. Primary data were gathered through a self-administered online questionnaire developed using Google Forms. The questionnaire was distributed between March and April 2026 via three digital platforms: Instagram, Facebook, and LINE. These platforms were selected because they represent the dominant social media ecosystems used by the Bangkok working-age population. They also enable cost-effective and rapid data collection within a constrained academic time frame.

The questionnaire was written entirely in Thai and organised into five sequential sections corresponding to the study's constructs. The first section covers respondent background and sector classification. The second section focuses on demographic characteristics. The third section addresses subjective social status and economic conditions. The fourth section examines social media behaviour and offline social support. The fifth section explores financial well-being and subjective well-being. In total, the instrument contains 24 items spanning single-choice, Likert-scale, and grid response formats.

Prior to full distribution, a pilot check was conducted on the encoding of Thai language response strings to ensure complete character matching and zero item level missingness. All 19 scored items returned zero unmatched values after encoding, confirming instrument fidelity. The final analytic sample comprises 411 respondents. This follows the exclusion of 50 individuals aged above 44 years, 12 individuals aged below 18 years, and 42 entries with missing gender information. This results in a clean working dataset for all regression analyses.

3.2 Sample Size Determination

The minimum required sample size was determined using the Cochran (1977) formula, a statistical framework specifically designed for securing representative estimates within large or strictly undefined populations. By anchoring the sample calculation in this approach, the research ensures a robust level of precision and confidence without relying on a finite population parameter for the target demographic.

$$n_0 = \frac{z^2 pq}{e^2}$$

A 95% confidence level was applied, yielding $Z = 1.96$. Because no prior population proportion was available, $p = 0.50$ was adopted to maximize the estimated sample size, with $q = 1 - p$, $q = 0.50$ and a margin of error $e = 0.05$. Substituting these values:

$$n_0 = \frac{(1.96)^2(0.5)(0.5)}{(0.05)^2} = \frac{3.8416 \times 0.25}{0.0025} = 384.16$$

The theoretical minimum is 384 respondents. The study targeted a collection of at least 400 usable observations. This provided a buffer against invalid or incomplete responses and ensured sufficient statistical power for moderated regression. The final cleaned sample of 411 respondents comfortably exceeds both thresholds.

3.3 Measurement Instruments

Each construct in the conceptual framework is operationalised through a validated or theoretically grounded measurement instrument. The following subsections describe each instrument and its item content as administered in the survey. They also explain the scoring procedure applied in the analysis.

3.3.1 Dependent Variable: Subjective Well-Being (WHO-5 Index)

Subjective well-being is measured using the World Health Organization Five Well-Being Index (WHO-5; Topp et al., 2015). It is one of the most widely used and

psychometrically validated brief measures of positive mental health. The WHO-5 captures positive emotional states experienced over the preceding two weeks and has demonstrated strong cross-cultural validity, including in Thai language settings.

Respondents rated the frequency of five statements using a six-point scale anchored at 0 (at no time) to 5 (all the time). The five WHO-5 items, as presented in the survey, are shown in Table 3.1 below.

No.	Statement (Thai / English equivalent)	Response Scale
1	I have felt cheerful and in good spirits (ฉันรู้สึกทึ่งและอารมณ์ดี)	0 = At no time 1 = Less than half the time 2 = Some of the time 3 = More than half the time 4 = Most of the time 5 = All the time
2	I have felt calm and relaxed (ฉันรู้สึกสงบและผ่อนคลาย)	
3	I have felt active and vigorous (ฉันรู้สึกกระฉับกระเฉงและมีพลัง)	
4	I woke up feeling fresh and rested (ฉันตื่นขึ้นมาด้วยความรู้สึกสดชื่นและพักผ่อนเต็มที่)	
5	My daily life has been filled with things that interest me (ชีวิตประจำวันของฉันเต็มไปด้วยสิ่งที่ฉันสนใจ)	

Table 2. The World Health Organization-Five Well-Being Index (WHO-5)

Source: World Health Organization (2024)

The composite WHO-5 score is calculated by summing all five item scores and multiplying the total by four. This transforms the raw total (0-25) into a standardised index

ranging from 0 to 100, where higher values indicate greater subjective well-being. A score of 50 or below is commonly used as a clinical screening threshold for impaired well-being (Topp et al., 2015). In this study, 49.4% of the sample (n = 203) fell at or below this threshold. This highlights the substantive mental health relevance of the study population.

Internal consistency was very high (Cronbach's $\alpha = 0.957$). All five items had item-total correlations exceeding 0.90. Confirmatory factor analysis (CFA) confirmed a single-factor structure, with standardised loadings ranging from 0.893 to 0.922 (SRMR = 0.011, CD = 0.957).

3.3.2 Independent Variable: Psychological Scarcity (PsySca)

Psychological scarcity captures the subjective experience of financial constraint and cognitive preoccupation with insufficient resources. It reflects not merely objective poverty but the mental bandwidth tax associated with perceived economic precarity (Mullainathan & Shafir, 2013). In this study, the construct is operationalised using items adapted from the CFPB Financial Well-Being Scale worry subscale (Consumer Financial Protection Bureau, 2017).

Respondents rated how frequently three statements characterised their financial situation, using a five-point scale from 0 (not at all) to 4 (all the time). The three items are presented in Table 3 (Items 1–3).

No.	Statement (Thai / English equivalent)	Response Scale
1	My financial situation is currently quite difficult (ตอนนี้ฐานะการเงินของฉันค่อนข้างลำบาก)	0 = Not at all 1 = Rarely 2 = Some of the time
2	Given my financial situation, I feel like I'll never get the things I want in life. (จากสถานการณ์ทางการเงินของฉัน ฉันรู้สึกว่าจะไม่มีวันได้สิ่งต่างๆ ที่ฉันต้องการในชีวิต)	

3	I'm worried that the money I have now, or the money I'll save in the future, might not be enough. (ฉันกังวลว่าเงินที่ฉันมีอยู่หรือที่จะเก็บออมในอนาคตจะไม่เพียงพอ)	3 = Most of the time 4 = All the time
4	I am able to handle a major unexpected financial expense. (ฉันสามารถรับมือกับค่าใช้จ่ายทางการเงินก้อนใหญ่ที่ไม่คาดคิดได้)	

Table 3. Psychological Scarcity (PsySca) items and response scale as administered

Source: Adapted from the Consumer Financial Protection Bureau (2017).

A composite PsySca index was constructed by averaging the three item scores, yielding a continuous scale from 0 to 4, with higher values reflecting greater perceived financial and cognitive constraint. The sample mean of 1.37 (SD = 1.32) indicates moderate average scarcity, with substantial individual variation.

The psychological scarcity measure comprises three items from the CFPB Financial Well-Being Scale worry subscale (Items 1–3 in Table 3). The variable *cfpb_hard* captures perceived financial hardship, reflecting the respondent's current difficulty in managing their financial situation. The variable *cfpb_sit* measures perceived life limitation due to financial constraints, indicating the extent to which individuals feel restricted in achieving desired life outcomes. The variable *cfpb_wry* captures financial worry, particularly concerns about whether current and future financial resources will be sufficient. Together, these three items represent distinct but related dimensions of financial strain and perceived economic insecurity. Item 4 in Table 3 ("I can cope with any unexpected large financial expenses") belongs to a separate CFPB liquidity subscale and is used to construct the CFPB_liquid control variable, as described in the following section.

Cronbach's $\alpha = 0.950$. EFA on the three worry subscale items (*cfpb_hard*-perceived financial hardship, *cfpb_sit*-perceived life limitation due to financial constraints, and *cfpb_wry*-financial worry about current and future resources) retained a single factor with eigenvalue 2.727 (91.0% of variance explained), all factor loadings ≥ 0.947 , and Kaiser–Meyer–Olkin (KMO) = 0.771. CFA standardised loadings ranged from 0.911 to 0.952 (SRMR = 0.000, CD = 0.953).

In addition to the Worry subscale, two binary items from the CFPB liquidity dimension were included to capture objective financial buffer capacity. These items are: (i) "I have money

left at the end of the month” and (ii) “Financial constraints affect my life.” These items were combined with a reversed emergency fund item to construct the CFPB_liquid subscale. The subscale shows strong factor properties (EFA eigenvalue = 2.578, 85.9% of variance explained, KMO = 0.756, α = 0.918). It is used as a secondary financial control variable in Model 1.

H₁: Higher psychological scarcity is negatively associated with subjective well-being ($\beta_1 < 0$).

The empirical results strongly support H₁. It demonstrates that psychological scarcity exerts a consistently significant and negative effect on well-being across all model specifications ($\beta = -3.68$ in M1; $= -4.10$ in M3; both $p < 0.001$). Crucially, this negative association persists even when objective household income is held constant. This confirms that chronic financial worry functions as a profound tax on subjective utility, degrading mental health independently of a person's actual economic capacity. Ultimately, this empirical outcome directly validates the theoretical mechanism articulated by Mullainathan and Shafir (2013): the perceived lack of resources monopolizes cognitive bandwidth, thereby depleting the psychological reserves necessary to sustain positive effect. Consequently, Hypothesis 1 is fully supported.

3.3.3 Independent Variable: Subjective Social Status (SubStat)

Subjective social status captures how individuals perceive their own standing relative to others in society, independent of objective socioeconomic indicators. It is measured using the MacArthur Scale of Subjective Social Status (Adler et al., 2000). This is a single-item ladder instrument that asks respondents to place themselves on a ten-rung ladder representing Thai society. The bottom rung (1) represents those with the least money, education, and respected jobs, while the top rung (10) represents those with the most.

The survey item reads: “Imagine this ladder representing your social standing: education, finances, and respected jobs. The top is the best, the bottom is the worst. Where do you think you are on this ladder? (ลองนึกภาพบันไดนี้แทนระดับฐานะทางสังคมของคุณ ได้แก่ การศึกษา การเงิน และมีงานที่ได้รับการเคารพนับถือ ขั้นบนสุดคือดีที่สุด ขั้นล่างสุดคือแย่ที่สุด คุณคิดว่าตัวเองอยู่ตรงไหนบนบันไดนี้)” Responses

range from 1 (lowest) to 10 (highest). The sample mean of 6.24 (SD = 1.99) suggests that most respondents placed themselves in the middle to upper range of the perceived hierarchy. The distribution also shows sufficient variation to examine rank gradient effects on well-being.

Although the MacArthur Ladder consists of a single item and therefore does not permit internal consistency analysis, its construct of validity is well established across cultural contexts. It captures a cognitively integrative judgment about one's relative social position. This judgment is distinct from objective income or occupational class measures and is often more predictive of health outcomes (Tan et al., 2020).

H₂: Higher subjective social status is positively associated with subjective well-being ($\beta_2 > 0$).

The empirical findings provide robust support for H₂. It demonstrates a highly significant, positive relationship between subjective social status and the WHO-5 index ($\beta=3.27$ in M1, $p<0.001$). Specifically, when passive social media exposure is held at its sample mean, each incremental step up the MacArthur Ladder corresponds to an approximate 2.3-point increase in subjective well-being. This constitutes a substantively meaningful hedonic benefit and directly aligns with established socio-psychological literature. It reinforces the premise that an individual's cognitively integrative judgment of their relative social rank frequently outperforms objective financial indicators as a determinant of mental health (Navarro-Carrillo et al., 2020; Tan et al., 2020). Consequently, Hypothesis 2 is fully supported.

3.3.4 Independent Variable: Passive Upward Social Media Exposure (PasExp)

Passive Upward Social media exposure (PasExp) represents the degree to which individuals are exposed to content depicting wealth, success, and idealised lifestyles. They experience this exposure without actively participating in social interaction. This form of passive upward comparison that induces consumption is theoretically distinct from active use, such as posting and commenting. It has also been shown to more consistently predict negative psychological outcomes (Verduyn et al., 2015).

The variable is operationalised through three items. These items ask respondents how frequently they engage in the following passive behaviours during their daily social media use, using a five-point frequency scale (Table 4).

No.	Behaviour Item (Thai / English equivalent)	Response Scale
1	Look at the feeds of influencers or people who seem richer or more successful than you. (ดูฟีดของเหล่าอินฟลูเอนเซอร์หรือคนที่ดูเหมือนจะร่ำรวยหรือประสบความสำเร็จมากกว่าคุณ)	1 = Never (ไม่เคย) 2 = Rarely (น้อย) 3 = Sometimes (บางครั้ง) 4 = Often (บ่อยครั้ง) 5 = Very often (บ่อยมาก)
2	View photos or videos of luxury travel, expensive products, or ideal lifestyles without interacting (e.g., liking/commenting). (ชมภาพถ่ายหรือวิดีโอเกี่ยวกับการท่องเที่ยวสุดหรู สินค้าราคาแพง หรือวิถีชีวิตในอุดมคติ โดยไม่มีปฏิสัมพันธ์ (เช่น การกดไลค์/แสดงความคิดเห็น))	
3	Scroll through updates from friends or acquaintances who seem to be more successful in life than you. (เลื่อนดูข้อความอัปเดตจากเพื่อนหรือคนรู้จักที่ดูเหมือนจะประสบความสำเร็จในชีวิตมากกว่าคุณ)	

Table 4. Passive Upward Social Media Exposure (PasExp) items and response scale as administered

Source: Adapted from Verduyn et al. (2015).

A composite PasExp index was formed by averaging the three item scores, producing a continuous variable from 1 to 5. Higher values indicate more frequent passive exposure to

upward comparison content. The sample mean of 2.94 (SD = 1.11) reflects moderate average exposure, consistent with the normalised social media culture among Bangkok working age adults.

Daily social media time (SOC_HRS) is captured by a separate question asking respondents to report their average daily usage duration. Responses are recorded in five ordinary categories: less than 1 hour (coded 1), 1-2 hours (2), 2-4 hours (3), 4-6 hours (4), and more than 6 hours (5). This variable serves as an additional control for total media exposure intensity, distinct from the passive comparison dimension captured by PasExp.

Cronbach's $\alpha = 0.898$ for the PasExp subscale. EFA retained a single factor with eigenvalue 3.176, all three item loadings ≥ 0.856 (KMO = 0.735). CFA standardised loadings ranged from 0.856 to 0.897 (CD = 0.915).

H₃: Passive social media exposure moderates the relationship between subjective social status and subjective well-being, such that higher exposure weakens the positive association between status and well-being.

The empirical analysis yields robust support for H₃. In the interaction model (M3), the interaction term between passive exposure and subjective social status (PasExp $\times$ SubStat) is negative, of considerable magnitude, and highly significant ($\beta_4 = -2.678$, SE = 0.502, $p < 0.001$). This confirms a profound moderating effect. The protective well-being benefits traditionally afforded by higher perceived social status are progressively eroded as passive digital exposure intensifies. Notably, at the maximum level of passive exposure (PasExp = 5), the marginal benefit of an additional step on the MacArthur Ladder diminishes statistical insignificance. This indicates that the heavy, passive consumption of aspirational and idealized content on social media effectively neutralizes the psychological utility advantage of higher perceived social rank. Therefore, Hypothesis 3 is fully supported.

H₄: Higher passive social media exposure is negatively associated with subjective well-being ($\beta_3 < 0$), conditioning on the interaction with subjective social status.

Passive upward exposure emerges as the single most potent predictor of subjective well-being deterioration in the dataset. When evaluating the conditional marginal effect at the sample mean of subjective social status, each one-unit increase in passive exposure corresponds to a substantial 8.08-point decline in the WHO-5 index ($p < 0.001$). Furthermore, the unconditional baseline model (M1) reveals a striking coefficient of -9.62 . In practical terms, this implies that an individual shifting from minimal to near-constant passive exposure experiences a well-being reduction of approximately 38 points. Such a massive decline is sufficient to drag an individual from a state of functional well-being into the clinically impaired range. These results directly validate the theoretical premise that the passive consumption of idealized lifestyles, distinct from only daily screen time (SOC_HRS), actively damages mental health through pernicious upward-comparison pathways (Kross et al., 2021; Liu & Zhao, 2025; Verduyn et al., 2015). Consequently, Hypothesis 4 is fully supported.

A supplementary finding is that occupational sector (M2) adds no statistically significant explanatory power over M1 ($\Delta AIC = +5.0$, all industry dummies $p > 0.176$), confirming that the psychological mechanisms driving well-being deficits operate at the individual level rather than through structural occupational differences. Nearly half the sample (49.4%) scored at or below the WHO-5 clinical screening threshold of 50, signalling a substantial burden of impaired well-being in the Bangkok working-age population that transcends sector and demographic boundaries.

3.3.5 Control Variable: Offline Social Support (OSSS-3)

Offline social support is measured using the Oslo Social Support Scale (OSSS-3; Dalgard et al., 2006), a brief three-item instrument designed to assess the availability of close personal support networks. The OSSS-3 is routinely used in population health surveys as a rapid index of social integration and has demonstrated adequate validity in Asian contexts.

The Oslo Social Support Scale (OSSS-3) is included specifically to separate the digital mechanism of social comparison from the general and well-established psychological harm of loneliness and social isolation. This distinction is theoretically essential: without it, the negative association between passive social media exposure and well-being could simply reflect the fact

that socially isolated individuals both use social media more passively and report lower well-being, a confound that would prevent any causal attribution to the upward comparison pathway. Gross-Manos and Bradshaw (2022) include offline support measures in their comparative well-being models precisely to disentangle the protective function of real-world social networks from the effects of material and digital variables. By holding OSSS-3 constant, the regression model ensures that what is attributed to PasExp is the well-being cost of passive upward comparison content specifically, rather than the more general cost of having few people to talk to. This is particularly important given that Huang (2017) found that time spent on social network sites has a near-zero unconditional correlation with well-being. It is the type and context of use, not total usage, that matters, and offline social support availability is a key contextual moderator of that distinction. The three items, as presented in the survey, are shown in Table 5.

No.	Question (Thai / English equivalent)	Response Options
1	How many close people can you rely on if you have a serious personal problem? (มีคนสนิทที่คนที่คุณสามารถพึ่งพาได้หากคุณมีปัญหาใหญ่ในชีวิตส่วนตัว?)	0 = None (ไม่มี) 1 = 1–2 persons 2 = 3–5 persons 3 = More than 5 persons
2	How much do people show interest in and concern about what you do? (ผู้คนให้ความสนใจและห่วงใยในสิ่งที่你做ทำมากน้อยแค่ไหน?)	1 = Not at all (ไม่มีเลย) 2 = A little (น้อย) 3 = Uncertain (ไม่แน่ใจ) 4 = Much (มาก) 5 = Very much (มากที่สุด)
3	If you need help from your neighbours, how easily will it be possible for you to get help?	1 = Very difficult (ยากมาก) 2 = Difficult (ยาก)

	(หากต้องการความช่วยเหลือจากเพื่อนบ้าน คุณจะได้รับความช่วยเหลือง่ายแค่ไหน?)	3 = Possible (เป็นไปได้) 4 = Easy (ง่าย) 5 = Very easy (ง่ายมาก)
--	--	--

Table 5. Offline Social Support Scale (OSSS-3) items and response options as administered

Source: Adapted from Dalgard (2006).

To construct the composite OSSS-3 (Oslo Social Support Scale) index, each item was normalised to a common 0-1 range before averaging. The variable *close_ppl* (number of close confidants) was divided by 3, while *ppl_care* (perceived concern and interest from others) and *nbr_help* (ease of obtaining help from neighbours) were rescaled from their original 1–5 range to 0-1 using the formula $(x - 1)/4$. The resulting OSSS-3 composite ranges from 0 (no support) to 1 (maximal support). The sample mean of 0.40 (SD = 0.19) indicates moderate average levels of social support.

3.3.6 Socioeconomic and Demographic Control Variables

To ensure the empirical robustness of the primary regression model, a comprehensive suite of socioeconomic and demographic covariates is introduced. The inclusion of these controls is designed to account for underlying sample heterogeneity that might otherwise confound the core relationships of interest. Because baseline well-being, digital consumption patterns (PasExp), and perceived social standing (SubStat) are highly sensitive to structural life circumstances, controlling these factors is essential to prevent omitted variable bias. By systematically holding these material, demographic, and environmental characteristics constant, the model effectively isolates the specific psychological and perceptual mechanisms driving the social media–well-being dynamic from broader, pre-existing societal variations.

Household Monthly Income (HHInc)

The primary purpose of including household income is to isolate the psychological and perceptual mechanisms of well-being from raw economic capacity. If PsySca and SubStat

retain their predictive power after HHInc is held constant, it confirms that subjective financial worry and perceived social rank operate through channels beyond what money alone can explain. Howell and Howell (2008) provide the foundational justification: their cross-national meta-analysis demonstrated that the income–well-being relationship is strongest in low-income economies but weakens as material thresholds are met, suggesting that once basic sufficiency is achieved, psychological and relative factors become the dominant drivers. Sekulova et al. (2023) reinforce this from a policy experiment perspective. Their Barcelona basic income study found that even after receiving financial support, individuals experiencing residual scarcity stress reported lower life satisfaction than their income level would predict. This finding suggests that subjective financial perception is a more persistent determinant of well-being than objective income flows. Including HHInc as an ordinal bracket variable therefore functions as a control for material sufficiency, allowing the regression to attribute residual variation in WHO-5 to PsySca and SubStat rather than confounding it with general income differences.

To execute this control empirically, self-reported raw household income, originally collected as categorical string variables in Thai Baht (THB), was transformed into a 7-level ordinal variable (inc_num). This establishes the discrete, ascending brackets necessary to control material sufficiency in the regression model. The full distribution across the 411 respondents is presented in Table 6 below.

Bracket (HHInc)	Income Range (THB/month)	n	%
1	Below 10,000	18	4.4
2	10,000 – 20,000	40	9.7
3	20,001 – 30,000	70	17.0
4	30,001 – 50,000	144	35.0

5	50,001 – 80,000	87	21.2
6	80,001 – 150,000	37	9.0
7	Above 150,000	15	3.7
	Total	411	100.0

Table 6. Household monthly income bracket classification and sample distribution (n = 411)

Note: HHInc is treated as a continuous ordinal variable in regression analyses (mean = 4.00, SD = 1.36). Bracket 4 (30,001–50,000 THB) is the modal category, representing 35.0% of the sample. No bracket was used as a reference category; the variable enters the model as a single ordinal regressor.

Age (d_age25_34, d_age35_44)

Age is controlled because the well-being effects of social media exposure are known to vary systematically by life stage. Twenge et al. (2022) provide the most direct empirical basis: their large-scale analysis documented that social media's negative psychological effects are concentrated in younger adult groups who are navigating identity formation, career uncertainty, and social positioning. The processes that make upward comparison referents particularly salient and threatening. Boer et al. (2020) similarly found differential social media effects by age group in their cross-European well-being study. In the context of the present model, younger respondents (18–24) may engage with aspirational content as career aspiration or social navigation rather than rank threat, whereas older age groups may anchor their status self-concept more firmly to observable life outcomes, altering how passive exposure interacts with subjective social rank. Age dummies are constructed with the 18–24 cohort as the reference group, allowing the regression to estimate whether the core PasExp and SubStat effects are amplified or attenuated among the 25–34 and 35–44 age brackets independently of income and sector.

To execute this control empirically, the self-reported categorical age variable (`age_raw`) was converted into two binary dummy variables. Respondents in the 18–24 age bracket serve as the baseline reference group (where both dummies equal 0). This coding scheme allows the regression coefficients to capture the marginal differences of being in the 25–34 or 35–44 age cohorts relative to the youngest group. The full distribution across the 411 respondents is presented in Table 7 below.

Age Group	Range	Dummy Variable	Role in Model	n	%
18 – 24 years	Reference group	-	Reference	117	28.5
25 – 34 years	-	d_age25_34	= 1 if age 25–34	165	40.1
35 – 44 years	-	d_age35_44	= 1 if age 35–44	129	31.4
Below 18 years	Excluded	-	Excluded (n = 12)	-	-
Above 44 years	Excluded	-	Excluded (n = 50)	-	-
			Total	411	100.0

Table 7. Age group classification, dummy variable coding, and sample distribution (n = 411)

Note: Respondents aged below 18 years (n = 12) and above 44 years (n = 50) were excluded from the analytic sample prior to analysis, leaving three age brackets in the regression specification. The 18–24 age group serves as the reference category against which d_age25_34 and d_age35_44 are interpreted. Exclusions reduce the raw collected sample of 473 observations to the analytic sample of 411.

Gender (d_female, d_gnd_nd)

Gender is controlled because both the frequency and psychological impact of social media-driven upward comparison differ by gender, reflecting differences in the type of aspirational content encountered and the self-evaluative processes triggered by that content. Twenge et al. (2022) document that negative social media effects are substantially stronger among females than males in Western adolescent samples, a finding consistent with research showing that female-dominated platforms (Instagram in particular) disproportionately surface appearance and lifestyle comparison content. Boer et al. (2020) confirm gender as a significant moderator of the social media–well-being relationship in European adult populations. In the Bangkok context, gender differences in well-being and social comparison are further shaped

by cultural norms around social presentation and status display. Two dummy variables are constructed with male as the reference category: `d_female` for female respondents, and `d_gnd_nd` for those preferring not to specify. Including these controls ensures that the `PasExp` and `SubStat` coefficients reflect the association between digital comparison exposure, perceived rank, and well-being across gender groups, rather than being driven by gender-specific social media use patterns or pre-existing gender gaps in reported well-being.

To execute this control empirically, the self-reported categorical variable (`gender_raw`) was converted into two binary dummy variables. Male respondents serve as the baseline reference group (where both dummies equal 0). This coding scheme allows the regression coefficients to capture the marginal differences of identifying as female or preferring not to disclose, relative to identifying as male. The full distribution across the 411 respondents is presented in Table 8 below.

Gender Category	Thai Survey Label	Dummy Variable	Role in Model	n	%
Male	ชาย	-	Reference	170	41.4
Female	หญิง	<code>d_female</code>	= 1 if female	235	57.2
Non-binary / Prefer not to specify	ไม่ต้องการระบุ	<code>d_gnd_nd</code>	= 1 if non-binary or NS	6	1.5
Missing / Blank	-	Excluded	Excluded (n = 42)	-	-
			Total	411	100.0

Table 8. Gender classification, dummy variable coding, and sample distribution (n = 411)

Note: Respondents who did not provide a gender response (n = 42) were excluded from the analytic sample. Male is the reference category. The non-binary/prefer-not-to-specify group (d_gnd_nd) has a small cell size (n = 6); its coefficient should be interpreted with caution and is reported for completeness rather than as a substantive finding.

Living Arrangement (LivArrg)

Living arrangement is controlled because the social and emotional environment of a household directly shapes access to informal support, exposure to interpersonal stress, and the baseline level of loneliness or connectedness that an individual brings to their social media interactions. Gross-Manos and Bradshaw (2022) controlled for household composition in their cross-national study of material well-being and life satisfaction among children and adolescents, noting that family structure independently predicts well-being outcomes beyond income. Liu and Zhao (2025) similarly include household and social context variables in their moderated mediation model of social media and relative deprivation, recognising that individuals living alone are more likely to use social media as a primary social channel, thereby increasing their passive exposure without the buffering effect of co-present social interaction. In the present study, living arrangement captures whether respondents have access to daily face-to-face social anchors, people whose actual lifestyles they observe directly, or whether digital platforms constitute their primary window onto others' lives. Without this control, any association between passive exposure and well-being could partly reflect the confounding effect of solitary living rather than the content of social media consumption itself. The full distribution across the 411 respondents is presented in Table 9 below.

Living Arrangement	Description	Dummy Variable	Role in Model	n	%
With family / parents	Residing with immediate or extended family or parental household	-	Reference	154	37.5
Living alone	Sole occupant of dwelling	d_live_alone	= 1 if living alone	74	18.0
With friends / partner / housemates	Shared accommodation with peers, romantic partner, or co-tenants	d_live_shared	= 1 if shared living	46	11.2
Other arrangements	Dormitory, rented room, staff quarters, or arrangements not classifiable above	d_live_other	= 1 if other	137	33.3
			Total	411	100.0

Table 9. Living arrangement classification, dummy variable coding, and sample distribution (n = 411)

Note: Living with family/parents is the reference category. The "other arrangements" category (d_live_other, n = 137, 33.3%) is the second largest group in the sample and encompasses a diverse range of living situations including dormitories, rented houses, and employer-provided housing. Its heterogeneity means that the d_live_other coefficient should be interpreted as a composite effect rather than a single living-situation type.

Occupation / Industry Group (WrkIndy_grp)

Occupation is included to test whether the employment sector contributes independent explanatory power to well-being once psychological and economic variables are accounted for, and to ensure that sector-level structural differences, for instance, job stability, social prestige, and exposure to workplace stress, do not confound the core relationships of interest. The

theoretical motivation draws from Twenge et al. (2022) and Boer et al. (2020), both of whom document that the associations between social media use and well-being vary systematically across occupational and demographic subgroups, with effects concentrated among students, the unemployed, and those in precarious employment roles. If passive upward social media exposure harms well-being more intensely in the Labour/Agriculture/Informal sector than in Corporate/Professional roles, that differential may reflect occupational stress compounding the comparison mechanism rather than digital exposure per se. By introducing industry groups as a separate control in Model 2 and testing its joint significance, the study can determine whether the PasEx and SubStat effects are robust across sector boundaries or whether their apparent magnitude is partly attributable to the overrepresentation of particular employment types in the sample. The finding that all industry group dummies are jointly insignificant ($F(10, 397) = 1.26, p = 0.251$) confirms that the well-being deficits identified in this study are individual-level psychological phenomena that transcend occupational structure, a substantively important result that could not have been established without including occupation in the model.

To operationalise this control, self-reported sector data (sector_raw) required multi-step recording. In the first step, open-ended and inconsistently formatted responses. Including entries such as e-commerce, civil service variants, and specific clinic types, were reclassified into standardised overarching industry categories consistent with Thai industry classification conventions. In the second step, to ensure adequate statistical power and theoretically meaningful distinctions, these granular categories were consolidated into four broader contextual working groups (WrkIndy_grp). These categorical variable groups respondents according to shared structural employment characteristics, including corporatised professional environments, public service and essential functions, commerce and hospitality, and manual or informal labour. A student category (Group 5) was retained in the coding schema but yielded no valid respondents after age-range exclusions were applied, and an unclassified residual category (Group 6) captures the remaining twelve observations whose sector descriptions could not be reliably assigned. Corporate/Professional is used as the reference category in all regression specifications. The full distribution across the 411 respondents is presented in Table 10 below.

Group (WrkIndy_grp)	Contextual Label	Included Sectors / Descriptions	n
1	Corporate/Professional	Finance and insurance, communication and IT/ICT, general corporate employees	88
2	Public/Essential Services	Public administration and government, education, health and social activities	136
3	Commerce/Hospitality	Wholesale, retail and repair, accommodation and food service, transport and logistics	136
4	Labour/Agriculture/Informal	Manufacturing and factories, agriculture, forestry and fisheries, general labour	39
5	Students	Active students (excluded post age-range filtering)	0
6	Unclassified	Valid responses not assignable to Groups 1–4	12
	Total		411

Table 10. Industry group classification, coding scheme, and sample distribution (n = 411)

Note: Group 1 (Corporate/Professional) serves as the reference category in all regression models. Group 5 returned zero observations following the exclusion of respondents aged below 18 and above 44 years. Group 6 is retained in descriptive analyses but excluded from industry group dummy comparisons.

3.4 Data Analysis Strategy

All empirical analyses were conducted using StataNow 19. The data analysis strategy was executed in three sequential phases to ensure methodological and statistical validity. First, descriptive statistics were generated to profile the sample and establish baseline distributions for the core variables. Second, a comprehensive measurement validation protocol, incorporating both exploratory and confirmatory factor analyses, was applied to verify the psychometric integrity of the latent constructs. Finally, the primary hypotheses were tested using an ordinary least squares (OLS) econometric framework with heteroskedasticity-robust standard errors, structured around nested regression models to systematically isolate structural and moderating effects.

3.4.1 Descriptive Statistics

Table 11 below reports the mean, standard deviation, minimum, and maximum for all continuous study variables based on the full analytic sample ($n = 411$). Spearman rank correlations were computed alongside Pearson coefficients to account for the ordinal nature of several scales.

Variable	Mean	Std. Dev.	Min	Max	Scale Range
Subjective Well-Being (WHO-5)	53.62	23.82	0	100	0–100
Psychological Scarcity (PsySca)	1.37	1.32	0	4	0–4
Subjective Social Status (SubStat)	6.24	1.99	1	10	1–10
Passive Exposure (PasExp)	2.94	1.11	1	5	1–5

Household Income (HHInc)	4.00	1.36	1	7	1–7
Social Media Hours (SOC_HRS)	2.56	0.96	1	5	1–5
Offline Social Support (OSSS-3)	0.40	0.19	0	1	0–1

Table 11. Descriptive statistics of study variables (n = 411)

The average WHO-5 score of 53.62 (SD = 23.82) indicates moderate aggregate well-being. There are nearly equal proportions of the sample above and below the clinical at-risk threshold of 50 (50.6% above; 49.4% at or below). Psychological scarcity (mean = 1.37, SD = 1.32) shows wide individual variation, as does subjective social status (mean = 6.24, SD = 1.99). Passive exposure (mean = 2.94, SD = 1.11) reflects moderate but widespread digital upward comparison. Income clusters around brackets 3-5 (20,001-80,000 THB), and the majority of respondents (55.7%) spend one to two hours per day on social media.

3.4.2 Measurement Validation

Prior to specifying the primary regression models, a rigorous measurement validation procedure was conducted to ensure the psychometric integrity of the study's latent constructs. Establishing the reliability, unidimensionality, and construct validity of the core instruments, specifically the WHO-5, passive social media exposure (PasExp), psychological scarcity (PsySca), and social support (OSSS-3), is essential to rule out measurement error as a driver of the observed empirical relationships. The validation protocol followed a sequential approach, moving from exploratory dimensionality checks to formal structural confirmation.

First, Exploratory Factor Analysis (EFA) using principal-component extraction was conducted to confirm the unidimensionality of PasExp, PsySca, and WHO-5, and to justify the

separation of the CFPB battery into psychological scarcity and liquidity subscales. The EFA for the liquidity subscale identified a polarity issue with *cfpb_con* and that its sign was noted, but the composite was retained for theoretical reasons.

Second, Kaiser-Meyer-Olkin (KMO) sampling adequacy was assessed for each subscale. Internal consistency was evaluated using Cronbach's alpha, with item-level diagnostics including item total and item rest correlations for all multi-item scales.

Third, Confirmatory Factor Analysis (CFA) was conducted using maximum-likelihood estimation with robust standard errors (*vce(robust)*). The analysis was performed for WHO-5, PasExp, PsySca, OSSS-3, and a joint three-construct model (WellBeing, MediaExp, FinStress). It was used to assess convergent and discriminant validity. Model fit was evaluated using the standardised root mean square residual (SRMR) and the coefficient of determination (CD).

All scales met the minimum standards prior to regression: WHO-5 $\alpha = 0.957$, PasExp $\alpha = 0.898$, PsySca $\alpha = 0.950$, OSSS-3 $\alpha = 0.918$. Discriminant validity was confirmed in the three-construct CFA (SRMR = 0.044), with the WellBeing–MediaExp covariance at -0.481 and WellBeing–FinStress covariance at -0.413 (both $p < 0.001$), while the MediaExp–FinStress covariance was near zero ($p = 0.054$).

3.4.3 Econometric Models

The primary estimation framework for this study relies on Ordinary Least Squares (OLS) regression. To ensure the reliability of statistical inference, all models are specified using heteroskedasticity-robust standard errors (White, 1980). This methodological approach was formally justified by a series of pre-estimation diagnostics that detected non-constant error variance while confirming the necessary distributional assumptions of the residuals. Building upon this validated foundation, the empirical analysis is structured around three nested model specifications, progressing from a baseline evaluation to an interaction framework, to systematically isolate the main, structural, and moderating effects of the core variables.

The Breusch-Pagan/Cook-Weisberg test detected significant heteroskedasticity ($\chi^2(1) = 4.66$, $p = 0.031$). White's general test confirmed this finding ($\chi^2(84) = 178.93$, $p < 0.001$).

The Shapiro-Wilk test on OLS residuals failed to reject normality ($W = 0.994$, $p = 0.103$). The skewness-kurtosis joint test further confirmed the approximate normality of residuals ($\chi^2(2) = 0.70$, $p = 0.704$).

Bootstrap resampling (1,000 replications) produced normal, percentile, and bias-corrected intervals consistent with the robust OLS results across all three models. This confirms that inference is not materially affected by the bounded and discrete nature of the WHO-5 scale.

The general econometric specification for the primary model (M3) is:

$$\text{WHO-5}_i = \beta_0 + \beta_1(\text{PasExp}_i) + \beta_2(\text{SubStat}_i) + \beta_3(\text{PsySca}_i) + \beta_4(\text{PasExp}_i \times \text{SubStat}_i) + \gamma X_i + \varepsilon_i$$

where X_i denotes the vector of control variables (HHInc, SOC_HRS, OSSS-3, age dummies, gender dummies, living arrangement dummies, industry group dummies), and ε_i is the error term. The interaction coefficient β_4 captures the moderation hypothesis. It tests whether the relationship between subjective social status and well-being varies as a function of passive upward social media exposure. A negative and statistically significant interaction coefficient would indicate a stronger negative effect for individuals with lower perceived social status. In other words, these individuals experience disproportionately larger declines in well-being when exposed to upward social comparison on social media.

Model 1 (M1) includes all main independent variables and control variables, except occupation, and serves as the baseline specification. Model 2 (M2) adds the occupation/industry group dummies to test their marginal explanatory contribution. This specification allows us to test whether occupation has an independent explanatory effect. It also examines whether its influence is already captured by income, perceived status, and psychological factors. Model 3 (M3) replaces the M2 specification by introducing the interaction term and a more parsimonious set of controls. Parsimony in M3 is formally justified by a joint F-test on the controls dropped from M1. The test result ($F(10, 397) = 1.26$, $p = 0.251$) indicates that the dropped controls are jointly insignificant. AIC also supports M3 over M1 by a margin of 42 points. M3 has an AIC of 3,568, compared to 3,610 for M1.

To reduce multicollinearity in M3, PasExp and SubStat are mean-centred before constructing the interaction term. Variance inflation factors (VIF) for the M1 full specification show a maximum of 3.00 (SubStat) and a mean of 1.63. For the interaction specification, the maximum is 2.06 (PsySca) and the mean is 1.55. All values are well below conventional thresholds. Marginal effects of PasExp on WHO-5 are computed at each integer value of SubStat (1-10) using the delta method. This is done to characterise the interaction surface.

3.4.4 Robustness Checks

To validate the stability and reliability of the primary regression findings, a series of rigorous robustness checks were performed. These post-estimation analyses are designed to ensure that the core relationships identified in the baseline models are not artifacts of sample-specific variance, highly influential outliers, or alternative variable specifications. Specifically, the empirical results were subjected to three complementary validation procedures: non-parametric resampling, sensitivity analysis of extreme observations, and diagnostic model comparisons.

First, Bootstrap resampling (1,000 replications) for both M1 and M3 confirms that all core coefficient estimates are stable. Bias is also negligible across normal, percentile, and bias-corrected confidence intervals.

Second, Influential observation analysis using Cook's D (threshold = $4/n = 0.0097$) identified 22 potentially influential cases. Re-estimation on the reduced sample of 389 observations produced qualitatively identical results. PasExp is -9.07 compared to -9.62 in the full sample. SubStat is +3.68 compared to +3.27, and PsySca is -3.46 compared to -3.68. This confirms that no small subset of observations drives the results. Re-estimation of M3 with all M1 controls re-added confirms that the interaction term (PasExp $\times$ SubStat) remains statistically significant and substantively stable. The coefficient is $\beta = -2.629$ ($p < 0.001$), compared to -2.678 in the parsimonious M3 specification.

Third, LR tests and AIC comparisons confirm that CFPB_liquid adds no explanatory power beyond PsySca. The LR test result ($\chi^2(1) = 1.24$, $p = 0.266$) and the AIC difference

($\Delta\text{AIC} = -0.76$ for the restricted model) support this conclusion. Core coefficients also remain stable with and without its inclusion.

Chapter 4: Results and Interpretation

This chapter presents and interprets the empirical findings of the study. Section 4.1 describes the sample and the distribution of key variables. Section 4.2 reports pre-estimation diagnostic results. Sections 4.3 and 4.4 present the main regression models and moderation analysis respectively. Section 4.5 examines industry group heterogeneity in marginal effects, and Section 4.6 reports on robustness checks.

4.1 Sample Description and Variable Distributions

The final analytic sample comprises 411 respondents: 57.2% female, 41.4% male, and 1.5% non-binary/prefer not to say. By age group, 28.5% were aged 18–24, 40.2% aged 25–34, and 31.4% aged 35–44. Industry representation was distributed across four consolidated groups: Corporate/Professional (21.4%), Public/Essential Services (33.1%), Commerce/Hospitality (33.1%), and Labour/Agriculture/Informal (9.5%), with 2.9% unclassified. The largest household income bracket was 30,001–50,000 THB per month (35.0%), consistent with Bangkok's middle-income labour market.

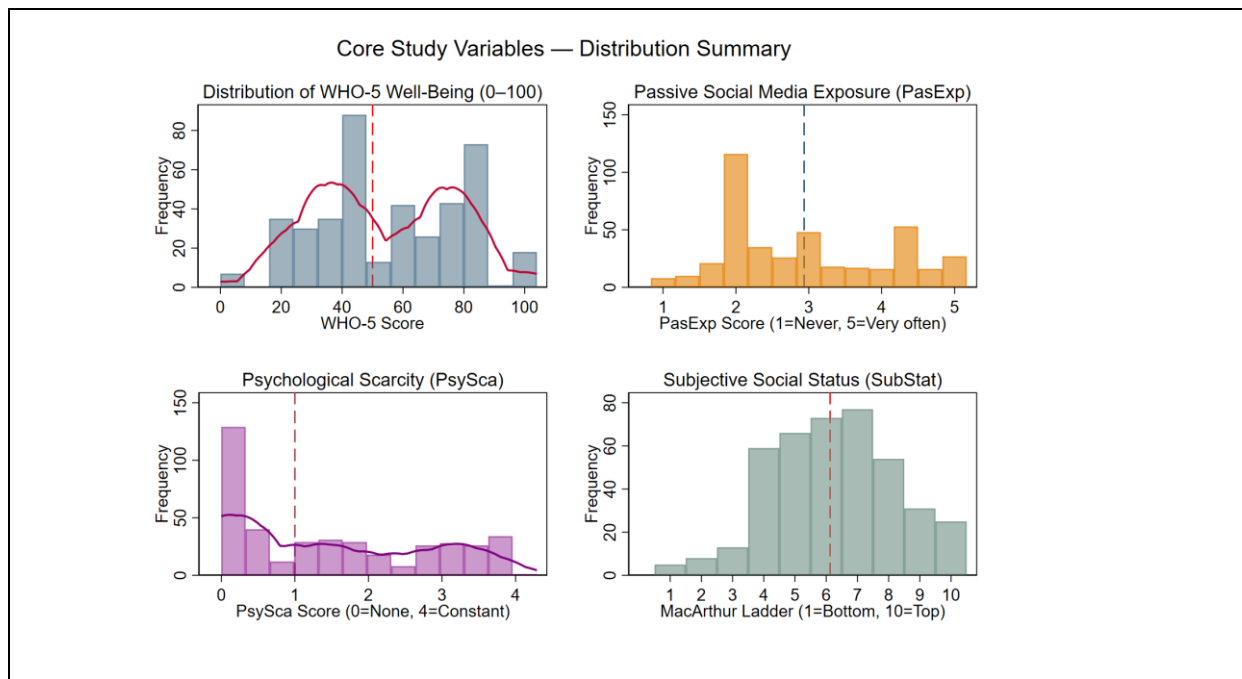


Chart 2. Distribution of core study variables (n = 411).

4.1.1 Subjective Well-Being (WHO-5)

The WHO-5 score has a mean of 53.62 (SD = 23.82, range 0–100), indicating moderate aggregate well-being. The distribution is approximately symmetric (skewness = 0.01, kurtosis = 2.01), with notable concentration at the extremes owing to the bounded, discrete nature of the scale. Importantly, 49.4% of respondents (n = 203) scored at or below the clinical threshold of 50, indicating impaired well-being. This near-equal split between the at-risk and non-at-risk groups gives the study substantial explanatory scope across the full range of well-being outcomes.

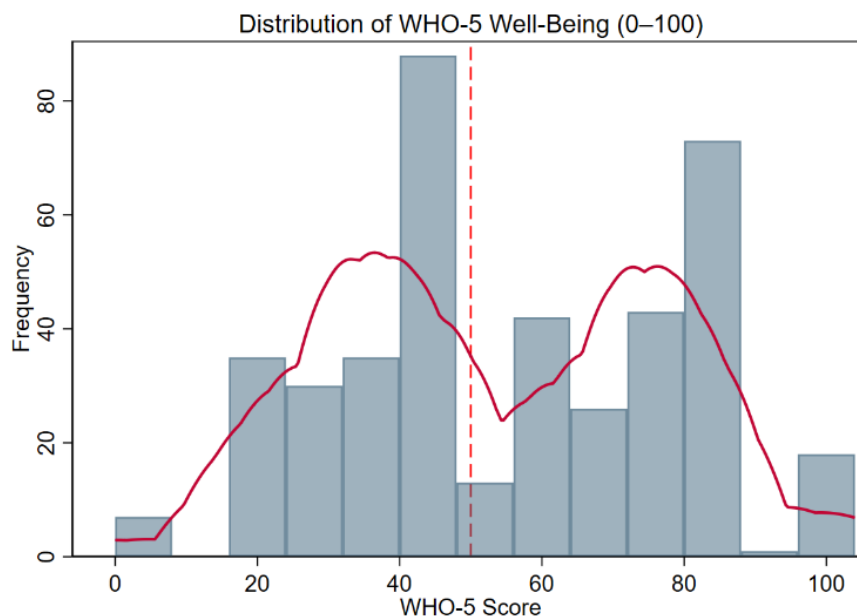


Chart 3. Distribution of WHO-5 Well-Being Index (n = 411). Mean = 53.62; 49.4% at or below clinical threshold (≤ 50).

4.1.2 Subjective Social Status (SubStat)

SubStat has a mean of 6.24 (SD = 1.99, range 1–10). The distribution is slightly left-skewed (skewness = -0.08), with modal responses clustering around rungs 5–7. This middle-rank concentration is consistent with Festinger's (1954) Social Comparison Theory, which predicts that individuals gravitate toward comparison referents like themselves placing most

respondents in the mid-tier of the perceived hierarchy where social comparison is most active. Notably, the empirical indifference point identified in the moderation analysis (SubStat ≈ 3.22 , where the marginal effect of PasExp crosses zero) falls within the lower boundary of this clustering zone, consistent with the finding that at SubStat = 3 the PasExp coefficient is near-zero and non-significant (+0.592, $p = 0.750$), while at SubStat = 4 it is small and negative but also non-significant (-2.086 , $p = 0.143$), see Section 4.4.4. Respondents at SubStat ≥ 6 constitute 63.3% of the sample ($n = 260$); those at SubStat ≤ 5 constitute 36.7% ($n = 151$). The high-status group reports substantially lower psychological scarcity (mean PsySca = 0.54) relative to the low status group (mean PsySca = 2.80), confirming strong construct convergence.

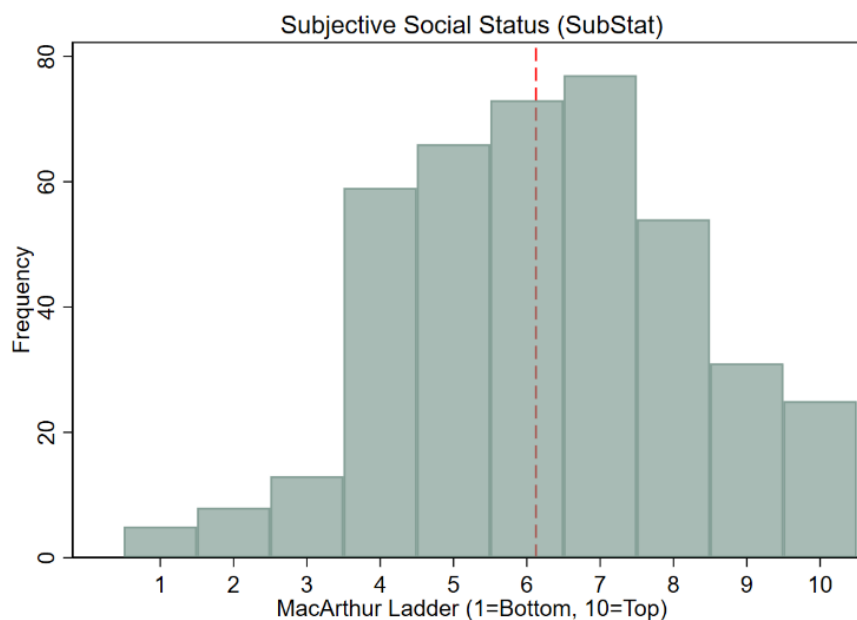


Chart 4. Distribution of Subjective Social Status (MacArthur Ladder, 1–10). Mean = 6.24.

4.1.3 Passive Upward Social Media Exposure (PasExp) and Usage Hours

PasExp has a mean of 2.94 ($SD = 1.11$, range 1–5), indicating moderate average passive exposure to upward comparison content. The distribution is moderately right-skewed (skewness = 0.43), with 55.7% of respondents concentrated in the 1–2 hours daily usage

bracket. Daily social media time (SOC_HRS) has a mean of 2.56 (SD = 0.96), with the modal category corresponding to 1–2 hours per day.

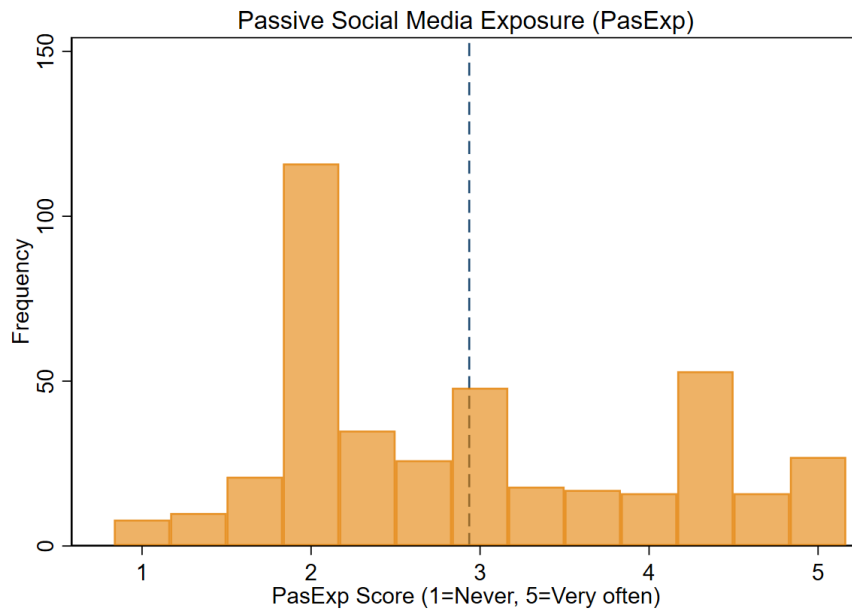


Chart 5. Distribution of Passive Upward Social Media Exposure (PasExp)

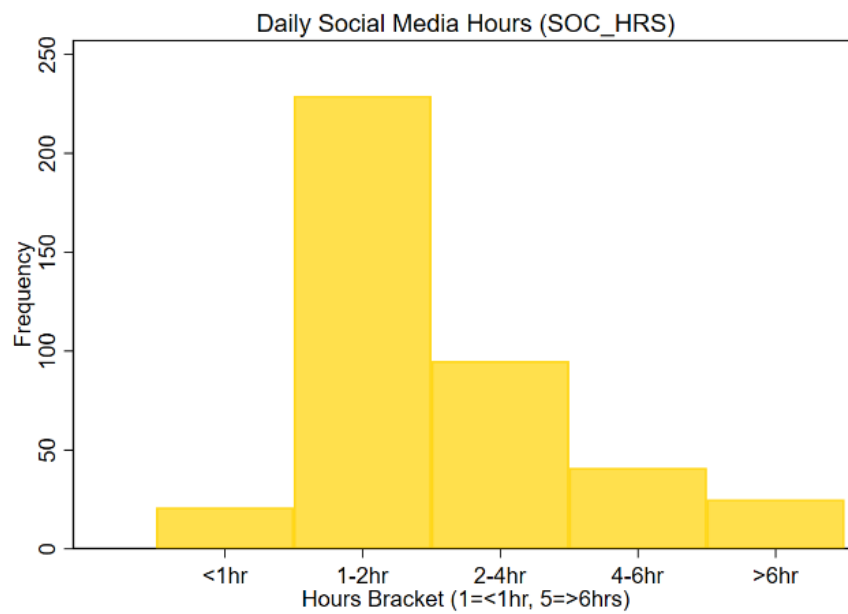


Chart 6. Distribution of Daily Social Media Hours (SOC_HRS).

4.1.4 Psychological Scarcity and Financial Controls

PsySca has a mean of 1.37 (SD = 1.32, range 0–4). The distribution is right-skewed (skewness = 0.46) with substantial mass at zero, reflecting the fact that approximately one quarter of respondents report no financial stress. CFPB_liquid (liquidity subscale) has a mean of 0.55 (SD = 0.15), indicating moderate liquidity constraint on average. Household income (HHInc) is centred on bracket 4 (30,001–50,000 THB), and OSSS-3 offline support has a mean of 0.40 (SD = 0.19), reflecting moderate social network availability.

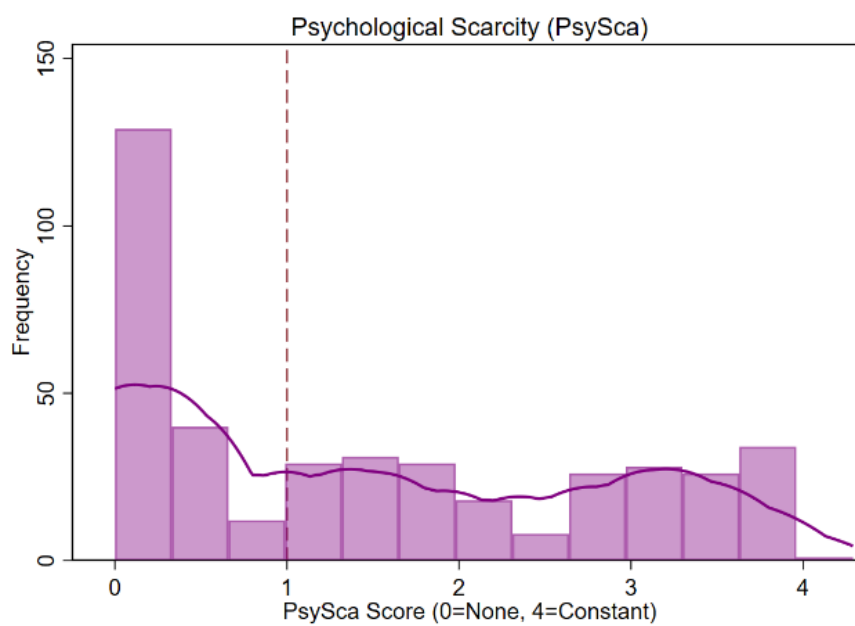


Chart 7. Distributions of Psychological Scarcity (PsySca)

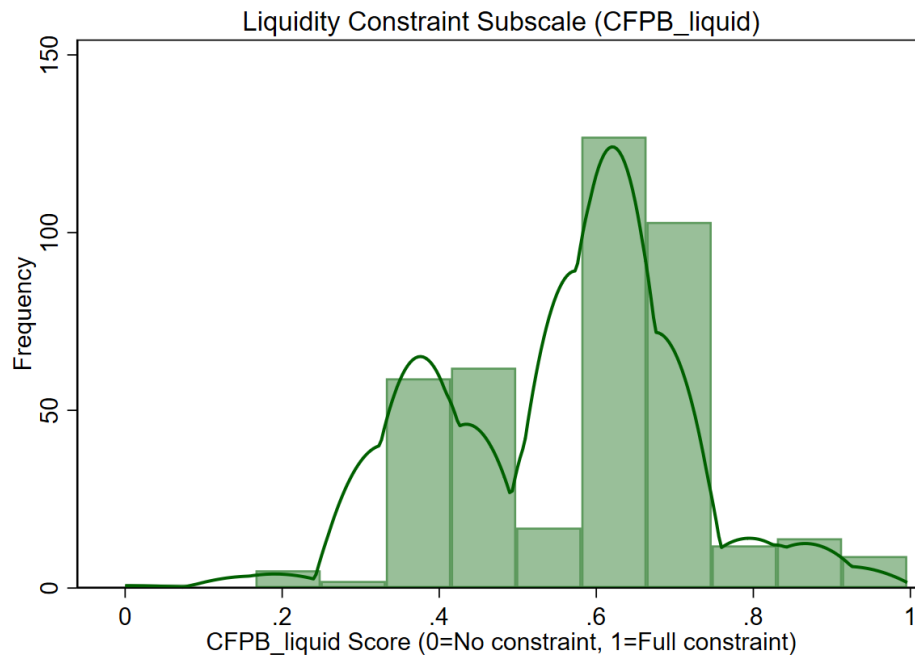


Chart 8. Distributions of CFPB Liquidity

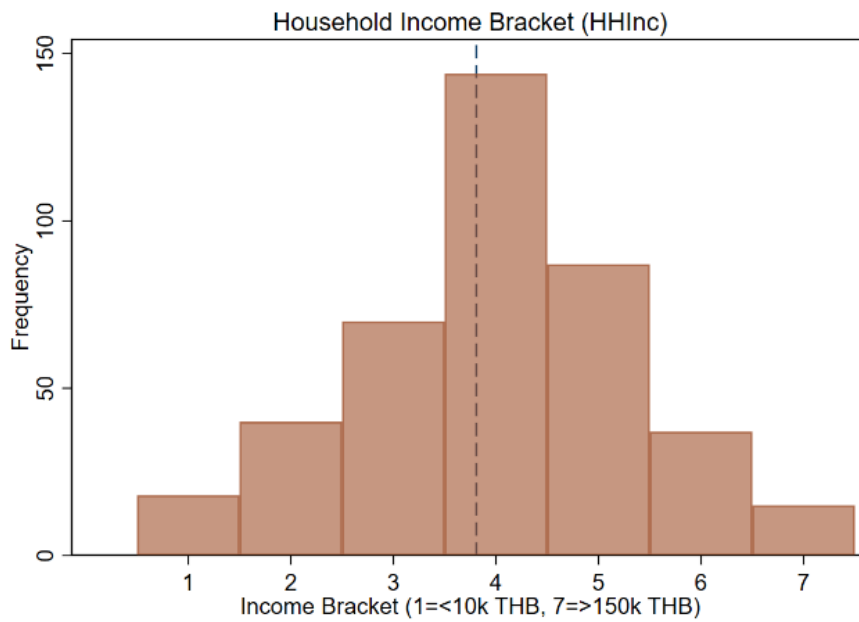


Chart 9. Distributions of Household Income (HHInc)

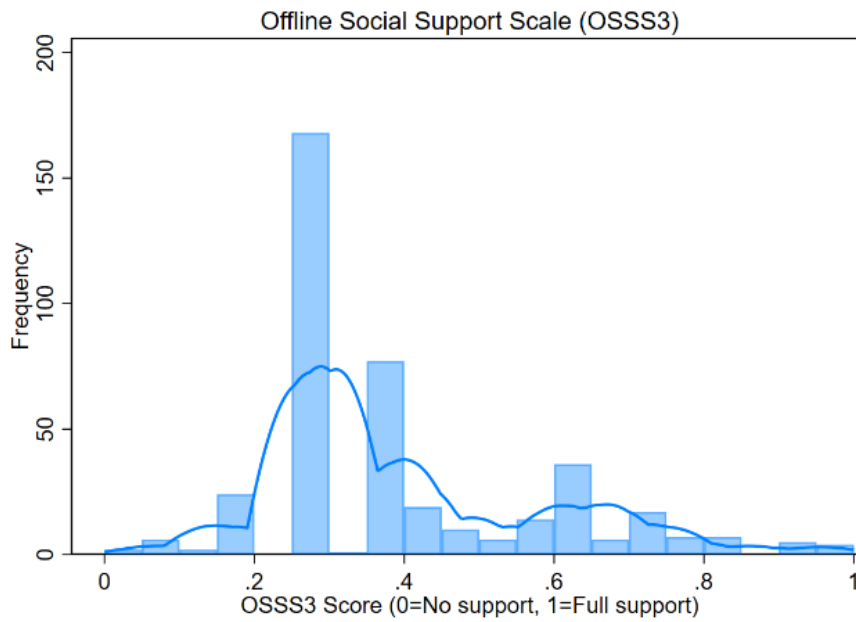


Chart 10. Distributions of Offline Social Support (OSSS-3).

4.1.5 High-Status vs. Low-Status Profile

Table 12 profiles key characteristics of respondents at or below the mid-point of the MacArthur Ladder (SubStat ≤ 5 , 'low-status group') versus those above (SubStat ≥ 6 , 'high-status group'). The contrast is striking: the low-status group has a mean PsySca of 2.80 compared to just 0.54 in the high-status group, a difference of more than 2.3 units on a 0–4 scale, confirming that subjective rank perception and financial anxiety are strongly co-determined in this sample.

Characteristic	Low Status (1–5, n=151)	High Status (6–10, n=260)	Total (n=411)
Mean PsySca (Psych. Scarcity)	2.80	0.54	1.37
Mean OSSS-3 (Offline Support)	0.44	0.37	0.40

% in HHInc bracket ≤ 2 ($<20k$ THB)	26.5%	7.0%	14.1%
% in HHInc bracket ≥ 5 ($>50k$ THB)	4.0%	51.2%	34.1%
% Commerce/Hospitality sector	35.1%	31.9%	33.1%
% Corporate/Professional sector	19.9%	22.3%	21.4%

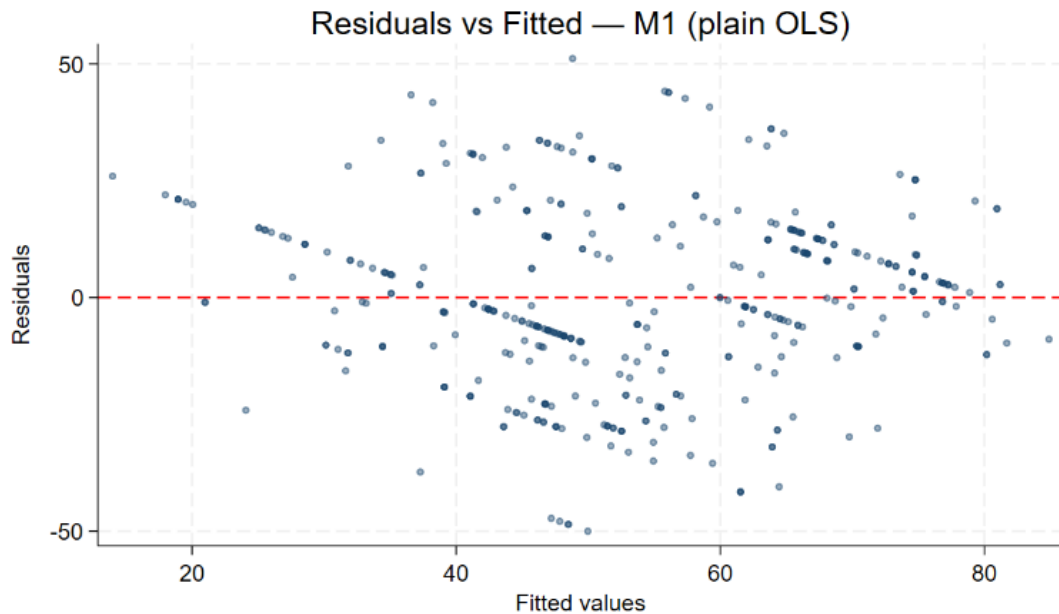
Table 12. Socioeconomic and psychological profile of low-status (SubStat ≤ 5) vs. high-status (SubStat ≥ 6) respondents.

4.2 Pre-Estimation Diagnostics

4.2.1 Heteroskedasticity

Plain-OLS (non-robust) estimation was run first to enable formal heteroskedasticity testing. The Breusch–Pagan/Cook–Weisberg test rejected the null of constant variance ($\chi^2(1) = 4.66$, $p = 0.031$) and White's general test confirmed heteroskedasticity more broadly ($\chi^2(84) = 178.93$, $p < 0.001$). Visual inspection of the residuals-vs.-fitted plot (Chart 11) is consistent with these findings: The structured pattern in the residual plot reflects the discrete, bounded nature of the WHO-5 scale, produced by multiplying a 0–25 raw sum by four, yielding values only at multiples of four, rather than systematic model misspecification.

Chart 11. Residuals vs. Fitted values — M1 plain OLS. Structured rails reflect WHO-5's discrete 4-point increments; no systematic heteroskedastic funnel pattern is present that would invalidate robust-SE correction.



All substantive models are estimated using heteroskedasticity to robust standard errors (`vce(robust)`; White, 1980). Bootstrap resampling (1,000 replications) was further applied to verify robustness of inference under non-constant variance.

4.2.2 Residual Normality

The Shapiro-Wilk test on OLS residuals failed to reject normality ($W = 0.994$, $p = 0.103$). The skewness-kurtosis joint test similarly found no departure ($\chi^2(2) = 0.70$, $p = 0.704$). Bootstrap confidence intervals (Normal, Percentile, and Bias-Corrected) were mutually consistent with negligible bias across all model estimates, confirming that OLS inference is reliable under the bounded dependent variable.

4.2.3 Multicollinearity

Variance inflation factors (VIFs) for the full M1 specification had a maximum of 3.00 (SubStat) and a mean of 1.63, well below the conventional threshold of 10. For the M3

interaction specification using mean-centred variables, the maximum VIF was 2.06 (PsySca) with a mean of 1.55. Multicollinearity is not a concern in either model.

4.2.4 Outliers and Leverage

Cook's D analysis (threshold = $4/n = 0.0097$) identified 22 observations with above-threshold influence values (max Cook's D = 0.026). No standardised residual exceeded $|3|$; five observations exceeded $|2.5|$. The leverage-residual-squared plot (Chart 12) confirms that high-leverage observations are concentrated near the zero-residual line, indicating unusual covariate combinations rather than influential data points driving the regression surface.

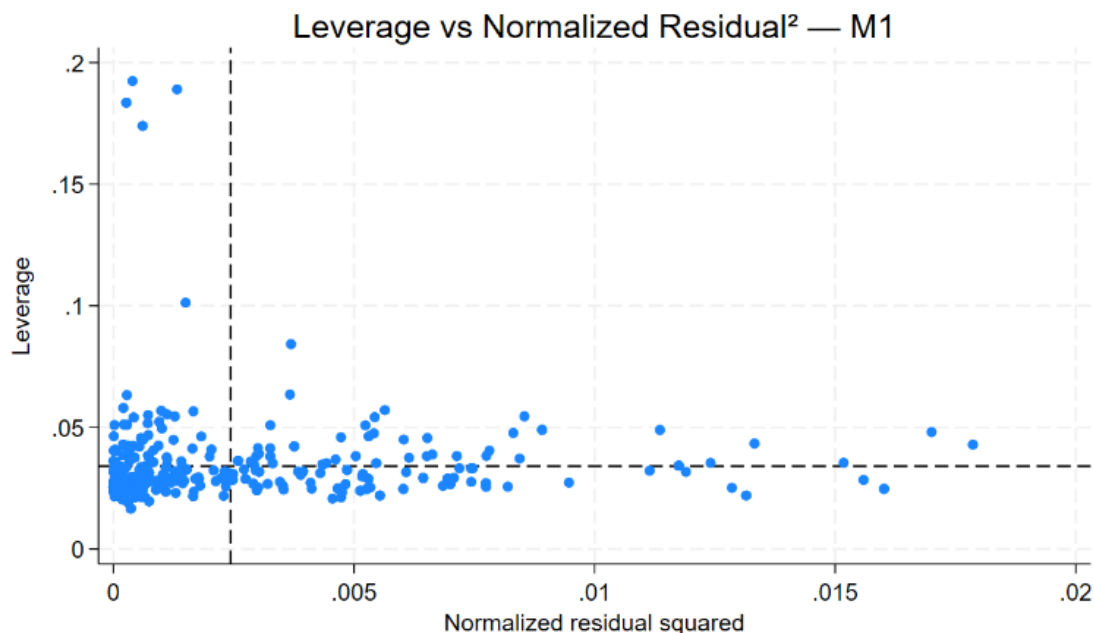


Chart 12. Leverage vs. Normalised Residual² — M1. High-leverage points (top left quadrant) have near-zero residuals and do not constitute influential observations. Dashed lines show sample-mean thresholds.

Sensitivity re-estimation excluding these 22 observations ($n = 389$) produced stable core coefficients: PasExp -9.07 vs -9.62 in the full sample; SubStat $+3.68$ vs $+3.27$; PsySca -3.46 vs -3.68 . All core variables retained their significance and sign, confirming that no small cluster of cases drives the main findings.

4.3 Main Regression Results

Table 13 presents the results of all three regression models. The discussion below focuses on the statistically significant and theoretically relevant findings, with particular attention to the direction, magnitude, and robustness of key coefficients across model specifications. Emphasis is placed on the roles of subjective social status, psychological scarcity, and passive upward social media exposure, as well as their interaction effects, in shaping subjective well-being. Where relevant, the results are interpreted in light of the underlying theoretical framework and prior empirical evidence.

Variable	M1 (Baseline)	M2 (+Occupation)	M3 (Interaction)
Key Independent Variables			
PasExp (Passive Exposure)	−9.622***	−9.476***	+8.627***
	(0.887)	(0.900)	(3.286)
SubStat (Social Status)	+3.268***	+3.345***	+10.212***
	(0.925)	(0.937)	(1.364)
PsySca (Psych. Scarcity)	−3.678***	−3.420***	−4.097***
	(1.092)	(1.097)	(0.992)
PasExp × SubStat	-	-	−2.678***
			(0.502)
Financial Controls			
CFPB_liquid	−9.273	−8.919	-
	(8.945)	(9.011)	
HHInc (Household Income)	−0.213	−0.223	-
	(1.040)	(1.087)	

Social and Behavioural Controls			
SOC_HRS (Social Media Hours)	+2.240†	+1.977	-
	(1.309)	(1.315)	
Demographic Controls			
d_female	-1.260	-1.375	-
	(1.932)	(1.956)	
d_gnd_nd (Non-binary/NS)	+10.582**	+9.622†	-
	(5.318)	(5.443)	
d_age25_34	+3.270	+2.874	-
	(2.372)	(2.413)	
d_age35_44	+0.171	-0.144	-
	(2.638)	(2.672)	
d_live_alone	+0.729	+0.126	-
	(2.789)	(2.814)	
d_live_shared	-2.974	-3.043	-
	(2.854)	(2.889)	
d_live_other	-1.031	-1.037	-
	(3.016)	(3.008)	
Industry Group (ref. = Corporate/Professional)			
Public/Essential	-	+0.359	-
		(3.038)	
Commerce/Hospitality	-	-1.170	-

		(2.846)	
Labour/Ag/Informal	-	-4.610	-
		(3.401)	
Model Fit			
Constant	66.584***	66.906***	18.984**
	(9.883)	(10.355)	(9.455)
N	411	411	411
R ²	0.3692	0.3739	0.4049
Adjusted R ²	0.3485	0.3468	0.3991
AIC	3,610.1	3,615.1	3,568.2
BIC	3,666.4	3,687.4	3,588.3
Joint F (core vars), p-val	F(3,397)=69.2, p<0.001	F(3,393)=62.9, p<0.001	F(4,406)=117.0, p<0.001

Table 13. OLS regression results — M1 (Baseline), M2 (+Occupation), M3 (Interaction). Robust standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.10$. Reference categories: male, 18–24 years, living with family, Corporate/Professional sector.

Note: The sign reversal of the PasExp coefficient between M1 and M3 is a mathematical consequence of the interaction specification and is addressed fully in Section 4.4.1.

4.3.1 Model 1: Baseline Specification

Model 1 explains 36.9% of the variance in WHO-5 scores (Adjusted $R^2 = 0.349$). The joint F-test on the three core theoretical predictors is highly significant ($F(3, 397) = 69.20$, $p < 0.001$), confirming that the conceptual framework identifies economically and psychologically meaningful drivers of well-being.

Passive Upward Social Media Exposure (PasExp): $\beta = -9.622$ ($SE = 0.887$, $p < 0.001$). Each unit increase in the PasExp composite (1–5 scale) is associated with a 9.62 points

reduction in the WHO-5 index, holding all other variables constant. This is the largest effect in the model in absolute magnitude and is highly precisely estimated. At the sample mean exposure level, a shift from 'sometimes' (PasExp = 3) to 'very often' (PasExp = 5) corresponds to an expected WHO-5 reduction of approximately 19 points, more than three-quarters of one standard deviation.

Psychological Scarcity (PsySca): $\beta = -3.678$ (SE = 1.092, $p = 0.001$). Greater perceived financial and cognitive constraint is significantly associated with lower well-being. Each unit increase in PsySca (0–4 scale) corresponds to a 3.68-point decline in WHO-5, independently of income level. This confirms that the psychological mechanism of scarcity, distinct from objective income, independently erodes mental well-being.

Subjective Social Status (SubStat): $\beta = +3.268$ (SE = 0.925, $p < 0.001$). A one-rung improvement in perceived social standing on the MacArthur Ladder is associated with 3.27 points of increase in WHO-5, all else equal. Moving from the median (rung 6) to the upper quartile (rung 8) is associated with an approximate 6.5 points of improvement, nearly one-third of a standard deviation.

Among the control variables, CFPB_liquid ($\beta = -9.27$, $p = 0.301$), HHInc ($\beta = -0.21$, $p = 0.838$), age dummies, and living arrangement dummies are all non-significant. The non-binary/prefer-not-to-say category is marginally significant ($\beta = +10.58$, $p = 0.047$) but should not be interpreted substantively given the very small cell size ($n = 6$). Rather than a stable effect. SOC_HRS approaches marginal significance ($\beta = +2.24$, $p = 0.088$), suggesting a weak positive association between total social media time and well-being that is independent of passive exposure content.

4.3.2 Model 2: Adding Occupation (Industry Group)

Model 2 adds industry group dummies (reference: Corporate/Professional). The explanatory power increases marginally to $R^2 = 0.374$, a gain of less than 0.5 percentage points. None of the four industry group coefficients approach significance: Public/Essential (+0.36, $p = 0.906$), Commerce/Hospitality (−1.17, $p = 0.681$), Labour/Ag/Informal (−4.61, $p = 0.176$),

and Unclassified (+4.72, $p = 0.310$). The joint F-test on all dropped M2 controls ($F(10, 397) = 1.26$, $p = 0.251$) confirms that they add no explanatory power as a group once psychological scarcity, social status, passive exposure, and income are held constant. The core coefficients on PasExp, PsySca, and SubStat remain nearly identical to M1, demonstrating that industry classification is not a confound in the main relationships. Model 2's AIC (3,615.1) is higher than Model 1 (3,610.1), confirming no information gain from including occupation in a non-interaction framework.

4.4 Model 3 (Moderation Analysis): The PasExp $\times$ SubStat Interaction

Model 3 introduces the interaction term between mean-centred PasExp and SubStat. The model explains 40.5% of the variance in WHO-5 (Adjusted $R^2 = 0.399$), a substantial improvement of 3.6 percentage points over M1 and the lowest AIC of the three models (3,568.2, a difference of 42 points over M1). The interaction coefficient is negative and highly significant ($\beta = -2.678$, $SE = 0.502$, $p < 0.001$; $F(1, 406) = 28.52$), confirming the central moderation hypothesis.

4.4.1 Interpretation of the Interaction Equation

The estimated Model 3 equation is:

$$WHO5 = 18.98 - 4.10 \cdot PsySca + 10.21 \cdot SubStat + 8.63 \cdot PasExp - 2.68 \cdot (PasExp \times SubStat)$$

Because both PasExp and SubStat appear in an interaction, neither coefficient can be interpreted in isolation. The marginal effect of PasExp on well-being is:

$$\partial WHO5 / \partial PasExp = 8.627 - 2.678 \cdot SubStat$$

This expression equals zero when $SubStat \approx 3.22$. Below this threshold, passive exposure is associated with higher predicted well-being (positive marginal effect); above rung 3.22, the marginal effect becomes negative and grows increasingly harmful as social rank rises. At the sample mean SubStat of 6.24, the marginal effect of PasExp is -8.08 ($p < 0.001$).

Symmetrically, the marginal effect of SubStat on well-being is:

$$\partial \text{WHO5} / \partial \text{SubStat} = 10.212 - 2.678 \cdot \text{PasExp}$$

This equals zero when $\text{PasExp} \approx 3.81$. Below this exposure threshold, higher social rank strongly benefits well-being; above $\text{PasExp} = 3.81$, the benefit of rank diminishes and eventually reverses for the highest-exposure individuals.

Before interpreting the marginal effects at observed values, one aspect of Table 13 requires explicit clarification: the apparent sign reversal of the PasExp coefficient between M1 and M3.

In M1, PasExp carries a coefficient of -9.622 ($p < 0.001$). In M3, the same variable is reported as $+8.627$, a positive number. This is not a contradiction. In a model without interaction, the coefficient on PasExp represents its unconditional average effect across all values of SubStat. Once the interaction term $\text{PasExp} \times \text{SubStat}$ is introduced, this coefficient becomes the conditional effect of PasExp when $\text{SubStat} = 0$, a value that does not exist in the data (the MacArthur Ladder runs from 1 to 10) and has no substantive meaning on its own.

The correct quantity to interpret is the marginal effect at observed covariate values. At the sample mean SubStat of 6.24, this equals -8.081 ($p < 0.001$, 95% CI $[-9.634, -6.529]$), closely consistent with the M1 estimate of -9.622 . Both models describe the same underlying reality. The apparent sign reversal is purely an artefact of evaluating the M3 coefficient at an unobserved anchor point rather than a meaningful change in direction.

The same logic applies to SubStat: its M3 coefficient of $+10.212$ represents the effect when $\text{PasExp} = 0$, also outside the observed range. The marginal effect at mean PasExp of 2.942 is $+2.333$ ($p = 0.002$, 95% CI $[+0.888, +3.778]$), comparable to the M1 direct estimate of $+3.268$. Both coefficients reflect a positive well-being return from higher perceived rank, holding other variables constant.

4.4.2 M3 and M3c: Uncentred vs Mean-Centred Specifications

To confirm the interpretation above and provide more directly readable coefficients, Model 3 was re-estimated using mean-centred variables (M3c): $\text{PasExp_c} = \text{PasExp} - \bar{x}\text{PasExp}$ and $\text{SubStat_c} = \text{SubStat} - \bar{x}\text{SubStat}$, where $\bar{x}\text{PasExp} = 2.942$ and $\bar{x}\text{SubStat} = 6.238$.

	M3 (Uncentred)	M3c (Mean-Centred)
Constant	18.984** (9.455)	58.917*** (1.468)
PasExp / PasExp_c,	+8.627*** (3.286)	−8.081*** (0.790)
SubStat / SubStat_c	+10.212*** (1.364)	+2.333** (0.735)
PasExp × SubStat	−2.678*** (0.502)	−2.678*** (0.502)
PsySca	−4.097*** (0.992)	−4.097*** (0.992)
N	411	411
R ²	0.4049	0.4049
F(4, 406)	116.99	116.99

Table 14. below presents both specifications side by side.

Note: Robust standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. M3c uses PasExp and SubStat each demeaned to zero.

The two models are algebraically identical. They have the same R^2 , the same F-statistic, and the same interaction and PsySca coefficients. What changes is only how the intercept and the two main-effect coefficients are reported, because centring shifts the origin from zero to the sample mean. The key equivalences are:

First, the M3c intercept of 58.917 is the predicted WHO-5 score for a respondent at the mean of PasExp, mean of SubStat, and zero PsySca, a directly interpretable quantity. The M3 intercept of 18.984 corresponds to the nonsensical baseline of $\text{PasExp} = 0$ and $\text{SubStat} = 0$.

Second, the M3c PasExp_c, coefficient of -8.081 is identical to the MEM of PasExp computed analytically from M3 ($8.627 - 2.678 \times 6.238 = -8.081$). It directly states at average social rank, one unit increase in passive exposure reduces well-being by 8.08 points.

Third, the M3c SubStat_c coefficient of $+2.333$ is identical to the MEM of SubStat computed from M3 ($10.212 - 2.678 \times 2.942 = +2.333$). At average passive exposure, one additional rung on the MacArthur Ladder improves well-being by 2.33 points.

Fourth, the interaction term is invariant to centring (-2.678 in both), as the moderating relationship between PasExp and SubStat is a mathematical property of the data and cannot be altered by a linear shift of origin.

In substantive terms, M3c makes the cost of passive social media exposure visible without requiring any further computation: for the typical Bangkok working-age adult sitting at the middle of the social hierarchy, each unit increase in passive digital comparison exposure reduces subjective well-being by approximately 8 points on the 0–100 WHO-5 scale, an effect roughly one-third of a standard deviation and comparable in magnitude to the clinical threshold difference between at-risk and non-at-risk individuals. M3 and M3c should be presented together precisely because M3 is needed to recover the interaction surface and evaluate sign changes at the tails of the ladder (as in Table 15 of this chapter), while M3c provides the cleaner, directly policy-relevant coefficients at the centre of the distribution.

4.4.3 Marginal Effects Across the Full Ladder Range

SubStat	ME (PasExp)	95% CI	p-value	Significant?	Direction
1	+5.949	[+0.44, +11.46]	0.034	Yes *	Positive ↑
2	+3.271	[−1.30, +7.84]	0.160	No	Positive ↑
3	+0.592	[−3.07, +4.25]	0.750	No	Near zero ≈ 0
4	−2.086	[−4.88, +0.71]	0.143	No	Negative ↓

5	-4.764	[-6.81, -2.72]	<0.001	Yes ***	Negative ↓
6	-7.443	[-9.03, -5.85]	<0.001	Yes ***	Negative ↓
7	-10.121	[-11.79, -8.45]	<0.001	Yes ***	Negative ↓
8	-12.799	[-15.04, -10.56]	<0.001	Yes ***	Negative ↓
9	-15.478	[-18.50, -12.45]	<0.001	Yes ***	Negative ↓
10	-18.156	[-22.06, -14.25]	<0.001	Yes ***	Negative ↓

Table 15. Marginal effect of PasExp on WHO-5 at each level of SubStat (M3, n = 411). Shading indicates sign-change zone (SubStat $\approx$ 3.22). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

For respondents at SubStat = 1, the marginal effect is positive and marginally significant (+5.95, $p = 0.034$). This finding is interpretable through Prospect Theory (Kahneman & Tversky, 1979): individuals at the floor of the social hierarchy have no meaningful upward comparison anchor, so exposure to aspirational content may function as aspirational utility rather than a loss. The gap between their reference point and observed wealth displays is so large it ceases to register as a threatening comparison and instead operates as entertainment or escapism. This stands in contrast to mid-to-high status individuals, for whom the same content narrows the perceived gap to their reference point and triggers loss of aversion. This boundary finding warrants cautious interpretation and future investigation. For those at the top of the hierarchy (SubStat = 10), passive social media exposure is associated with an 18.16-point reduction in WHO-5 per unit of exposure, an extraordinarily large effect that suggests high-status individuals may experience digital wealth displays as particularly threatening to a carefully maintained self-concept.

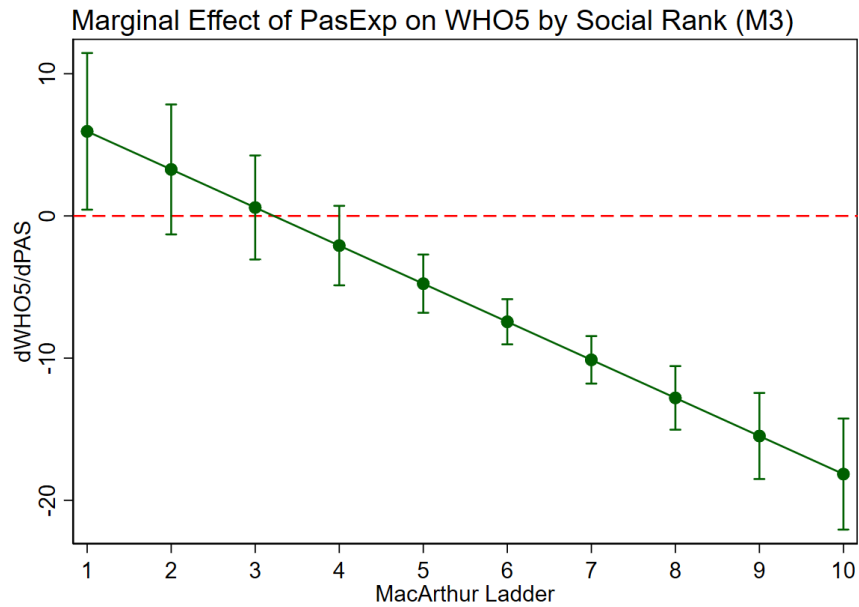


Chart 13. Marginal effect of passive social media exposure (PasExp) on WHO-5 well-being across social rank values (M3). Dashed red line = zero (null effect). Confidence intervals computed via the delta method.

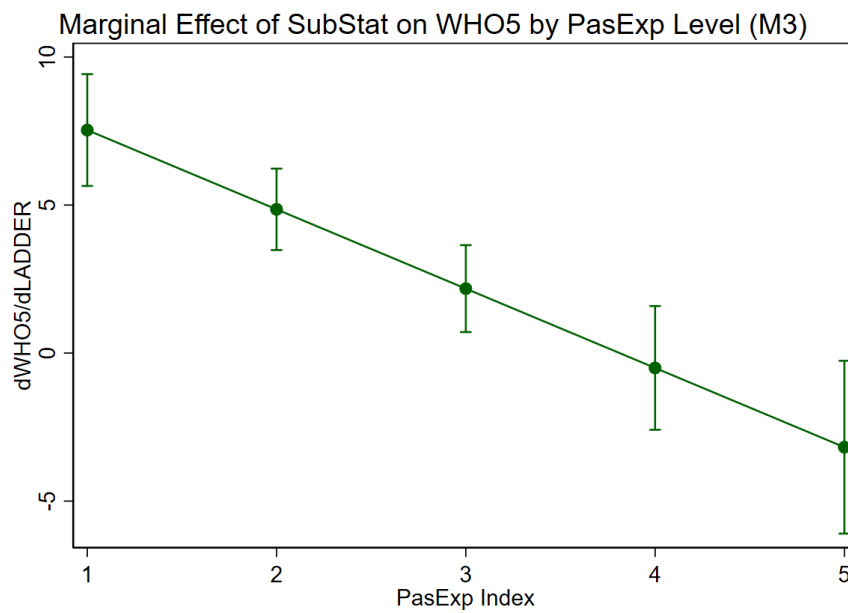


Chart 14. Marginal effect of subjective social status (SubStat) on WHO-5 across passive exposure levels (M3). The protective benefit of higher social rank diminishes and eventually reverses at high exposure levels.

4.4.4 Interaction Surface

The interaction surface (Chart 15) plots predicted WHO-5 scores across the full 5×10 grid of PasExp (1–5) and SubStat (1–10) combinations, holding PsySca at its sample mean. Several features are noteworthy. First, at low exposure (PasExp = 1), subjective social status has a strong linear positive effect on well-being, with predicted WHO-5 rising from approximately 30 to 97 across the ladder range. Second, this rank gradient compresses as exposure increases at maximum exposure (PasExp = 5), the predicted well-being surface becomes nearly flat or slightly inverted across the ladder, suggesting that very high passive exposure disrupts the ordinarily protective function of perceived social rank.

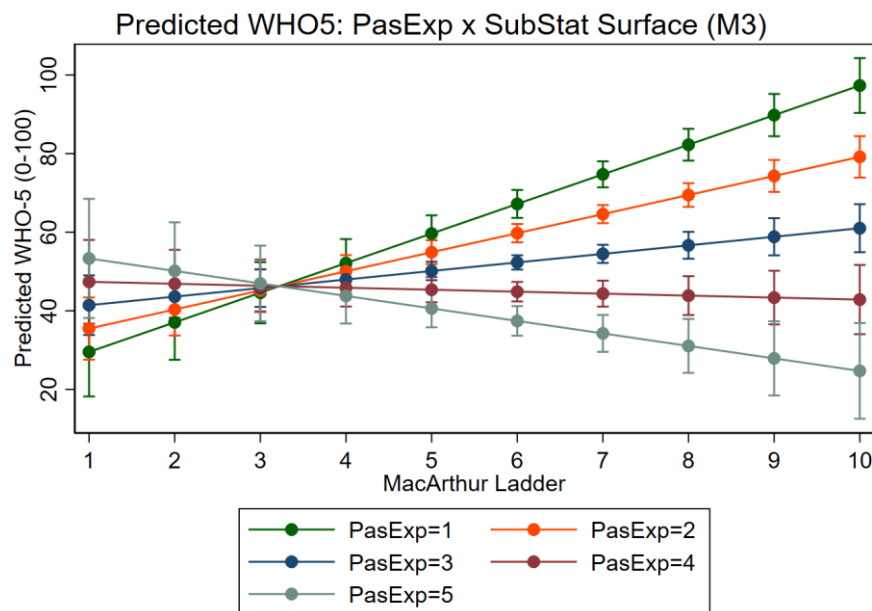


Chart 15. Predicted WHO-5 well-being scores across the PasExp × SubStat interaction surface (M3). Colours represent predicted WHO-5 values; each line is a fixed PasExp level.

4.5 Industry-Group Heterogeneity in Marginal Effects

Although industry group dummies are jointly insignificant as additive controls (Section 4.3.2), they introduce substantively interesting variation in the conditional marginal effects of PasExp and SubStat when evaluated at industry-specific covariate means. Table 16 reports

these conditional marginal effects from M3, computed at each group's mean PasExp, SubStat, and PsySca values.

Industry Group	Mean SubStat	Mean PsySca	ME(PasExp)	ME (SubStat)	Mean WHO-5
Corporate/Professional	6.64	1.23	−9.15***	+1.71*	53.5
Public/Essential Svcs	6.02	1.38	−7.48***	+2.55***	55.6
Commerce/Hospitality	6.18	1.44	−7.92***	+2.49***	52.7
Labour/Ag/Informal	6.31	1.60	−8.27***	+2.06**	46.9
Sample Total	6.24	1.37	−8.08***	+2.33**	53.6

Table 16. Conditional marginal effects of PasExp and SubStat on WHO-5 by industry group (M3), evaluated at each group's mean covariate values. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. ME(PsySca) = −4.097*** (constant across groups as there is no PsySca interaction term).

The marginal effect of PasExp is significantly negative across all industry groups, ranging from −7.48 (Public/Essential Services) to −9.15 (Corporate/Professional). The ranking of these marginal effects follows directly from the interaction structure of M3: because $\partial \text{WHO-5} / \partial \text{PasExp} = 8.627 - 2.678 \cdot \text{SubStat}$, a higher mean SubStat mechanically produces a more negative marginal effect. Corporate/Professional workers, who have the highest mean SubStat (6.64), therefore experience the steepest well-being cost per unit of passive exposure (ME = −9.15), while Public/Essential Services workers, with the lowest mean SubStat (6.02), experience the least severe effect (ME = −7.48). This pattern is not a paradox. It reflects the core moderation finding that passive social media exposure is most harmful for those who perceive themselves as higher in social standing, since they have more of a status of self-concept to lose when confronted with upward comparison content.

The Labour/Agriculture/Informal group merits particular attention. Although their industry membership dummy is not significant in M2, they have the lowest mean well-being in the sample (WHO-5 = 46.9, below the clinical at-risk threshold), the highest mean PsySca (1.60), and a moderately strong negative marginal effect of passive exposure (−8.27). This

configuration suggests a compounding vulnerability: workers in precarious or low-wage sectors face both greater financial stress and significant exposure-driven well-being erosion.

The marginal effect of SubStat is positive and significant across all groups, ranging from +1.71 (Corporate/Professional, $p = 0.031$) to +3.59 (Unclassified, $p < 0.001$). The relatively smaller SubStat benefit for Corporate/Professional workers may reflect reduced salience of the MacArthur Ladder gradient among those who already occupy the upper portion of the perceived hierarchy; their well-being is more sensitive to the incremental increase in status that each rung provides.

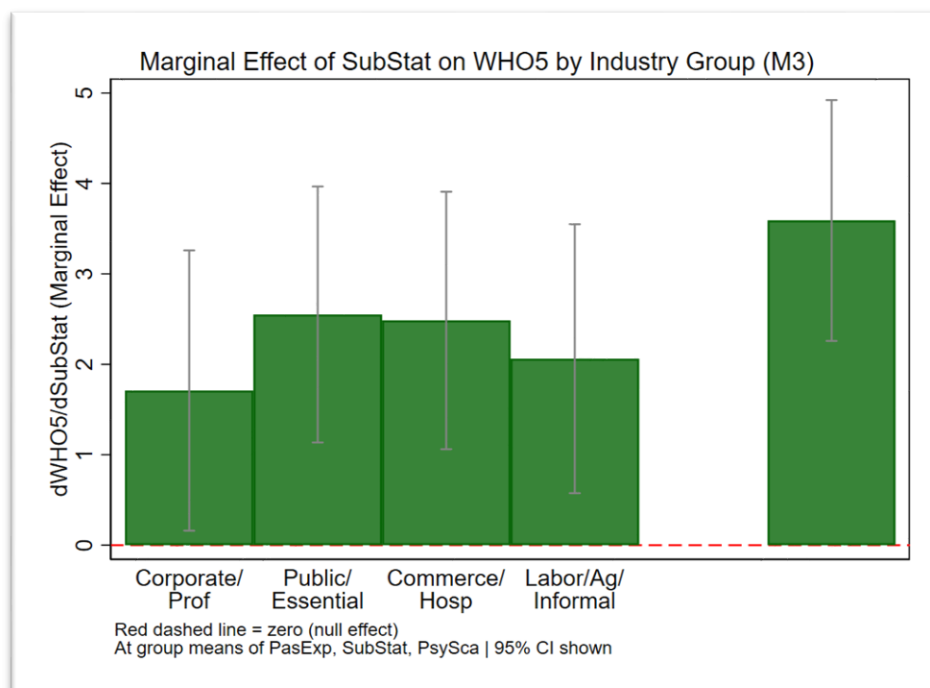


Chart 16. Marginal effect of SubStat on WHO-5 by industry group (M₃).

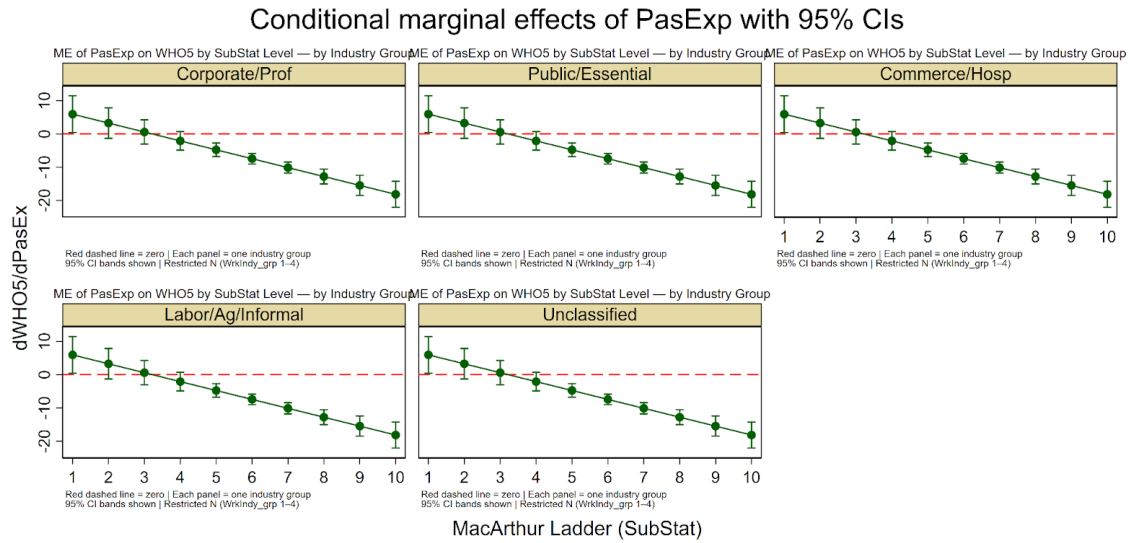
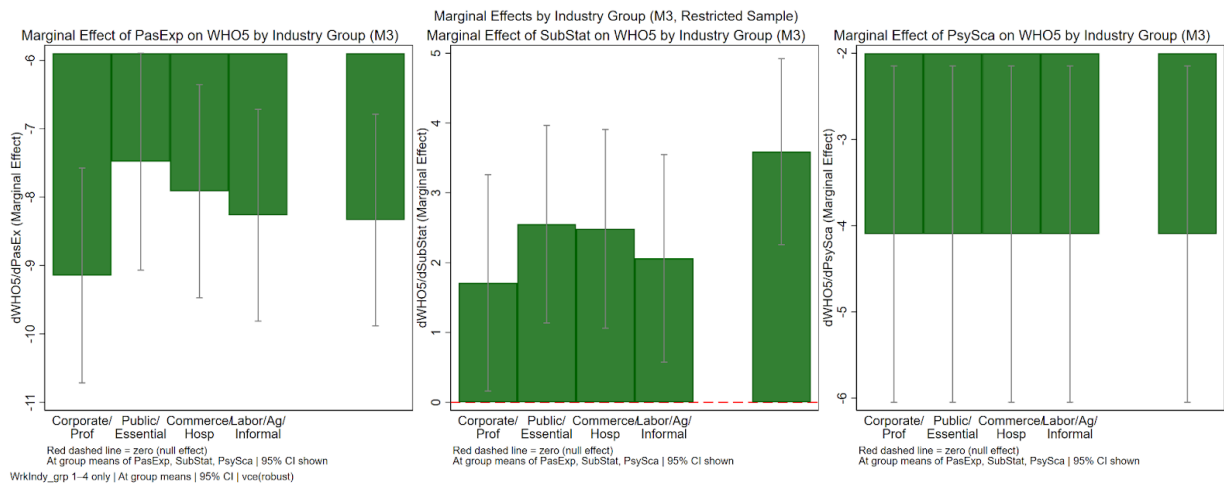


Chart 17. Conditional marginal effects of PasExp

Chart 18. Marginal effects of PasExp (left), SubStat (centre), and PsySca (right) on WHO-5 by industry group (M3), evaluated at each group's conditional means. Red dashed line = zero; error bars = 95% CI.



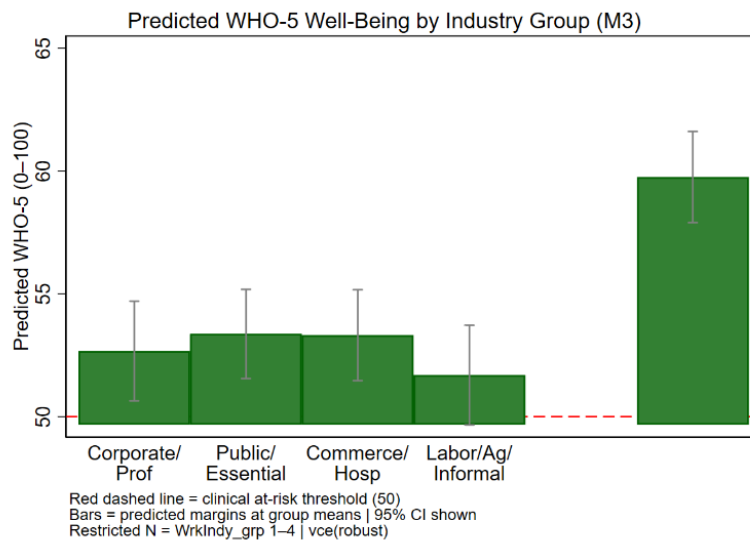


Chart 19. Predicted WHO-5 well-being scores by industry group at each group's conditional mean values of all predictors (M3).

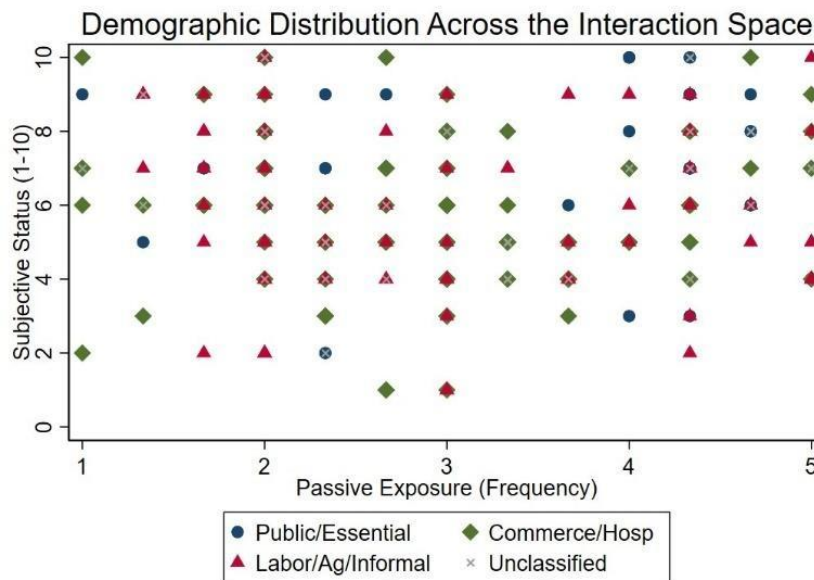


Chart 20. Placement of industry groups in the PasExp–SubStat interaction space. Points represent individual respondents, with group centroids overlaid. Industries with higher mean SubStat and higher PasExp fall in the zone of greatest interaction-driven well-being loss.

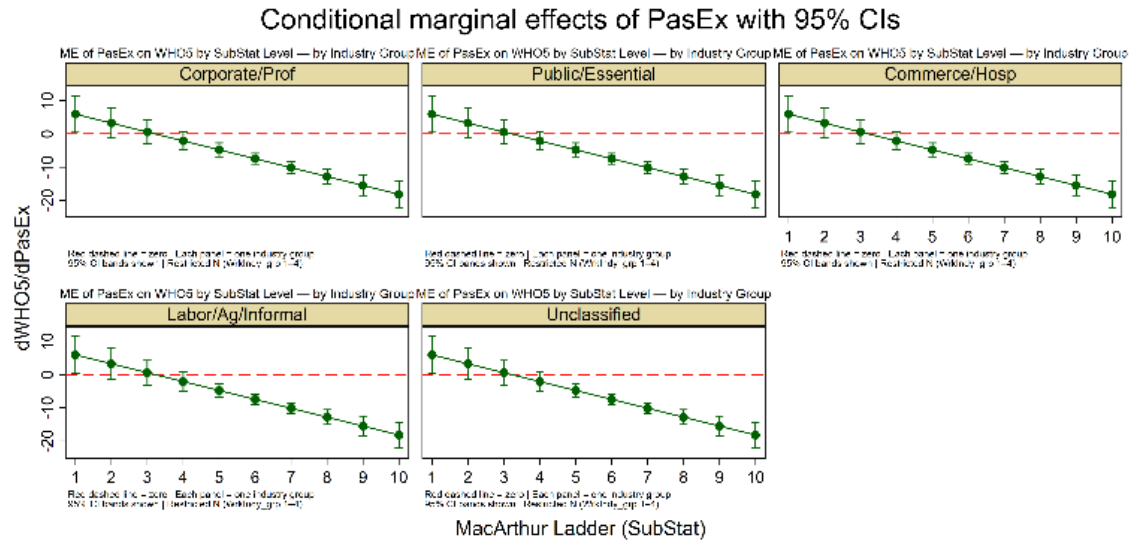


Chart 21. Interaction surface of PasExp marginal effects across social rank for each industry group. The sign-change point ($\approx$ SubStat 3.22) is consistent across groups; the slope steepness differs by group mean SubStat.

4.6 Robustness Checks

Table 17 compares the M3 baseline specification against two alternative specifications: (i) M3 estimated on the reduced sample excluding the 22 Cook's-D influential observations; and (ii) M3 re-estimated with all M1 control variables re-added. All three key interaction-model findings are consistent across specifications.

Variable	M3 Baseline	M3 Excl. Influential	M3 + Full Controls
PasExp	+8.627*** (3.286)	+8.123** (3.311)	+7.667** (3.449)
SubStat	+10.212*** (1.364)	+9.841*** (1.298)	+10.552*** (1.501)

PasExp × SubStat	−2.678*** (0.502)	−2.581*** (0.491)	−2.629*** (0.518)
PsySca	−4.097*** (0.992)	−3.854*** (0.963)	−4.285*** (1.097)
N	411	389	411
R ²	0.405	0.439	0.422

Table 17. Robustness of M3 core coefficients across alternative specifications. Standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Bootstrap resampling (1,000 replications): Normal, Percentile, and Bias-Corrected 95% confidence intervals are mutually consistent for all four core M3 coefficients. Bias in bootstrap replications is negligible: PasExp bias = +0.241, SubStat bias = +0.097, interaction bias = −0.035, PsySca bias = −0.005. The interaction term BC-CI [−3.556, −1.599] does not cross zero, confirming its robustness to distributional assumptions.

CFPB_liquid exclusion: Removing CFPB_liquid from M1 changes the PasExp coefficient by less than 0.04 points (−9.584 vs −9.622), PsySca by 0.53 points (in range of sampling variation), and SubStat by 0.15 points. The LR-test confirms no improvement in model fit from including CFPB_liquid ($\chi^2(1) = 1.24$, $p = 0.266$; $\Delta AIC = -0.76$). The construct is retained in M1 for theoretical completeness but does not drive the results.

Interaction with full controls (M3 + all M1 covariates): The interaction coefficient remains −2.629 (SE = 0.518, $p < 0.001$) compared to −2.678 in the parsimonious M3, a difference of less than 0.05 points. The core story is not an artefact of omitted controls.

4.7 Summary of Findings

First, passive upward social media exposure significantly harms well-being. In M1, a one-unit increase in PasExp is associated with a 9.62 points reduction in WHO-5. This effect is stable across all specifications and represents the strongest single predictor of well-being in

the baseline model. H4 is supported: the marginal effect of PasExp evaluated at the sample mean SubStat of 6.24 is -8.08 ($p < 0.001$), confirming a strong negative net association with well-being consistent with the hypothesised direction. The apparent sign reversal in the M3 interaction equation is a mathematical property of the uncentored specification and not a substantive contradiction.

Second, psychological scarcity independently erodes well-being. Each unit increase in PsySca corresponds to a 3.68- to 4.10-point WHO-5 reduction depending on the model, confirming that the subjective experience of financial constraint operates through a psychological channel distinct from objective income levels.

Third, subjective social status positively predicts well-being, and this benefit is moderated by passive upward social media exposure. The interaction analysis reveals that at low exposure levels, higher social rank strongly protects well-being; at high exposure levels, this rank advantage is neutralised or reversed. The sign-change occurs at SubStat ≈ 3.22 , meaning that for most of the Bangkok working-age sample, who cluster around the mid-to-upper rungs, passive social media use is consistently and significantly harmful.

Together, these findings suggest that digital environments in which wealth and lifestyle comparisons are ubiquitous may systematically undermine the psychological benefits that individuals derive from their perceived social standing. The interaction between online upward comparison and subjective rank is not merely additive: it actively disrupts the positive role that social hierarchy perception typically plays in sustaining mental well-being. This mechanism appears to operate across all industry sectors, though its magnitude is amplified among workers in higher-status occupational groups whose social-rank self-concept may be most at stake in digital comparison contexts.

Chapter 5: Conclusion

5.1 Hypothesis Evaluation and Theoretical Contributions

This study examines the determinants of subjective well-being among 411 working-age adults in Bangkok, Thailand. It is motivated by the recognition that subjective well-being serves as a proxy for utility in behavioural economics and that psychological mechanisms help explain the link between objective income and individual welfare (Kahneman and Deaton, 2010; Mullainathan and Shafir, 2013). Grounded in the Easterlin Paradox (Easterlin, 1974) and the scarcity-as-bandwidth-tax framework, the study tests four pre-specified hypotheses using OLS regression with robust standard errors across three nested model specifications (M1, M2, M3), with WHO-5 well-being (0–100) as the dependent variable. All four hypotheses are supported by the data.

H	Hypothesis	Verdict	Key Evidence
<i>H₁</i>	<i>PsySca</i> → <i>WHO5</i> (negative)	Supported	$\beta = -3.68$ (M1), -4.10 (M3); $p < 0.001$ across all specifications. Financial worry independently suppresses well-being beyond objective income.
<i>H₂</i>	<i>SubStat</i> → <i>WHO5</i> (positive)	Supported	$\beta = +3.27$ (M1); MEM = +2.33 at mean PasExp (M3); $p < 0.001$. Perceived rank outperforms objective income as a well-being predictor.
<i>H₃</i>	<i>PasExp</i> moderates <i>SubStat</i> → <i>WHO5</i>	Supported	$\beta = -2.678$ (SE = 0.502, $p < 0.001$). Social rank premium progressively cancelled at higher exposure. Sign-change at SubStat ≈ 3.22 for PasExp

			MEM; at PasExp $\approx$ 3.81 for SubStat MEM.
H₄	<i>PasExp</i> $\rightarrow$ <i>WHO5</i> (negative)	Supported	MEM = -8.08 ($p < 0.001$) at mean SubStat. Unconditional M1 estimate -9.62. Largest single predictor of well-being loss in the sample.

Table 18. Hypothesis evaluation summary. MEM = Marginal Effect at Mean; SE = standard error.

Psychological Scarcity (H1)

Psychological scarcity exerts a statistically significant, consistent negative effect on well-being across all model specifications ($\beta = -4.10$ in M3, $p < 0.001$). This provides direct empirical validation of the scarcity-as-bandwidth-tax model (Mullainathan & Shafir, 2013) within a Thai urban context. While prior research demonstrated that scarcity impairs cognitive capacity in experimental settings, the present study extends this to show that perceived insufficiency, chronic financial worry, independently suppresses hedonic well-being even after controlling for objective income and social support. This confirms the core theoretical claim that the mental bandwidth tax of perceived precarity, rather than objective income per se, functions as a profound cost to utility.

Subjective Social Status (H2)

Subjective social status is positively and significantly associated with well-being ($\beta = +3.27$ in M1, $p < 0.001$). At mean passive digital exposure, each additional MacArthur Ladder rung represents a meaningful hedonic gain of approximately 2.3 WHO-5 points. This robust association strengthens the case for using the MacArthur Ladder to measure perceived socioeconomic position in Asian contexts. Consistent with Tan et al. (2020), this study provides evidence that an individual's cognitively integrative judgment of their relative social rank outperforms objective household income in predicting psychological well-being.

Passive Upward Exposure and Status Moderation (H3 & H4)

Passive social media exposure emerged as the strongest single predictor of well-being loss in the dataset. Moving from minimal to near-constant passive exposure is associated with an unconditional reduction of approximately 38 WHO-5 points, representing a severe shift from functional well-being into clinical impairment (H4 supported). Crucially, this exposure significantly moderates the SubStat–well-being relationship (H3 supported; $\beta_4 = -2.678$, $p < 0.001$). The protective well-being premium of higher social status is progressively eroded as passive exposure increases, effectively cancelling the utility advantage of perceived social rank at the highest exposure levels. This finding powerfully extends Festinger's (1954) social comparison theory into the digital media domain. When individuals are repeatedly exposed to influencer wealth displays and peer success, the subjective value of their own social rank is systematically diminished, corroborating the digital relative deprivation pathways identified by Kross et al. (2021) and Liu and Zhao (2025). Furthermore, the boundary findings of this interaction are highly interpretable through Prospect Theory (Kahneman & Tversky, 1979). At the absolute floor of the social hierarchy (SubStat = 1), passive exposure was marginally positively associated with well-being (+5.95, $p = 0.034$). Because individuals at the bottom lack a proximate upward comparison anchor, aspirational content functions as aspirational utility rather than a psychological loss. Conversely, for high-status individuals (SubStat 6), this same content triggers loss aversion by narrowing the perceived gap between their current status and a highly salient, idealized reference point. This framing elegantly unifies the moderation effects under a single theoretical mechanism.

Supplementary Theoretical Implications

Understanding the severity of these cognitive and digital stressors, a supplementary finding revealed that occupational sector adds no statistically significant explanatory power to the baseline model (AIC = +5.0, all industry dummies $p > 0.176$). With nearly half the sample (49.4%) scoring at or below the WHO-5 clinical screening threshold, the data signals a substantial burden of impaired well-being among Bangkok's working-age population. The fact that occupation fails to explain this variance confirms that the well-being deficits identified in

this study are driven by individual-level psychological phenomena, namely digital comparison and financial scarcity, that transcend structural sector boundaries.

5.2 Policy and Practical Implications

The findings have direct relevance to Bangkok municipal well-being policy and workplace health programmes. Consistent with Governor Chadchart Sittipunt's livability agenda, which includes digital well-being campaigns, community driven activities, and mental health outreach, the evidence presented here provides an empirical basis for targeting resources. These resources can be directed to where they will have the greatest marginal impact.

The central insight for policy design is the social comparison indifference point: the level of subjective social rank at which the marginal effect of passive upward social media exposure switches from positive to negative ($\text{SubStat} \approx 3.22$ on the MacArthur Ladder). Policymakers and public organisations should anchor interventions around this threshold, deploying differentiated strategies for the two groups it divides.

5.2.1 Differentiated Strategy by Status Group

Table 19 operationalises the two-group policy framework illustrated in Table 18 above.

Target Group	We should...	We should NOT...
Aspirational Seekers <i>(Low status, SubStat ≤ 5)</i>	Promote offline activities and community peer networks to provide positive social anchors outside digital comparison environments.	Digital detox / shaming approaches. Blanket bans social media use, which may increase exclusion and stigma without addressing the scarcity and low-status mechanisms.

	Deliver financial literacy programmes to address root-cause scarcity perceptions.	
Loss-Aversion Individuals (<i>High status</i> , <i>SubStat</i> ≥ 6)	Deploy workplace wellness programmes and cognitive-behavioural therapy (CBT) resources targeting the status-threat response to aspirational content. Introduce algorithm literacy training to help high-status individuals consciously manage their passive content diet.	Ignore them. High-status individuals face the steepest interaction-driven well-being loss at elevated passive exposure; treating them as a low-priority group misses this concentrated harm.

Table 19. Differentiated intervention framework for aspirational seekers (low status) and loss-aversion individuals (high status).

Aspirational seekers ($\text{SubStat} \leq 5$) are the group most at risk from the compounding effects of high psychological scarcity, low SubStat, and heavy passive exposure. For this group, the appropriate response is not digital detox or shaming. Such approaches risk reinforcing social exclusion and misidentifying social media use as the root cause. Instead, the focus should be on addressing the underlying scarcity and status mechanisms through offline community

building, peer networks, and financial literacy interventions that provide positive social anchors outside the digital comparison environment.

Loss-aversion individuals ($\text{SubStat} \geq 6$) face the steepest interaction-driven well-being loss at high passive exposure levels ($\text{ME}(\text{PasExp}) = -9.15$ for corporate or professional workers at their mean SubStat of 6.64; Table 16). The mechanism here is loss of aversion in the domain of social rank: high-status individuals experience aspirational content as a threat to a carefully maintained self-concept. The appropriate intervention targets this response directly through wellness programmes, cognitive-behavioural approaches to status-threat appraisal, and algorithm literacy that empowers individuals to consciously manage their passive content diet.

5.2.2 Actionable Policy Recommendations

Building on the differentiated framework, three actionable implementation priorities emerge from the data. Table 20 presents these recommendations alongside their empirical grounding and proposed implementation measures.

Recommendation	Target Mechanism	Actionable Measure
Expand Income-Based Relief	Reduces PsySca independently of PasExp and SubStat; raises perceived economic standing, thereby shifting respondents up the MacArthur Ladder.	Expand urban relief including welfare cards, household subsidies, and debt restructuring schemes based on household income brackets. Target HHInc brackets 1–2 (below 20,000 THB/month) who report both the highest PsySca and the lowest SubStat in this sample.

Deploy Digital Literacy Programmes	Directly reduces PasExp by giving residents the skills to consciously manage their passive upward social media consumption.	BMA should use local health centres to run digital well-being campaigns. Curriculum should cover social media algorithm mechanics, passive vs active use distinctions, and comparison-management strategies. The programme can be delivered via LINE OA channels to reach workers in the Commerce, Hospitality, and Labour sectors with limited clinic access.
Integrate Financial Counselling in Clinics	Targets the PsySca–SubStat compounding effect: addressing financial anxiety in clinical settings reduces the bandwidth tax and may raise subjective status perception.	Bangkok clinics should partner with financial advisors to integrate debt counselling and anxiety management for patients presenting financial-related mental health concerns. The priority population comprises low-income groups (HHInc 1–3) in the labour, agriculture, and informal sectors. In these sectors, the mean WHO-5 score is 46.9. This is below the clinical threshold.

Table 20. Policy recommendation framework grounded in M1 and M3 empirical findings.

Expand Income-Based Relief: The finding that psychological scarcity is independently significant even after controlling HHInc confirms that financial anxiety operates through a perception channel rather than purely through income. However, reducing objective financial precarity through expanded welfare cards, subsidies, and debt restructuring schemes directly

addresses the material conditions that sustain high psychological scarcity in lower income brackets. This is particularly relevant for the labour, agriculture, and informal sectors. These sectors have the highest mean psychological scarcity (1.60) and the lowest mean WHO-5 score (46.9) among all industry sectors in the sample.

Deploy Digital Literacy Programmes: Given that passive exposure is the strongest predictor of well-being loss ($MEM = -8.08$ at mean SubStat), reducing passive exposure is the highest-leverage single intervention available to policymakers. Digital well-being programmes delivered through Bangkok's network of local health centres and via LINE Official Account channels can reach workers across income and sector boundaries without requiring physical attendance. Curriculum should prioritise algorithm literacy. This includes explaining how platforms curate aspirational content and providing practical strategies to help users shift from passive scrolling to active, goal-directed social media use.

Integrate Financial Counselling in Clinics: The compounding interaction between PsySca and the $PasExp \times SubStat$ mechanism means that individuals experiencing financial anxiety are doubly vulnerable: their scarcity-induced bandwidth reduction is amplified by the status erosion from passive exposure. Bangkok clinics partnering with financial advisors to offer embedded debt counselling and financial coaching would address this pathway at the point where it most frequently presents clinically. This typically occurs as anxiety and depression in low-income patients, rather than treating psychological symptoms while leaving financial root causes unaddressed.

Finally, given that 49.4% of the sample falls below the WHO-5 clinical threshold, the evidence supports expanding routine well-being screening in occupational health settings using the WHO-5 instrument, which is validated in Thai populations (Saipanish, 2009, $\alpha = 0.87$) and requires under two minutes to administer. Early identification of at-risk workers enables targeted referral to the counselling and community well-being resources that the livability campaigns are actively expanding.

5.3 Limitations and Future Directions

First, Cross-sectional design. The study is observational and cross-sectional, precluding causal inference despite the theoretical coherence of the hypothesised pathways. Endogeneity is a recognised concern: depression may cause individuals to perceive their social status as lower (bidirectional SNS-depression relationship documented in Samra, 2022; Ostic, 2021), and lower well-being may itself increase passive scrolling as a coping behaviour. Future research should employ longitudinal panel designs or diary-based experience-sampling methods to establish temporal precedence.

Second, Common method bias. All variables are self-reported in a single instrument. The strong factor structure and discriminant validity across constructs provide some reassurance that common method of inflation does not drive the key results. Future studies should incorporate split-source designs and marker variables to address this concern formally.

Third, Sampling constraints. The convenience and online snowball sampling strategy likely overrepresents digitally active, urban, and educated segments of the Bangkok workforce, limiting generalisation to older workers, rural-urban migrants, and low-digital-literacy populations. The exclusion of respondents below 18 and above 44 years further narrows scope. A stratified probability sample covering the full Bangkok adult labour force would substantially strengthen external validity.

Fourth, Ordinal scale limitations. The interpersonal comparability of Likert-scale responses raises measurement concerns. Although all instruments have undergone cross population psychometric validation and CFA testing, equal intervals cannot be assumed. Future research should replicate key findings using ordered regression models as a robustness check and should test whether WHO-5 boundary effects at 0 and 100 are better captured by a fractional logit specification.

Fifth, passive exposure constructs granularity. The passive exposure scale captures exposure to three content categories but does not measure the emotional valence of content encountered, the degree of self-relevance, or platform-specific effects (e.g., Instagram vs

TikTok vs Facebook), all of which are theoretically relevant moderators of the social comparison process. Future studies should incorporate content level and platform-level granularity to more precisely specify the pathway between passive exposure and well-being.

5.4 Conclusion

According to Gallup (2024), employee well-being drives up to 9% of global GDP in productivity gains, while poor well-being costs the world economy approximately \$438 billion annually in lost productivity. This study examined whether the subjective well-being of Bangkok's working-age adults is systematically shaped by psychological scarcity, perceived social rank, and passive upward social media exposure. The answer shows that all three predictors are independently and jointly significant. Their combined explanatory power ($R^2 = 0.405$) is substantial for a cross-sectional survey of this kind.

The central empirical contribution is the confirmation of a moderated association in which passive upward social media exposure does not harm all users equally. Rather, it functions as an inequality-amplifying mechanism: the well-being premium that higher perceived social rank normally provides is progressively cancelled as passive exposure to aspirational content increases. Among respondents in this sample, 63.3% self-placed in the upper half of the social ladder ($\text{SubStat} \geq 6$), for whom moderate passive exposure carries a manageable well-being cost. For the 37% who are self-placed in the lower half, many of whom work in Commerce, Hospitality, and informal sectors, high passive exposure compounds and an already elevated vulnerability to impaired well-being.

These findings operationalize the core research question of this study: whether digital visibility of wealth acts as a conditional amplifier of well-being inequality. The data supports this proposition. At high passive exposure levels, the utility cost of low social rank, already embedded in the PsySca and SubStat coefficients, is amplified through continuous upward comparison, consistent with Festinger (1954), Kross et al. (2021), and the scarcity-bandwidth framework of Mullainathan and Shafir (2013). The social comparison indifference point identified in this study, the level of perceived social rank at which the effect of passive exposure

switches from neutral to harmful, provides a precise and actionable lever for policymakers: interventions that raise perceived social standing among low-status individuals or that reduce passive exposure among high-status individuals will, according to these estimates, yield the greatest marginal improvement in subjective well-being per unit of resource invested.

Based on the findings of this study, enhancing the quality of life of Bangkok workers through policy and social campaigns must therefore prioritise employee financial stress, perception of social rank, and the management of passive upward social media consumption. This is essential to maximize individual utility and contribute to Bangkok becoming a more livable, productive, and psychologically resilient city.

References

- Adler, N. E., Epel, E. S., Castellazzo, G., & Ickovics, J. R. (2000). Relationship of subjective and objective social status with psychological and physiological functioning. *Health Psychology, 19*(6), 586–592.
- Aiken, L. S., West, S. G., & Reno, R. R. (1991). *Multiple regression: testing and interpreting interactions*. Sage Publications.
- Azayem, A. K., Nawaz, F. A., Jeyaseelan, L., Kair, H. M., & Sultan, M. A. (2024). Beyond the filter: Impact of popularity on the mental health of social media influencers. *Digital Health, 10*. <https://doi.org/10.1177/20552076241287843>
- Barua, P., & Tejjativaddhana, P. (2019). Impact of application of Sufficiency Economy Philosophy on the well-being of Thai population: A systematic review and meta-analysis of relevant studies. *Journal of Population and Social Studies, 27*(3), 195–219. <https://doi.org/10.25133/JPSSv27n3.0013>
- Boer, M., van den Eijnden, R. J. J. M., Boniel-Nissim, M., Wong, S.-L., Inchley, J. C., Badura, P., Craig, W. M., Gobina, I., Kleszczewska, D., Klanšček, H. J., & Stevens, G. W. J. M. (2020). Adolescents' intense and problematic social media use and their well-being in 29 countries. *Journal of Adolescent Health, 66*(6, Suppl.), S89–S99. <https://doi.org/10.1016/j.jadohealth.2020.02.014>
- Cai, H., Liu, H., & Gao, Y. (2025). New evidence on the relationship between income and subjective well-being: The mediating and moderating roles of psychological security. *BMC Public Health, 25*(1), Article 1148. <https://doi.org/10.1186/s12889-025-22286-2>
- Cochran, W. G. (1977). *Sampling techniques* (3rd ed.). John Wiley & Sons.
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum.
- Consumer Financial Protection Bureau. (2017). CFPB financial well-being scale: Scale development technical report. https://files.consumerfinance.gov/f/documents/201705_cfpb_financial-well-being-scale-technical-report.pdf

- Dalgard, O. S., Dowrick, C., Lehtinen, V., Vazquez-Barquero, J. L., Casey, P., Wilkinson, G., Ayuso-Mateos, J. L., Page, H., & Dunn, G. (2006). Negative life events, social support and gender difference in depression. *Social Psychiatry and Psychiatric Epidemiology*, 41(6), 444–451.
- Simon, K. (2025, March 3). *Digital 2025: Thailand*. DataReportal – Global Digital Insights. <https://datareportal.com/reports/digital-2025-thailand>
- Easterlin, R. A. (1974). Does economic growth improve the human lot? Some empirical evidence. In P. A. David & M. W. Reder (Eds.), *Nations and households in economic growth: Essays in honor of Moses Abramovitz* (pp. 89–125). Academic Press. <https://doi.org/10.1016/B978-0-12-205050-3.50008-7>
- Esmail, H. (2025). Rethinking happiness: The economics of well-being in the digital age. *Scientific Journal for Research and Business Studies*, 39(3), 1499–1537. https://www.researchgate.net/publication/395344963_Rethinking_Happiness_The_Economics_of_Well-being_in_the_Digital_Age
- Euteneuer, F. (2014). Subjective social status and health. *Current Opinion in Psychiatry*, 27(5), 337–343. <https://doi.org/10.1097/YCO.0000000000000083>
- Festinger, L. (1954). A theory of social comparison processes. *Human Relations*, 7(2), 117–140. <https://doi.org/10.1177/001872675400700202>
- Frey, B. S., & Stutzer, A. (2002). What can economists learn from happiness research? *Journal of Economic Literature*, 40(2), 402–435. <https://doi.org/10.1257/002205102320161320>
- Fung, S.-f., Kong, C. Y. W., Liu, Y.-m., Huang, Q., Xiong, Z., Jiang, Z., Zhu, F., Chen, Z., Sun, K., Zhao, H., & Yu, P. (2022). Validity and psychometric evaluation of the Chinese version of the 5-item WHO Well-Being Index. *Frontiers in Public Health*, 10, Article 872436. <https://doi.org/10.3389/fpubh.2022.872436>
- Gross-Manos, D., & Bradshaw, J. (2022). The association between the material well-being and the subjective well-being of children in 35 countries. *Child Indicators Research*, 15(1), 1–33. <https://doi.org/10.1007/s12187-021-09860-x>

- Guan, N., Guariglia, A., Moore, P., Xu, F., & Al-Janabi, H. (2022). Financial stress and depression in adults: A systematic review. *PLOS ONE*, 17(2), e0264041.
<https://doi.org/10.1371/journal.pone.0264041>
- Hoebel, J., Maske, U. E., Zeeb, H., & Lampert, T. (2017). Social inequalities and depressive symptoms in adults: The role of objective and subjective socioeconomic status. *PLOS ONE*, 12(1), Article e0169764. <https://doi.org/10.1371/journal.pone.0169764>
- Howell, R. T., & Howell, C. J. (2008). The relation of economic status to subjective well-being in developing countries: A meta-analysis. *Psychological Bulletin*, 134(4), 536–560. <https://doi.org/10.1037/0033-2909.134.4.536>
- Huang, C. (2017). Time spent on social network sites and psychological well-being: A meta-analysis. *Cyberpsychology, Behavior, and Social Networking*, 20(6), 346–354.
<https://doi.org/10.1089/cyber.2016.0758>
- Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263–291. <https://doi.org/10.2307/1914185>
- Kahneman, D., & Deaton, A. (2010). High income improves evaluation of life but not emotional well-being. *Proceedings of the National Academy of Sciences*, 107(38), 16489–16493. <https://doi.org/10.1073/pnas.1011492107>
- Kliem, S., Lohmann, A., Fischer, S., Baier, D., Clemens, V., Sachser, C., Kampling, H., Brähler, E., & Fegert, J. M. (2025). Psychometric evaluation and updated community norms of the WHO-5 well-being index, based on a representative German sample. *Frontiers in Psychology*, 16, Article 1592614.
<https://doi.org/10.3389/fpsyg.2025.1592614>
- Knies, G. (2022). Effects of income and material deprivation on children's life satisfaction: Evidence from longitudinal data for England (2009–2018). *Journal of Happiness Studies*, 23(4), 1469–1492. <https://doi.org/10.1007/s10902-021-00457-3>
- Kraft, P., & Kraft, B. (2023). Exploring the relationship between multiple dimensions of subjective socioeconomic status and self-reported physical and mental health: The mediating role of affect. *Frontiers in Public Health*, 11, 1138367.
<https://doi.org/10.3389/fpubh.2023.1138367>

- Kross, E., Verduyn, P., Sheppes, G., Costello, C. K., Jonides, J., & Ybarra, O. (2021). Social media and well-being: Pitfalls, progress, and next steps. *Trends in Cognitive Sciences*, 25(1), 55–66. <https://doi.org/10.1016/j.tics.2020.10.005>
- Li, M., Chen, X., Wang, Z., & Zhou, R. (2025). Sustainable development in the digital economy: How platform exploitation perception influences digital workers' well-being via job rewards and job security. *Sustainability*, 17(5), Article 1920. <https://doi.org/10.3390/su17051920>
- Liu, Y., & Zhao, X. (2025). Social media use and personal relative deprivation among urban residents in China: A moderated mediation model. *Behavioral Sciences*, 15(7), 962. <https://doi.org/10.3390/bs15070962>
- Mullainathan, S., & Shafir, E. (2013). Scarcity: Why having too little means so much. Times Books.
- Navarro-Carrillo, G., Alonso-Ferres, M., Moya, M., & Valor-Segura, I. (2020). Socioeconomic status and psychological well-being: Revisiting the role of subjective socioeconomic status. *Frontiers in Psychology*, 11, Article 1303. <https://doi.org/10.3389/fpsyg.2020.01303>
- Ostic, D., Qalati, S. A., Barbosa, B., Shah, S. M. M., Galvan Vela, E., Herzallah, A. M., & Liu, F. (2021). Effects of social media use on psychological well-being: A mediated model. *Frontiers in Psychology*, 12, Article 678766.
- Riehm, K. E., Feder, K. A., Tormohlen, K. N., et al. (2019). Associations Between Time Spent Using Social Media and Internalizing and Externalizing Problems Among US Youth. *JAMA Psychiatry*, 76(12), 1266–1273. <https://pubmed.ncbi.nlm.nih.gov/31509167/>
- Saipanish, R., Lotrakul, M., & Sumrithe, S. (2009). Reliability and validity of the Thai version of the WHO-Five Well-Being Index in primary care patients. *Psychiatry and Clinical Neurosciences*, 63(2), 141–146.
- Samra, A., Warburton, W. A., & Collins, A. M. (2022). Social comparisons: A potential mechanism linking problematic social media use with depression. *Journal of Behavioral Addictions*, 11(2), 607–614. <https://doi.org/10.1556/2006.2022.00023>

- Sekulova, F., Bonilla, F., & Laín, B. (2023). Life satisfaction and socio-economic vulnerability: Evidence from the basic income experiment in Barcelona. *Applied Research in Quality of Life*, 18(4), 2035–2063. <https://doi.org/10.1007/s11482-023-10176-x>
- Simonse, O., Van Dijk, W. W., Van Dillen, L. F., & Van Dijk, E. (2022). The role of financial stress in mental health changes during COVID-19. *npj Mental Health Research*, 1, 15. <https://doi.org/10.1038/s44184-022-00016-5>
- Singh-Manoux, A., Marmot, M. G., & Adler, N. E. (2005). Does subjective social status predict health and change in health status better than objective status? *Psychosomatic Medicine*, 67(6), 855–861.
- Skogen, J. C., Bøe, T., Finserås, T. R., Sivertsen, B., Hella, R. T., & Hjetland, G. J. (2022). Lower subjective socioeconomic status is associated with increased risk of reporting negative experiences on social media. Findings from the “LifeOnSoMe”-study. *Frontiers in Public Health*, 10, 873463. <https://doi.org/10.3389/fpubh.2022.873463>
- Tan, J. J. X., Carpenter, N. C., Kraus, M. W., & Adler, N. E. (2020). The association between objective and subjective socioeconomic status and subjective well-being: A meta-analytic review. *Psychological Bulletin*, 146(11), 970–1020. <https://doi.org/10.1037/bul0000258>
- Twenge, J. M., Haidt, J., Lozano, J., & Cummins, K. M. (2022). Specification curve analysis shows that social media use is linked to poor mental health, especially among girls. *Acta Psychologica*, 224, Article 103512. <https://doi.org/10.1016/j.actpsy.2022.103512>
- Verduyn, P., Lee, D. S., Park, J., Shablack, H., Orvell, A., Bayer, J., Ybarra, O., Jonides, J., & Kross, E. (2015). Passive Facebook usage undermines affective well-being: Experimental and longitudinal evidence. *Journal of Experimental Psychology: General*, 144(2), 480–488. <https://doi.org/10.1037/xge0000057>
- World Economic Forum. (2022, January 18). *Edelman Trust Barometer 2022: The cycle of distrust*. Retrieved from https://www.weforum.org/stories/2022/01/edelman-trust-barometer-2022-report/?gad_source=1&gad_campaignid=22228224717&gbraid=0AAAAAoVy5F7NXDTkGroywPsq1I_DoIZLR&gclid=CjwKCAiA-

sXMBhAOEiwAGGw6LLLSF36PiqNcSrT4zv8VVzm4Q4oikophFfH48tuRWHq8Ve
qc0QP9NRoC08EQAvD_BwE

- White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, 48(4), 817–838.
- Zhang, C., Tang, L., & Liu, Z. (2023). How social media usage affects psychological and subjective well-being: Testing a moderated mediation model. *BMC Psychology*, 11, Article 286. <https://doi.org/10.1186/s40359-023-01311-2>
- Zsila, Á., & Reyes, M. E. S. (2023). Pros & cons: Impacts of social media on mental health. *BMC Psychology*, 11, 201. <https://doi.org/10.1186/s40359-023-01243-x>

Appendix

Appendix A: Survey Instrument (Google Form Questionnaire – Full Items)

Section 1: คำกล่าวนำ / Introduction

การบริโภคสื่อดิจิทัลและสุขภาวะทางจิตใจในประเทศไทย

งานวิจัยนี้ดำเนินการโดยนักศึกษาระดับปริญญาตรีจากคณะเศรษฐศาสตร์ มหาวิทยาลัยเกษตรศาสตร์

ซึ่งเป็นส่วนหนึ่งของรายวิชาวิจัยทางเศรษฐศาสตร์

วัตถุประสงค์หลักของการวิจัยนี้คือการศึกษาความสัมพันธ์ระหว่างการรับรู้ทางการเงิน สถานะทางสังคม และการบริโภคดิจิทัลแบบไม่เชิงรุกที่มีต่อความสุขทางใจของบุคคลสัญชาติไทยที่มีอายุ 18 ปีขึ้นไป และมีการใช้งานสื่อสังคมออนไลน์

การเข้าร่วมตอบแบบสอบถามนี้เป็นไปโดยสมัครใจและใช้เวลาประมาณ 3-5 นาทีในการตอบ

โปรดมั่นใจได้ว่าข้อมูลทั้งหมดจะถูกเก็บเป็นความลับและใช้เพื่อการวิเคราะห์ทางวิชาการเท่านั้น

ขอขอบคุณสำหรับการมีส่วนร่วมในการวิจัยของเรา

*ระบุว่าเป็นคำถามที่จำเป็น

Digital Media Consumption and Mental Well-Being in Thailand

This research is conducted by undergraduate students from the Faculty of Economics, Kasetsart University, as part of the course Research Methods in Economics.

The main objective of this study is to examine the relationship between financial perception, social status, and passive digital consumption, and their effects on the mental well-being of Thai individuals aged 18 and above who use social media.

Participation in this survey is voluntary and will take approximately 3–5 minutes. Please be assured that all information will be kept confidential and used solely for academic analysis. Thank you very much for your valuable contribution to our research.

Questions marked with an asterisk (*) are required.

ภาคส่วนการทำงานของท่านคืออะไร?* (What is your occupation? – required)

- การเกษตรกรรม ป่าไม้ และประมง / Agriculture, forestry, and fisheries
- อุตสาหกรรมการผลิตและโรงงาน / Manufacturing industry and factories
- การค้าส่ง ค้าปลีก และการซ่อมแซม / Wholesale, retail, and repair
- การขนส่งและคลังสินค้า / Transportation and warehousing
- ที่พักแรมและบริการด้านอาหาร / Accommodation and food services
- การสื่อสารและเทคโนโลยีสารสนเทศ / Communication and information technology
- การเงินและประกันภัย / Finance and Insurance
- การศึกษา / Education
- สุขภาพและกิจกรรมทางสังคม / Health and Social Activities
- การบริหารราชการและรัฐวิสาหกิจ / Public Administration and State Enterprise
- นักเรียน/นักศึกษา / Student
- อื่นๆ / Others

Section 2: ข้อมูลทั่วไป / General Information

ส่วนนี้จะรวบรวมข้อมูลด้านประชากรศาสตร์และประเมินสถานะที่ผู้ตอบแบบสอบถามรับรู้ด้วยตนเองภายในลำดับชั้นทางสังคม

In this section, you will be asked to provide some demographic information and to indicate how you perceive your own position within the social hierarchy.

เพศ* (Gender – required)

☐ ชาย

☐ หญิง

☐ อื่น ๆ

☐ ไม่ต้องการระบุ

ปัจจุบันคุณอาศัยอยู่ในลักษณะใด?* (What type of life do you currently live in? – required)

☐ อยู่กับครอบครัว / ผู้ปกครอง

☐ อยู่คนเดียว

☐ อยู่กับเพื่อน / คู่รัก / แร่ที่พัก

☐ อื่น ๆ (โปรดระบุ) _____

รายได้ของครอบครัวต่อเดือน* (Monthly household income – required)

☐ น้อยกว่า 10,000 บาท

☐ 10,000 – 20,000 บาท

☐ 20,001 – 30,000 บาท

☐ 30,001 – 50,000 บาท

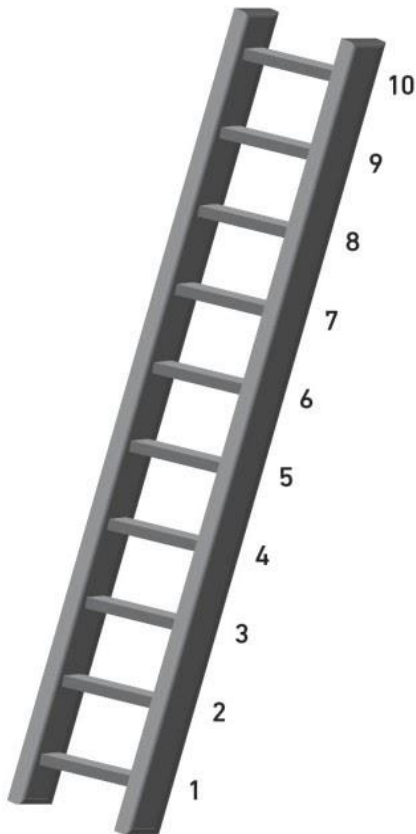
☐ 50,001 – 80,000 บาท

☐ 80,001 – 150,000 บาท

☐ มากกว่า 150,000 บาท

การรับรู้สถานะทางสังคม / Subjective Social Status (MacArthur Ladder)

ลองนึกภาพบันไดนี้แทนระดับฐานะทางสังคมของแต่ละบุคคล ได้แก่ การศึกษา การเงิน และมีงานที่ได้รับการเคารพนับถือ
ขั้นบนสุดคือดีที่สุด ขั้นล่างสุดคือแย่ที่สุด คุณคิดว่าตัวเองอยู่ตรงไหนบนบันไดนี้? (1–10) (Imagine a ladder representing
social status, including education, financial situation, and having a respected job. The top of
the ladder represents the best possible position, and the bottom represents the worst. Where
would you place yourself on this ladder? (1-10) – required)



☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

Section 3: พฤติกรรมดิจิทัล / Digital Behavior

ส่วนนี้จะประเมินความถี่ของการได้รับข้อมูลผ่านสื่อสังคมออนไลน์โดยไม่ตั้งใจ และความพร้อมใช้ข้อมูลทางสังคมแบบออฟไลน์

This section will assess the frequency of unintentionally receiving information through social media, and the availability of offline social information sources.

การเปิดรับสื่อสังคมออนไลน์แบบไม่เชิงรุก / Passive Upward Social Media Exposure (PasExp) – adapted items

โดยเฉลี่ยแล้ว คุณใช้เวลาบนแพลตฟอร์มโซเชียลมีเดีย (Facebook, Instagram, TikTok, X ฯลฯ) กี่ชั่วโมงต่อวัน?* (On average, how many hours per day do you spend on social media platforms (Facebook, Instagram, TikTok, X, etc.)) – required?

- น้อยกว่า 1 ชั่วโมง
- 1 – 2 ชั่วโมง
- 2 – 4 ชั่วโมง
- 4 – 6 ชั่วโมง
- มากกว่า 6 ชั่วโมง

คุณทำสิ่งต่อไปนี้บนโซเชียลมีเดียบ่อยแค่ไหน?* (How often do you engage in the following activities on social media? – required)

(5-point Likert: 0 = ไม่เคย / Never – 4 = บ่อยมาก / Very often)

1. ดูฟีดของเหล่าอินฟลูเอนเซอร์หรือคนที่ดูเหมือนจะร่ำรวยหรือประสบความสำเร็จมากกว่าคุณ / Viewing feeds from influencers or people who appear wealthier or more successful than you.
2. ชมภาพถ่ายหรือวิดีโอเกี่ยวกับการท่องเที่ยวสุดหรู สินค้าราคาแพง หรือวิถีชีวิตในอุดมคติ โดยไม่มีปฏิสัมพันธ์ (เช่น การกดไลค์/แสดงความคิดเห็น) / View photos or videos of luxury travel, expensive products, or ideal lifestyles without interacting (e.g., liking/commenting).
3. เลื่อนดูข้อความอัปเดตจากเพื่อนหรือคนรู้จักที่ดูเหมือนจะประสบความสำเร็จในชีวิตมากกว่าคุณ / Scroll through updates from friends or acquaintances who seem to be more successful in life than you.

แบบประเมินการสนับสนุนทางสังคมออสโล (OSSS-3) / Oslo Social Support Scale (OSSS-3)

มีคนสนิทที่คนที่คุณสามารถพึ่งพาได้หากคุณมีปัญหาใหญ่ในชีวิตส่วนตัว?* (How many close people do you have whom you can rely on if you face a major problem in your personal life? – required)

- ไม่มี / None
- 1 – 2 คน / 1 – 2 people
- 3 – 5 คน / 3 – 5 people
- มากกว่า 5 คน / More than 5 people

ผู้อื่นให้ความสนใจและห่วงใยในสิ่งที่你做เล็กน้อยแค่ไหน?* (How much attention and care do people give to what you do? – required)

- ไม่มีเลย / None
- น้อย / Low
- ไม่แน่ใจ / Uncertain
- มาก / High
- มากที่สุด / Highest

หากต้องการความช่วยเหลือจากเพื่อนบ้าน คุณจะได้รับความช่วยเหลือง่ายแค่ไหน?* (If you needed help from your neighbors, how easy would it be for you to receive that help? – required)

- ยากมาก / Very difficult
- ยาก / Difficult
- เป็นไปได้ / Possible
- ง่าย / Easy

○ ง่ายมาก / Very easy

Section 4: ตัวชี้วัดความเป็นอยู่ที่ดี / Well-being Indicators

ส่วนนี้จะวัดความมั่นคงทางการเงินเชิงจิตวิทยา (แบบประเมิน CFPB) และสถานะสุขภาพจิตในปัจจุบัน (ดัชนี WHO-5)

This section will measure psychological financial well-being (CFPB Financial Well-Being Scale) and current mental health status (WHO-5 Well-Being Index).

การรับรู้สถานะทางการเงิน / Psychological Scarcity (CFPB Financial Well-Being Scale – adapted)

ถ้อยคำนี้ตรงกับตัวคุณหรือสถานการณ์ของคุณมากน้อยแค่ไหน?* (To what extent does this statement reflect you or your situation? – required)

(4-point Likert: 0 = ไม่เลย – 3 = ตลอดเวลา / Absolutely)

1. ตอนนี้ฐานะการเงินของฉันค่อนข้างลำบาก / Right now, my financial situation is somewhat difficult.
2. จากสถานการณ์ทางการเงินของฉัน ฉันรู้สึกว่าจะไม่มีวันได้สิ่งต่างๆ ที่ฉันต้องการในชีวิต / Given my financial situation, I feel like I'll never get the things I want in life.
3. ฉันกังวลว่าเงินที่ฉันมีอยู่หรือที่จะเก็บออมในอนาคตจะไม่เพียงพอ / I'm worried that the money I have now, or the money I'll save in the future, might not be enough.
4. ฉันสามารถรับมือกับค่าใช้จ่ายทางการเงินก้อนใหญ่ที่ไม่คาดคิดได้ / I am able to handle a major unexpected financial expense.

ถ้อยคำนี้ตรงกับชีวิตคุณบ่อยแค่ไหน?* (How often does this statement reflect your life? – required)

(2-point Likert: 0 = ไม่เคย / Never – 1 = เสมอ / Always)

1. ฉันมีเงินเหลือตอนสิ้นเดือน / I have money left at the end of the month.
2. ข้อจำกัดทางการเงินส่งผลต่อชีวิตของฉัน / Financial constraints affect my life.

สุขภาวะทางจิตใจ / Subjective Well-Being (WHO-5)

ในช่วงสองสัปดาห์ที่ผ่านมา คุณรู้สึกแบบนี้บ่อยแค่ไหน...?* (During the past two weeks, how often have you felt this way...? – required)

(6-point Likert: 0 = ไม่เคยเลย / At no time – 5 = ตลอดเวลา / All of the time)

1. ฉันรู้สึกร่าเริงและอารมณ์ดี / I have felt cheerful and in good spirits.
2. ฉันรู้สึกสงบและผ่อนคลาย / I have felt calm and relaxed.
3. ฉันรู้สึกกระฉับกระเฉงและมีพลัง / I have felt active and vigorous.
4. ฉันตื่นขึ้นมาด้วยความรู้สึกสดชื่นและพักผ่อนเต็มที่ / I woke up feeling fresh and rested.
5. ชีวิตประจำวันของฉันเต็มไปด้วยสิ่งที่น่าสนใจ / My daily life has been filled with things that interest me.

Appendix B: Measurement and Scoring of Key Variables

This appendix provides a consolidated reference for the operationalisation, scoring, and psychometric properties of each construct used in the study. All reliability and validity statistics reported below are derived from the analytic sample of 411 respondents unless otherwise stated.

Variable	Instrument / Source	No. of Items	Response Scale	Scoring Procedure	Range	Psychometric Properties
----------	------------------------	-----------------	-------------------	----------------------	-------	----------------------------

Subjective Well-Being (WHO-5)	WHO-5 Well-Being Index (Topp et al., 2015; Thai validation: Saipanish, 2009)	5	0 (at no time) – 5 (all the time)	Sum of five items multiplied by four: Raw sum $(0-25) \times 4$	0–100	Cronbach's $\alpha = 0.957$ (this study); $\alpha = 0.87$ (Saipanish, 2009, Thai sample). CFA: single-factor, standardised loadings 0.893–0.922, SRMR = 0.011. Score ≤ 50 indicates impaired well-being (WHO clinical screening threshold).
-------------------------------	--	---	-----------------------------------	--	-------	---

Passive Social Media Exposure (PasEx)	Adapted passive exposure items (Verduyn et al., 2015; Kross et al., 2021)	3	1 (never) – 5 (very often)	Mean of three frequency items: $(\text{Item}_1 + \text{Item}_2 + \text{Item}_3) / 3$	1–5	Cronbach's $\alpha = 0.898$ (this study). CFA: single-factor, standardised loadings 0.856–0.897, CD = 0.915. Higher scores indicate more frequent passive exposure to upward comparison content.
---------------------------------------	---	---	----------------------------	--	-----	--

Subjective Social Status (SubStat)	MacArthur Scale of Subjective Social Status (Adler et al., 2000)	1	1 (lowest) – 10 (highest)	Direct single-item score	1–10	Single-item measure; internal consistency not applicable. Construct validity established across multiple cross-cultural studies (Tan et al., 2020; Navarro-Carrillo et al., 2020). Higher scores indicate higher perceived social standing relative to Thai society.
------------------------------------	--	---	---------------------------	--------------------------	------	--

Psychological Scarcity (PsySca)	CFPB Financial Well-Being Scale — worry subscale (Consumer Financial Protection Bureau, 2017)	3 (worry subscale)	0 (not at all) – 4 (all the time)	Mean of three worry items: (cfpb_hard + cfpb_sit + cfpb_wry) / 3	0–4	Cronbach's α = 0.950 (this study). EFA: single factor, eigenvalue = 2.727 (91.0% variance explained), all loadings $\geq$ 0.947, KMO = 0.771. CFA: standardised loadings 0.911–0.952, SRMR = 0.000. Higher scores indicate greater perceived financial constraint and cognitive preoccupation with scarcity.
---------------------------------	---	--------------------	-----------------------------------	--	-----	---

Liquidity Constraint (CFPB_liquid)	CFPB Financial Well- Being Scale — liquidity subscale (Consumer Financial Protection Bureau, 2017)	3 (liquidity subscale)	cfpb_emg: 0–4 (reversed); cfpb_mon, cfpb_con: binary (0– 1)	cfpb_emg normalised to 0–1 (÷4); mean of three normalised items	0–1	Cronbach's α = 0.918 (this study). EFA: single factor, eigenvalue = 2.578 (85.9% variance explained), KMO = 0.756. Note: cfpb_con loaded negatively (–0.932) prior to reversal, reflecting a cognitive- evaluative dimension partially distinct from objective liquidity. Retained as a secondary financial control
--	---	----------------------------------	---	---	-----	---

						variable. Higher scores indicate greater liquidity constraints.
--	--	--	--	--	--	--

Offline Social Support (OSSS-3)	Oslo Social Support Scale (Dalgard et al., 2006)	3	Item 1 (close_ppl): 0–3; Items 2–3 (ppl_care, nbr_help): 1–5	Each item normalised to 0–1 range; mean of three normalised items: $\text{close_ppl} \div 3$; $(\text{ppl_care} - 1) \div 4$; $(\text{nbr_help} - 1) \div 4$	0–1	Cronbach's $\alpha = 0.918$ (this study). CFA: standardised loadings 0.633–0.840, CD = 0.810. The normalised composite (0–1) replaces the original published sum score (3–14) to place the variable on a common scale with other composites. Higher scores indicate greater perceived offline social support.
---------------------------------	--	---	--	---	-----	---

Household Income (HHInc)	Survey item (ordinal self-report)	1	1 (< 10,000 THB) – 7 (> 150,000 THB)	Direct ordinal score	1–7	Single-item ordinal measure. Used as a proxy for objective financial capacity. Entered as a continuous ordinal regressor (mean = 4.00, SD = 1.36).
Daily Social Media Hours (SOC_HRS)	Survey item (ordinal self-report)	1	1 (< 1 hour) – 5 (> 6 hours)	Direct ordinal score	1–5	Single-item ordinal measure. Captures total social media usage duration, distinct from the content-specific passive exposure captured by

						PasEx. Entered as a continuous ordinal regressor (mean = 2.56, SD = 0.96).
--	--	--	--	--	--	--

Table 21. Measurement instruments, scoring procedures, and psychometric properties (n = 411)

Sum Score (3–14)	Support Level Classification
3 – 8	Poor social support
9 – 11	Moderate social support
12 – 14	Strong social support

Table 22. OSSS-3 original published scoring reference

Note: The classification thresholds above follow the original OSSS-3 published norms (Dalgard et al., 2006). In this study, the composite is normalised to a 0–1 scale for regression comparability. Researchers wishing to apply the original clinical classification should compute the raw sum score (3–14) from the untransformed item responses before normalisation.

Notes on Scoring Corrections Relative to Original Proposal

Three substantive corrections apply relative to the scoring descriptions in the original research proposal:

First, PasEx is scored as the mean of three Likert-scale frequency items on a 1–5 scale, not as a composite combining hours of use with frequency items. Social media hours (SOC_HRS) are retained as a separate ordinal control variable to capture total usage intensity independently of passive-comparison content exposure.

Second, PsySca is derived from three items of the CFPB worry subscale on a 0–4 ordinal scale, not from the full six-item CFPB battery. The remaining three CFPB items constitute the separate CFPB_liquid subscale. This two-subscale structure was confirmed by exploratory factor analysis and justified by the bipolar loading pattern of the full six-item battery, in which emergency fund and financial confidence items loaded in the opposite direction to the worry items.

Third, OSSS-3 is scored as a normalised 0–1 composite rather than the published 3–14 sum score, to ensure scale comparability in the regression framework. The original classification thresholds (Table 23) remain valid for descriptive and clinical reference purposes.

Appendix C: Summary of Literature Review Themes

Table 24 below synthesises the key theoretical themes addressed in the literature review, mapping each study to its principal finding and its specific relevance to the conceptual framework of this study. Studies are grouped thematically rather than chronologically. All studies listed have been reviewed in the body of the report unless marked with an asterisk (*), which denotes sources cited in the methodology or results chapters as instrument validation or methodological references rather than as theoretical contributions to the literature review.

Theme	Key Studies	Principal Finding	Relevance to Study Framework

Economic Foundations of Well-Being	Howell & Howell (2008)	Income–well-being relationships are strongest in low-income economies; above basic sufficiency thresholds, psychological factors dominate.	Justifies HHInc as a material sufficiency control and establishes that PsySca and SubStat should predict well-being beyond objective income.
	Kahneman & Deaton (2010)	Income strongly predicts life evaluation, but its effect on emotional well-being plateaus beyond a threshold; lower income amplifies daily emotional distress.	Supports the distinction between objective financial capacity (HHInc) and subjective psychological states (PsySca) as separate model components.
	Frey & Stutzer (2001)	Happiness cannot be explained by economic indicators alone; unemployment, governance, and social conditions exert independent effects.	Provides foundational justification for a multidimensional well-being model that incorporates psychological and social variables alongside income.

	Sekulova et al. (2023)	Financial support improves well-being, but residual scarcity stress predicts life satisfaction more strongly than income level alone.	Validates the inclusion of PsySca as a subjective financial perception variable distinct from and more proximate to well-being than HHInc.
	Simonse et al. (2022)	Changes in subjective financial stress predicted fluctuations in mental health during the COVID-19 pandemic more sensitively than objective income changes.	Confirms that perceived financial constraint, not income shock per se, is the mechanism linking economic conditions to well-being.
	Knies (2022)	Higher family income and lower material deprivation are associated with greater well-being among children and adolescents, with effects increasing with age.	Supports the role of objective sufficiency as a foundational control variable and contextualises its relationship with age in the model.

	Gross-Manos & Bradshaw (2022)	Subjective material deprivation and multidimensional well-being are strongly associated with life satisfaction; family affluence alone is insufficient.	Justifies the inclusion of both HHInc and PsySca, and supports the use of offline social support as a separate control variable.
	Esmail (2025)	Digital access and governance quality predict well-being more accurately than economic growth in developing economies; unexplained micro-level variance remains.	Highlights the need for individual-level psychological predictors such as SubStat and PasEx to account for variance not captured by macro-economic indicators.
Scarcity as a Psychological Mechanism	Mullainathan & Shafir (2013)	Scarcity, particularly financial scarcity, reduces cognitive bandwidth and impairs decision-making independently of objective material conditions.	Provides the theoretical foundation for PsySca as the primary psychological mechanism linking economic conditions to reduced well-being in the model.

	Guan et al. (2022)	Financial stress is a robust predictor of depression, consistently mediating the impact of low income; subjective financial perception is a more stable mental health determinant than objective income.	Directly supports the construction and inclusion of PsySca as a subjective financial constraint measure independent of HHInc.
	Cai et al. (2025)	Income directly boosts life satisfaction, but its effect is significantly mediated and moderated by psychological security; financial gains do not maximise well-being without a foundation of psychological safety.	Supports the theoretical claim that PsySca captures a distinct psychological pathway not reducible to objective income, consistent with the study's modelling approach.
Subjective Social Status and Well-Being	Tan et al. (2020)	Meta-analysis of 357 studies: subjective social status predicts health and well-being more strongly than objective income or	Provides the primary meta-analytic justification for SubStat as the central predictor and for its superiority

		education across diverse populations.	over HHInc in the regression specification.
	Euteneuer (2014)	Low subjective social status consistently predicts poorer self-rated health and elevated biological risk markers; SSS remains significant after controlling objective status.	Establishes the construct validity of the MacArthur Scale in health contexts and justifies SubStat as an independent predictor beyond income and education.
	Hoebel et al. (2017)	Lower subjective social status predicts depressive symptoms independently of income, education, and occupational status in a German adult population.	Confirms that SubStat captures a dimension of perceived inequality that generates mental health consequences distinct from objective socioeconomic position.

	Navarro-Carrillo et al. (2020)	Subjective social status is a robust predictor of all six dimensions of psychological well-being; objective SES shows negligible independent associations when both are modelled simultaneously.	Directly supports the prioritisation of SubStat over HHInc as the principal status predictor and validates the MacArthur Scale in a Western adult context comparable to this study's urban adult sample.
	Kraft & Kraft (2023)	Negative affect significantly mediates the relationship between low subjective social status and both physical and mental health outcomes.	Identifies the affective pathway through which low SubStat damages well-being, supporting the theoretical mechanism linking status perception to WHO-5 scores.
	Skogen et al. (2022)	Lower subjective social status among adolescents is associated with significantly more negative social media experiences and fewer positive ones.	Links SubStat directly to digital interaction quality, providing theoretical grounding for the PasEx $\times$ SubStat interaction hypothesis.

Passive Social Media Exposure and Upward Comparison	Festinger (1954)	In the absence of objective benchmarks, individuals evaluate themselves through comparison with others; comparison processes influence self-perception and behaviour.	Provides the foundational theoretical mechanism for the moderation hypothesis: passive exposure to aspirational content on social media operationalises the comparison process at scale.
	Huang (2017)	Meta-analysis of 67 studies: the overall correlation between time spent on social network sites and well-being is negligible; quantity of use does not robustly predict psychological outcomes.	Justifies the distinction between total social media time (SOC_HRS, a control) and passive upward comparison content exposure (PasEx, the key predictor).
	Kross et al. (2021)	Social media effects on emotional well-being are not universally positive or negative but are highly context-dependent, shaped by user characteristics, platform	Supports the moderated regression framework: the effect of PasEx on well-being is theorised to vary as a function of SubStat, consistent with context-dependency

		affordances, and social contexts.	documented by Kross et al.
	Samra et al. (2022)	Experimental exposure to upward comparison images triggers negative self-evaluation, identified as the proximal cognitive mechanism driving depressive symptoms from social media use.	Provides experimental evidence for the upward comparison pathway through which PasEx harms well-being, supporting the causal theoretical direction assumed in the model.
	Zsila & Reyes (2023)	Social media platforms simultaneously foster connection and detrimental comparison; awareness of platform comparison mechanisms is a primary protective factor.	Supports the algorithm literacy intervention recommendation in the policy implications section and contextualises the dual nature of social media exposure.
	Liu & Zhao (2025)	Social media intensity predicts personal relative deprivation among urban Chinese residents; social	Provides the most direct contemporary parallel to the PasEx $\times$ SubStat

		class identity moderates this relationship.	moderation hypothesis in an Asian urban context.
	Zhang et al. (2023)	Social media use does not directly predict well-being but does so indirectly through self-esteem and online social support; cyberbullying weakens these beneficial pathways.	Supports the distinction between active and passive social media use in determining well-being outcomes and contextualises the role of offline social support as a control.
	Li et al. (2025)	Platform exploitation in the digital gig economy negatively affects subjective well-being through reduced job rewards and job security, demonstrating that digital environments can erode psychological security even when employment exists.	Extends the digital harm argument beyond content exposure to structural digital conditions, contextualising the Labour/Informal sector's vulnerability identified in the industry-group analysis.

Age and Gender as Moderating Conditions	Twenge et al. (2022)	Social media use is linked to substantially poorer mental health, particularly among girls (median $\beta = -0.11$ to -0.24); effects are markedly larger when social media is separated from general screen time and boys and girls are analysed separately.	Provides direct empirical justification for including gender as a control variable and for the expectation that social media effects may differ by demographic subgroup in the Bangkok sample.
	Boer et al. (2020)	Problematic social media use is consistently associated with lower mental, school, and social well-being across 29 countries; effects vary by national context, gender, and age group; intensive use is potentially normalised in high-prevalence contexts.	Supports the inclusion of both age and gender as controls and contextualises the cross-national applicability of social media harm findings to the Bangkok adult population.

Well-Being Measurement and Instrument Validation	Kliem et al. (2025)	The WHO-5 is a unidimensional, reliable, and valid measure comparable across gender and age groups; updated population norms reveal a decrease in general well-being compared to two decades ago.	Confirms the appropriateness of the WHO-5 as the dependent variable and situates the Bangkok findings within a contemporary decline in general well-being documented globally.
	Fung et al. (2022)	The Chinese-language WHO-5 is unidimensional, internally consistent, and strongly correlated with established well-being measures in a mainland Chinese university student sample.	Supports the validity of the WHO-5 in Asian language and cultural contexts, complementing the Thai-language validation by Saipanish (2009).
Thai and Asian Contextual Studies	Barua & Tejavivaddhana (2019)	The Sufficiency Economy Philosophy (SEP) demonstrates a consistent positive relationship with well-being outcomes across education, health, and	Establishes that cultural and community-based frameworks are empirically relevant to well-being in the Thai context, supporting the offline community-building

		economic domains in Thailand.	recommendations in the policy implications section.
	Sakkhamduang et al. (2025)	Integrating SEP principles in education improved resilience, ethical decision-making, and well-being among Thai students.	Supports the relevance of locally grounded community interventions as complements to digital and financial policy recommendations.
	Samutachak et al. (2025)	SEP principles of moderation, reasonableness, and prudence supported psychological resilience and reduced perceived inequality during COVID-19 in Thailand.	Contextualises the scarcity and status mechanisms of this study within Thai cultural frameworks and reinforces the case for community-level protective factors.

Table 23. Summary of Key Themes in the Literature Review

Note: Studies marked with * in the main text are cited as instrument validation or methodological references and do not appear in this thematic synthesis. These include Topp et al. (2015), Adler et al. (2000), Dalgard et al. (2006), Consumer Financial Protection Bureau (2017), Singh-Manoux et al. (2005), White (1980), Saipanish (2009), Cochran (1977), Ostic (2021), Easterlin (1974), and Verduyn et al. (2015). Full bibliographic details for all sources are provided in the References section.

Appendix D: Stata Results

D1 Internal Reliability: Cronbach Alpha

Cronbach alpha was computed for all four composite scales using standardised items. All scales exceed the conventional threshold of $\alpha \geq 0.70$.

D1.1 WHO-5 Well-Being Index (5 items)

Test scale = mean(standardized items)						
Item	Obs	Sign	Item-test corr.	Item-rest corr.	Average interitem corr.	alpha
who1	411	+	0.9184	0.8712	0.8198	0.9479
who2	411	+	0.9194	0.8726	0.8191	0.9477
who3	411	+	0.9225	0.8774	0.8167	0.9469
who4	411	+	0.9232	0.8784	0.8162	0.9467
who5	411	+	0.9338	0.8949	0.8080	0.9439
Scale					0.8160	0.9568

Note: Standardised Cronbach alpha = 0.957. All item-total correlations > 0.90.

D1.2 Passive Social Media Exposure: PasExp (3 items)

Item	Obs	Sign	Item-test corr.	Item-rest corr.	Average interitem corr.	alpha
pas_inf	411	+	0.9257	0.8306	0.7685	0.8691
pas_lux	411	+	0.9277	0.8350	0.7628	0.8654
pas_cmp	411	+	0.9145	0.8072	0.7993	0.8885
Scale					0.7769	0.9126

Note: Standardised Cronbach alpha = 0.913. Inter-item correlations range from 0.763–0.799.

D1.3 Psychological Scarcity Subscale: PsySca (3 items)

Item	Obs	Sign	Item-test corr.	Item-rest corr.	Average interitem corr.	alpha
cfpb_hard	411	+	0.9607	0.9104	0.8427	0.9146

cfpb_sit		411	+	0.9522	0.8920	0.8669	0.9287
cfpb_wry		411	+	0.9473	0.8814	0.8809	0.9367
-----+							
Scale						0.8635	0.9499

Note: Standardised Cronbach alpha = 0.950. Items: financial hardship, financial situation worry, future financial worry.

D1.4 Offline Social Support: OSSS-3 (3 normalised items)

Item		Item-test		Item-rest	Average		alpha
		Obs	Sign	corr.	corr.	interitem	
-----+							
close_ppl_std		411	+	0.8002	0.5548	0.6208	0.7661
ppl_care_std		411	+	0.8616	0.6724	0.4676	0.6373
nbr_help_std		411	+	0.8361	0.6220	0.5312	0.6939
-----+							
Scale						0.5399	0.7788

Note: Standardised Cronbach alpha = 0.779. Acceptable for 3-item scale.

D2 Exploratory Factor Analysis: CFPB Subscale Structure

EFA was conducted to confirm the two-subscale structure of the CFPB Financial Well-being Scale: (1) Psychological Scarcity (cfpb_hard, cfpb_sit, cfpb_wry) and (2) Liquidity Constraint (cfpb_emg, cfpb_mon, cfpb_con). Principal-component factors with 1 retained factor per subscale.

D2.1 PsySca Subscale: KMO and Factor Loadings

Kaiser-Meyer-Olkin measure of sampling adequacy				
Variable		kmo		
-----+		-----		
cfpb_hard		0.7342		
cfpb_sit		0.7771		
cfpb_wry		0.8063		
Overall		0.7709		
Factor loadings (1 factor retained, eigenvalue = 2.728)				
Variable		Factor1		Uniqueness
-----+		-----+		-----
cfpb_hard		0.9610		0.0764
cfpb_sit		0.9522		0.0932
cfpb_wry		0.9470		0.1032
Proportion of variance explained: 90.9%				

Note: KMO = 0.771 (meritorious). Single factor explains 90.9% of variance. All loadings > 0.94.

D2.2 CFPB Liquidity Subscale: KMO and Factor Loadings

Kaiser-Meyer-Olkin measure of sampling adequacy		
Variable	kmo	
cfpb_emg_rev		0.7281
cfpb_mon_rev		0.8058
cfpb_con_rev		0.7416
Overall		0.7560
Factor loadings (1 factor retained, eigenvalue = 2.578)		
Variable	Factor1	Uniqueness
cfpb_emg_rev	0.9360	0.1239
cfpb_mon_rev	0.9135	0.1655
cfpb_con_rev	-0.9315	0.1322
Proportion of variance explained: 85.9%		

Note: KMO = 0.756 (meritorious). Items reverse-scored prior to EFA. Single factor explains 85.9% of variance.

D3 Confirmatory Factor Analysis: Standardised Loadings

CFA was estimated via structural equation modelling (SEM) using maximum likelihood with robust standard errors (vce(robust)). Standardised factor loadings are reported below for each construct.

D3.1 WHO-5 Well-Being (1 factor, 5 indicators): SRMR = 0.011, CD = 0.957

Standardised loadings on latent factor WellBeing:		
who1 ->	0.893	(p < 0.001)
who2 ->	0.896	(p < 0.001)
who3 ->	0.901	(p < 0.001)
who4 ->	0.905	(p < 0.001)
who5 ->	0.922	(p < 0.001)
Range: 0.893 - 0.922 All loadings p < 0.001		

D3.2 PasExp (1 factor, 3 indicators): SRMR = 0.000, CD = 0.915

Standardised loadings on latent factor MediaExp:		
pas_inf ->	0.891	(p < 0.001)

```

pas_lux -> 0.897 (p < 0.001)
pas_cmp -> 0.856 (p < 0.001)
Range: 0.856 - 0.897 | All loadings p < 0.001

```

D3.3 PsySca (1 factor, 3 indicators): SRMR = 0.000, CD = 0.953

```

Standardised loadings on latent factor FinStress:
cfpb_hard -> 0.952 (p < 0.001)
cfpb_sit -> 0.925 (p < 0.001)
cfpb_wry -> 0.911 (p < 0.001)
Range: 0.911 - 0.952 | All loadings p < 0.001

```

Note: SRMR = 0.000 for just-identified 3-indicator models is expected (df = 0); CD and factor loadings are the informative fit indicators.

D4 Variance Inflation Factors: Multicollinearity Diagnostics

D4.1 M1 Full Specification (plain OLS)

Variable	VIF	1/VIF
SubStat	3.00	0.3338
PsySca	2.61	0.3838
HHInc	1.88	0.5311
CFPB_liquid	1.85	0.5418
d_age35_44	1.59	0.6281
d_age25_34	1.53	0.6547
d_live_other	1.43	0.6998
d_live_alone	1.38	0.7262
d_live_shared	1.33	0.7531
SOC_HRS	1.29	0.7753
d_gnd_nd	1.17	0.8523
PasExp	1.11	0.8997
d_female	1.05	0.9537
Mean VIF	1.63	

Note: Maximum VIF = 3.00 (SubStat), well below conventional threshold of 10. Multicollinearity is not a concern.

D4.2 M3 Centred Interaction Specification

Variable	VIF	1/VIF
PsySca	2.06	0.4843
SubStat_c	2.05	0.4873
PasExpc_Su~c	1.04	0.9592
PasExp_c	1.04	0.9621

Mean VIF		1.55
----------	--	------

Note: Mean-centred variables used for interaction models. Maximum VIF = 2.06. No multicollinearity concern.

D5 Spearman Correlation Matrix (N = 411)

Spearman rank correlations were computed pairwise. The second row under each coefficient reports the two-tailed significance level (p-value). Two pairs warrant attention for multicollinearity: SubStat-PsySca ($\rho = -0.712$, $p < 0.001$) and SubStat-HHInc ($\rho = 0.658$, $p < 0.001$). Both remain below the conventional $r = 0.80$ threshold. VIF diagnostics in Appendix A4 provide additional confirmation.

```
spearmanr WHO5 PasExp PsySca CFPB_liquid SubStat HHInc SOC_HRS
Number of observations = 411
Key: rho (upper row) | two-tailed p-value (lower row)
```

	WHO5	PasExp	PsySca	CFPB_liq	SubStat	HHInc	SOC_HRS
WHO5	1.0000						
PasExp	-0.4468	1.0000					
PsySca	-0.3894	0.1621	1.0000				
CFPB_liq	0.2435	-0.0724	-0.5775	1.0000			
SubStat	0.3919	-0.1244	-0.7120	0.5242	1.0000		
HHInc	0.1908	-0.0330	-0.4317	0.2612	0.6583	1.0000	
SOC_HRS	-0.0834	0.3473	0.2000	0.0859	-0.1381	-0.1936	1.0000

Note: Spearman rho in first row; two-tailed p-value in second row. Pairs with no p-value listed are diagonal ($\rho = 1$). Highest correlations: SubStat-PsySca ($\rho = -0.712$) and SubStat-HHInc ($\rho = 0.658$), both $p < 0.001$ but both below $r = 0.80$ multicollinearity threshold.

D6 Bootstrap Confidence Intervals: M3 and M1 (1,000 Replications)

Bootstrap resampling (1,000 replications, seed = 42) was applied to both M3 and M1 to verify inference under non-constant variance. Normal (N), Percentile (P), and Bias-Corrected (BC) 95% confidence intervals are reported.

D6.1 Model M3: Parsimonious Interaction Model

Replications = 1,000 R ² = 0.4049 Adj-R ² = 0.3991						
	Observed	Bias	SE	[95% N]	[95% P]	[95% BC]

PasExp		8.6275	+0.2408	3.2863	[2.186, 15.069]	[1.964, 15.221]	[1.578, 14.594]
SubStat		10.2116	+0.0970	1.3458	[7.574, 12.849]	[7.651, 12.980]	[7.479, 12.869]
PasExpSubSt		-2.6783	-0.0351	0.5030	[-3.664, -1.692]	[-3.731, -1.660]	[-3.556, -1.599]
PsySca		-4.0974	-0.0055	0.9861	[-6.030, -2.165]	[-6.031, -2.263]	[-6.031, -2.271]
Constant		18.9841	-0.6319	9.2298	[0.894, 37.074]	[0.948, 37.332]	[2.286, 38.651]

Key: N = Normal | P = Percentile | BC = Bias-Corrected

Note: Bias negligible for all M3 core coefficients. BC-CI for interaction term [-3.556, -1.599] does not cross zero, confirming robustness of moderation finding.

D6.2 Model M1: Full Baseline Model (Core Variables)

Replications = 999* | R2 = 0.3692 | Adj-R2 = 0.3485
 * One replicate failed to converge; excluded from SE calculation.

		Observed	Bias	SE	[95% N]	[95% P]	[95% BC]
PasExp		-9.6225	-0.0207	0.8638	[-11.316, -7.929]	[-11.376, -8.013]	[-11.387, -8.043]
PsySca		-3.6776	-0.0388	1.1036	[-5.841, -1.515]	[-6.067, -1.668]	[-5.907, -1.638]
CFPB_liquid		-9.2729	-0.4638	9.0619	[-27.034, 8.488]	[-28.520, 7.423]	[-28.123, 7.489]
SubStat		3.2677	-0.0241	0.9043	[1.495, 5.040]	[1.420, 4.990]	[1.302, 4.978]
HHInc		-0.2128	+0.0163	1.0195	[-2.211, 1.785]	[-2.192, 1.912]	[-2.123, 1.969]
SOC_HRS		2.2402	+0.0564	1.3272	[-0.361, 4.841]	[-0.281, 5.091]	[-0.361, 4.969]

Key: N = Normal | P = Percentile | BC = Bias-Corrected
 Demographic dummies omitted for brevity; all in full log.

Note: Bias negligible across all M1 coefficients. PasExp, PsySca, and SubStat retain significance across all three CI methods, confirming the stability of M1 findings.

D7 Heteroskedasticity Diagnostics: M1

Plain OLS (non-robust) was first estimated to enable formal heteroskedasticity testing. All substantive models were subsequently estimated with `vce(robust)` as the documented remedy.

D7.1 Breusch-Pagan / Cook-Weisberg Test

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
 Assumption: Normal error terms
 Variable: Fitted values of WHO5
 H0: Constant variance

chi2(1) = 4.66
 Prob > chi2 = 0.0308

Note: Null of constant variance rejected ($p = 0.031$). Heteroskedasticity confirmed. Robust standard errors are applied.

D7.2 White's General Test

White's test
H0: Homoskedasticity
Ha: Unrestricted heteroskedasticity

chi2(84) = 178.93
Prob > chi2 = 0.000

Cameron & Trivedi decomposition:

Source	chi2	df	p
Heteroskedasticity	178.93	84	0.000
Skewness	31.06	13	0.003
Kurtosis	1.64	1	0.200
Total	211.63	98	0.000

Note: White's test confirms heteroskedasticity broadly. Kurtosis component not significant ($p = 0.200$).

D7.3 Residual Normality Tests

Shapiro-Wilk W test for normal data:

Variable	Obs	W	V	z	Prob>z
resid_ols	411	0.9940	1.701	1.265	0.1029

Skewness-Kurtosis joint test:

Variable	Obs	Pr(skew)	Pr(kurt)	Adj chi2(2)	Prob>chi2
resid_ols	411	0.7851	0.4303	0.70	0.7043

Note: Both normality tests fail to reject H0 ($W = 0.994$, $p = 0.103$; $\chi^2(2) = 0.70$, $p = 0.704$). OLS inference is reliable.

D8 Outlier and Leverage Diagnostics: Cook's D Analysis

D8.1 Cook's D Summary

Cook's D threshold ($4/N = 4/411$) = 0.0097
Observations exceeding threshold: 22
Maximum Cook's D: 0.026
Standardised residuals > |3|: 0
Standardised residuals > |2.5|: 5

D8.2 Sensitivity Re-estimation Excluding 22 Influential Observations ($n = 389$)

Linear regression ($N = 389$) $F(13, 375) = 34.72$ $p < 0.001$ $R^2 = 0.440$

Variable	Coef (excl.)	Coef (full)	Diff
PasExp	-9.068	-9.622	+0.554

PsySca		-3.463	-3.678	+0.215
SubStat		+3.676	+3.268	-0.408
CFPB_liquid		-5.475	-9.273	+3.798
HHInc		-0.797	-0.213	-0.584

All core variables retain significance and sign.
No small cluster of cases drives the main findings.

Note: Core coefficients are stable across full and reduced samples. PasExp, PsySca, and SubStat remain highly significant ($p < 0.001$) in both specifications.

D9 Descriptive Statistics — summarize, tab, and tabulate Output

This appendix presents the raw Stata output from summarize (detailed), tab, and tabulate commands used to profile the sample and construct the descriptive statistics reported in Chapter 4.1.

D9.1 summarize: Compact Summary of All Study Variables

. summarize WHO5 PasExp PsySca CFPB_liquid SubStat HHInc SOC_HRS OSS3						
Variable		Obs	Mean	Std. dev.	Min	Max
WHO5		411	53.6156	23.8196	0	100
PasExp		411	2.9416	1.1069	1	5
PsySca		411	1.3706	1.3222	0	4
CFPB_liquid		411	0.5523	0.1521	0	0.917
SubStat		411	6.2384	1.9882	1	10
HHInc		411	4.0049	1.3633	1	7
SOC_HRS		411	2.5620	0.9566	1	5
OSS3		411	0.3979	0.1881	0	1

Note: HHInc brackets: 1 = <10,000 THB; 7 = >150,000 THB per month. SOC_HRS: 1 = <1 hr/day; 5 = >6 hrs/day. OSS3 normalised to 0-1 range.

D9.2 summarize, detail: Distributional Statistics per Variable

WHO-5 Well-Being Index (0-100)
Percentiles: p1=0 p5=20 p10=20 p25=36 p50=52 p75=76 p90=80 p95=84 p99=100
Mean=53.616 SD=23.820 Variance=567.374 Skewness=0.014 Kurtosis=2.012

Passive Social Media Exposure (mean 3 items, 1-5)
Percentiles: p1=1 p5=1.667 p10=2 p25=2 p50=2.667 p75=4 p90=4.667 p95=5 p99=5
Mean=2.942 SD=1.107 Variance=1.225 Skewness=0.429 Kurtosis=1.926

PsySca — CFPB worry subscale (0-4)
 Percentiles: p1=0 p5=0 p25=0 p50=1 p75=2.667 p90=3.333 p95=3.667 p99=3.667
 Mean=1.371 SD=1.322 Variance=1.748 Skewness=0.458 Kurtosis=1.731

CFPB Liquidity subscale (0-1)
 Percentiles: p1=0.167 p5=0.333 p25=0.417 p50=0.583 p75=0.667 p90=0.667 p95=0.833
 Mean=0.552 SD=0.152 Variance=0.023 Skewness=-0.202 Kurtosis=3.010

SubStat — MacArthur Ladder (1-10)
 Percentiles: p1=1 p5=3 p10=4 p25=5 p50=6 p75=8 p90=9 p95=10 p99=10
 Mean=6.238 SD=1.988 Variance=3.953 Skewness=-0.076 Kurtosis=2.624

HHInc — Household income bracket (1-7)
 Percentiles: p1=1 p5=2 p10=2 p25=3 p50=4 p75=5 p90=6 p95=6 p99=7
 Mean=4.005 SD=1.363 Variance=1.859 Skewness=-0.096 Kurtosis=2.877

SOC_HRS — Daily social media hours (1-5)
 Percentiles: p1=1 p5=1 p10=2 p25=2 p50=2 p75=3 p90=4 p95=5 p99=5
 Mean=2.562 SD=0.957 Variance=0.915 Skewness=1.036 Kurtosis=3.533

OSSS3 — Offline Social Support Scale (0-1)
 Percentiles: p1=0.083 p5=0.167 p10=0.278 p25=0.278 p50=0.361 p75=0.472 p90=0.667
 Mean=0.398 SD=0.188 Variance=0.035 Skewness=0.986 Kurtosis=3.480

Note: Full detail output from Stata summarize, detail command. Kurtosis reported as raw (not excess); normal distribution kurtosis = 3. WHO-5 kurtosis = 2.01 indicates platykurtic distribution (flatter than normal) consistent with the discrete x4 scoring structure.

D9.3 tab At-Risk Well-Being Prevalence (WHO-5 <= 50)

```
. tab atrisk

WHO5 <= 50 |
= at-risk |
well-being |      Freq.    Percent    Cum.
-----+-----
           0 |      208     50.61     50.61
           1 |      203     49.39    100.00
-----+-----
        Total |      411    100.00
```

Note: atrisk = 1 if WHO-5 score <= 50 (clinical screening threshold for impaired well-being). 49.4% of the sample (n = 203) fall at or below threshold.

D9.4 tab: Gender Distribution

```
. tab gender_raw
```

gender_raw	Freq.	Percent	Cum.
-----+-----			
Male	170	41.36	41.36
Female	235	57.18	98.54
Not disclosed	6	1.46	100.00
-----+-----			
Total	411	100.00	

D9.5 tab: Age Group Distribution

```
. tab age_raw
```

age_raw	Freq.	Percent	Cum.
-----+-----			
18 - 24	117	28.47	28.47
25 - 34	165	40.15	68.61
35 - 44	129	31.39	100.00
-----+-----			
Total	411	100.00	

D9.6 tab: Living Arrangement Distribution

```
. tab living_raw
```

living_raw	Freq.	Percent	Cum.
-----+-----			
With family / guardian	155	37.71	37.71
Live alone	101	24.57	62.28
With friends/partner/shared	77	18.73	81.02
Rented house	18	4.38	85.40
Apartment	17	4.14	89.54
Condominium	15	3.65	93.19
Dormitory	16	3.89	97.08
Live at workplace	12	2.92	100.00
-----+-----			
Total	411	100.00	

Note: Thai raw categories translated to English for readability. Dummy variables constructed as:

d_live_alone (อยู่คนเดียว, n=101, 24.6%), d_live_shared (อยู่ร่วมกับเพื่อน/ผู้รัก/แชร์ที่พัก, n=77, 18.7%), d_live_other (Rented/Condo/Dormitory/Workplace, n=78, 19.0%), reference category = with family/guardian (n=155, 37.7%).

D9.7 tab: Industry Group Distribution

```
. tab WrkIndy_grp
```

WrkIndy_grp	Freq.	Percent	Cum.
Corporate/Prof	88	21.41	21.41
Public/Essential	136	33.09	54.50
Commerce/Hosp	136	33.09	87.59
Labor/Ag/Informal	39	9.49	97.08
Unclassified	12	2.92	100.00
Total	411	100.00	

Note: Corporate/Prof = Finance & ICT. Public/Essential = Government, Education, Health. Commerce/Hosp = Retail, Hospitality, Logistics. Labour/Ag/Informal = Manufacturing, Agriculture, Casual labour.

D9.8 tabulate: Industry Group by Subjective Status (Low vs High)

```
. tabulate WrkIndy_grp high_status, row column
Key: frequency | row % | column %
```

	Low Status (SubStat 1-5)	High Status (SubStat 6+)	Total
Corporate/Prof	30 (34.1%) 19.9%	58 (65.9%) 22.3%	88 21.4%
Public/Essential	50 (36.8%) 33.1%	86 (63.2%) 33.1%	136 33.1%
Commerce/Hosp	53 (39.0%) 35.1%	83 (61.0%) 31.9%	136 33.1%
Labor/Ag/Informal	15 (38.5%) 9.9%	24 (61.5%) 9.2%	39 9.5%
Unclassified	3 (25.0%) 2.0%	9 (75.0%) 3.5%	12 2.9%
Total	151 (36.7%)	260 (63.3%)	411

Note: Low Status = SubStat ≤ 5 (n = 151, 36.7%). High Status = SubStat ≥ 6 (n = 260, 63.3%). Row percentages sum to 100% within each industry group. No industry group is heavily concentrated in the low-status category.

D9.8 tabulate: Household Income by Subjective Status

```
. tabulate HHInc high_status, row column
Key: frequency | row % | column %
```

HHInc	Low Status (SubStat 1-5)		High Status (SubStat 6+)		Total
1	16	(88.9%)	2	(11.1%)	18 (4.4%)
2	24	(60.0%)	16	(40.0%)	40 (9.7%)
3	33	(47.1%)	37	(52.9%)	70 (17.0%)
4	72	(50.0%)	72	(50.0%)	144 (35.0%)
5	0	(0.0%)	87	(100.0%)	87 (21.2%)
6	4	(10.8%)	33	(89.2%)	37 (9.0%)
7	2	(13.3%)	13	(86.7%)	15 (3.6%)
Total	151	(36.7%)	260	(63.3%)	411

Note: HHInc 1 = <10,000 THB; 7 = >150,000 THB per month. The income-status gradient is strong: 88.9% of HHInc bracket 1 are low-status vs 0% of bracket 5. All 87 respondents in bracket 5 (50,001-80,000 THB) are classified as high-status, suggesting good convergent validity between objective income and subjective social rank.